The Unification of Matter, Forces, and Spacetime from the Bost–Connes System

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Abstract

Starting from the Bost–Connes system, we perform the p-adic reduction at a prime p and construct the state space

$$\mathcal{H}_p = L^2(\mathbb{Z}/p^K\mathbb{Z}), \qquad K = p(p-1)-1,$$

and argue that its structural properties naturally describe the physical world. By analyzing the structure of the state space, we derive general-p formulas for spacetime, matter, forces, and cosmic evolution; when the characteristic quantities of the p channels are compared with observation– three generations of fermions, five elementary particles, three gauge forces, and the Standard Model gauge group– the unique compatible value is $p = 7$: the physical world we inhabit is described by the state space over $\mathbb{Z}/7\mathbb{Z}$. Fixing $p = 7$ in the general-p formulas and computing the outputs one by one: the electroweak scale $246.19\,\text{GeV}$ (-0.011%), the Planck mass $1.221 \times 10^{19}\,\text{GeV}$ ($+0.04\%$), the proton mass $937\,\text{MeV}$ (-0.17%), the nine charged-fermion masses all within 1.5% (the three charged leptons at experimental precision), the electromagnetic coupling $\alpha_{\text{EM}}^{-1} = 137.036$ ($+0.0003\%$), the color coupling $\alpha_s^{-1} = 8.49$ ($+0.08\%$), the weak coupling $\alpha_W^{-1} = 29.60$ (-0.004%), the nuclear coupling $\alpha_{\text{nuclear}} = 14.10$ ($+0.7\%$), and the gravitational constant $G = 6.70 \times 10^{-39}\,\text{GeV}^{-2}$ (-0.09%). Neutrinos are of Majorana type, with absolute masses 5.07, 9.96, $50.6\,\text{meV}$ for the three generations and mass-squared-difference ratio 34.5 (experiment 33.8 ± 0.9), a decisive prediction.

Contents

§1 The Bost–Connes System

The Bost–Connes system is a quantum statistical mechanical system constructed in 1995[1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

$$\mathcal{A}_{\mathrm{BC}} = C^*(\mathbb{Q}/\mathbb{Z}) \rtimes \mathbb{N}^*.$$

This paper starts from its local decomposition: prime-by-prime reduction yields finite-dimensional state spaces; the geometric properties of the state space are examined, and a physical description is then established on it. This section first introduces the definition of the system.

§1.1 Definition of the System

The BC system is completely determined by three items of arithmetic data:

(i) the object set $\mathbb{N}^* = \{1, 2, 3, \ldots\}$;

(ii) unique factorization: $n = \prod_p p^{a_p(n)}$ (the fundamental theorem of arithmetic);

(iii) the Hamiltonian: $H(n) = \log n$ –the nonnegative, regular, completely additive function on $\mathbb{N}^*$, unique up to positive scaling.

The following five structures are all determined by (i)–(iii).

(iv) C^*-algebra: $\mathcal{A}_{\mathrm{BC}} = C^*\langle e(r), \mu_n\rangle$, with generators satisfying

$$\begin{aligned} &e(r_1)e(r_2) = e(r_1 + r_2), \quad \mu_n\mu_m = \mu_{nm}, \quad \mu_n^*\mu_n = 1, \\ &e(r)\,\mu_n = \mu_n\, e(nr) \text{ (covariance relation)}, \quad \mu_n\, e(r)\,\mu_n^* = \frac{1}{n}\sum_{ns=r} e(s) \text{ (transfer relation)}. \end{aligned} \tag{1}$$

Uniquely determined by the arithmetic structure of (i)(ii).

(v) Time evolution: $\sigma_t(\mu_n) = n^{it}\mu_n$, $\sigma_t(e(r)) = e(r)$ (uniquely determined by (iii): $n^{it} = e^{it\log n}$).

(vi) Partition function:

$$Z(\beta) = \sum_n n^{-\beta} = \zeta(\beta) = \prod_p (1 - p^{-\beta})^{-1}, \quad \beta > 1. \tag{2}$$

Obtained by direct summation from (iii); the Euler product[19] is given by (ii).

(vii) KMS states:

$$\phi_\beta(|n\rangle\langle m|) = \delta_{nm}\,\frac{n^{-\beta}}{\zeta(\beta)}, \qquad \beta > 1. \tag{3}$$

The main theorem of Bost–Connes gives the complete classification of KMS states: for $\beta \le 1$ the KMS state is unique; for $\beta > 1$ the extremal KMS states are parametrized by $\hat{\mathbb{Z}}^* \cong \mathrm{Gal}(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q})$–symmetry breaking. Equation (3) is the Gibbs diagonal value shared by the extremal KMS states with $\beta > 1$ on matrix units.

(viii) Tensor product decomposition:

$$\ell^2(\mathbb{N}^*) \cong \widehat{\bigotimes}_p \ell^2(\mathbb{Z}_{\ge 0}), \quad |n\rangle \mapsto \bigotimes_p |a_p\rangle_p. \tag{4}$$

Uniquely determined by (ii); under this decomposition σ_t factorizes over the prime factors.

Uniqueness of $H = \log n$. Let $f : \mathbb{N}^* \to \mathbb{R}$ satisfy (a) $f(mn) = f(m) + f(n)$, (b) $f(n) \geq 0$, (c) $f(n+1) - f(n) \to 0$. Then $f = c \log n$, $c \geq 0$.

The argument proceeds in three steps.

Step 1. By (a), f is completely determined by its values on the primes: $f(n) = \sum_p a_p(n) f(p)$. (a)(b) alone do not suffice to fix f–each $f(p) \geq 0$ can be chosen independently.

Step 2. Regularity (c) is the key: the Erdős theorem (1946)[22] asserts that a completely additive function satisfying $f(n+1)-f(n) \to 0$ must be $f = c \log n$. The core mechanism: (c) forces the values at adjacent integers to agree asymptotically, while the prime-factor structures of adjacent integers are mutually independent, so the ratios $f(p)/\log p$ in all prime directions are locked to the same constant c.

Step 3. Condition (b) gives $c \geq 0$; $H = \log n$ corresponds to the scale $c = 1$ (the scale is absorbed into the time unit). □

The above completely defines the BC system. The tensor product decomposition (viii) shows that the full system decomposes over primes into a product of local factors. The next section uses this decomposition to extract from the global system the finite-dimensional structure at each prime p–the full group $G_p^{(K)} = \mathbb{Z}/p^K\mathbb{Z}$.

§2 Local Decomposition and the Full Group

The tensor product decomposition (4) shows that the BC system can be analyzed prime by prime. The mathematical basis of this decomposition is Ostrowski's theorem (1916)[23]: the nontrivial absolute values on $\mathbb{Q}$ fall into exactly two classes:

(i) the Archimedean absolute value $|\cdot|_\infty$, giving the completion $\mathbb{R}$;

(ii) for each prime p, the p-adic absolute value $|\cdot|_p$, giving the completion $\mathbb{Q}_p$.

Under this classification, $H(n) = \log n$ is precisely the logarithm of the Archimedean absolute value; the p-adic directions carry the local structure of the BC system at each prime p. This section extracts these local structures and truncates them to the finite-dimensional full group $G_p^{(K)} = \mathbb{Z}/p^K\mathbb{Z}$.

§2.1 Local Decomposition

The phase operators $e(r)$ of the BC algebra act on $\mathbb{Q}/\mathbb{Z}$. $\mathbb{Q}/\mathbb{Z}$ decomposes naturally over primes.

Direct sum decomposition of $\mathbb{Q}/\mathbb{Z}$.

$$\mathbb{Q}/\mathbb{Z} = \bigoplus_p \mathbb{Z}[1/p]/\mathbb{Z}, \tag{5}$$

where $\mathbb{Z}[1/p] = \{a/p^k : a \in \mathbb{Z},\, k \geq 0\}$ is the localization of $\mathbb{Z}$ at p, and the direct sum runs over all primes p.

The argument proceeds in three steps.

Step 1. Any $r \in \mathbb{Q}/\mathbb{Z}$ can be written as $r = a/n$ ($a \in \mathbb{Z}$, $n \geq 1$, $0 \leq a < n$). Let $n = p_1^{k_1} \cdots p_s^{k_s}$ be the unique factorization. By the Chinese remainder theorem[25],

$$\frac{a}{n} = \frac{a}{p_1^{k_1} \cdots p_s^{k_s}} \equiv \frac{a_1}{p_1^{k_1}} + \cdots + \frac{a_s}{p_s^{k_s}} \pmod{\mathbb{Z}},$$

where $a_i \in \{0, 1, \ldots, p_i^{k_i} - 1\}$ is uniquely determined by CRT (using the modular inverse of $\prod_{j \neq i} p_j^{k_j}$). Hence r is uniquely written in $\mathbb{Q}/\mathbb{Z}$ as a sum of p-components $r = \sum_{i=1}^{s} r_{p_i}$, $r_{p_i} \in \mathbb{Z}[1/p_i]/\mathbb{Z}$.

Step 2. If $\sum_p s_p = 0$ ($s_p \in \mathbb{Z}[1/p]/\mathbb{Z}$, finitely many nonzero), then after clearing denominators the p-parts of the numerator are all 0, so each $s_p = 0$. Each $\mathbb{Z}[1/p]/\mathbb{Z}$ is a subgroup (closed under addition), and $\mathbb{Q}/\mathbb{Z} = \bigoplus_p \mathbb{Z}[1/p]/\mathbb{Z}$ holds as a direct sum of Abelian groups.

Step 3. $\mathbb{Z}[1/p]/\mathbb{Z} = \varinjlim(\mathbb{Z}/p^k\mathbb{Z})$, where the direct limit is given by $\mathbb{Z}/p^k\mathbb{Z} \hookrightarrow \mathbb{Z}/p^{k+1}\mathbb{Z}$ (the multiplication-by-p map). □

Concrete construction of $\widehat{\mathbb{Z}}$. The elements of the projective limit $\widehat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n\mathbb{Z}$ are compatible sequences:

$$\widehat{\mathbb{Z}} = \Big\{(a_n)_{n \geq 1} \in \prod_n \mathbb{Z}/n\mathbb{Z} \ : \ m \mid n \Rightarrow a_n \equiv a_m \pmod{m}\Big\}.$$

The argument proceeds in four steps.

Step 1. $\varprojlim \mathbb{Z}/n\mathbb{Z}$ is given by the limit cone of the directed system $\{\pi_{mn} : \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/m\mathbb{Z}\}_{m|n}$ (the natural reduction maps). Concretely, consider the positive integers partially ordered by divisibility $m|n$; for each pair $m|n$ there is a homomorphism $\pi_{mn}(a \bmod n) = a \bmod m$. The elements of the projective limit are exactly the sequences satisfying the compatibility condition above.

Step 2. Componentwise, $(a_n)+(b_n) = (a_n+b_n)$; compatibility is verified directly: if $a_n \equiv a_m$ and $b_n \equiv b_m \pmod{m}$, then $(a_n + b_n) \equiv (a_m + b_m) \pmod{m}$.

Step 3. Similarly, $(a_n)(b_n) = (a_n b_n)$, yielding a ring.

Step 4. Equipped with the subspace topology of the product topology, $\widehat{\mathbb{Z}}$ is compact (Tychonoff's theorem). Typical elements: an integer $m \in \mathbb{Z}$ gives the constant sequence $(m \bmod n)_n$, so $\mathbb{Z} \hookrightarrow \widehat{\mathbb{Z}}$ is a dense embedding. □

Product decomposition of the full profinite integer ring.

$$\widehat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n\mathbb{Z} \ \cong \ \prod_p \mathbb{Z}_p, \tag{6}$$

where $\mathbb{Z}_p = \varprojlim_k \mathbb{Z}/p^k\mathbb{Z}$ is the ring of p-adic integers (the full ring, containing all p-adic integers, not just the units). The isomorphism holds both in the sense of rings and of topological groups.

The argument proceeds in three steps.

Step 1. For $n = \prod_i p_i^{k_i}$ (a finite product), the Chinese remainder theorem gives a ring isomorphism:

$$\mathbb{Z}/n\mathbb{Z} \ \xrightarrow{\sim} \ \prod_i \mathbb{Z}/p_i^{k_i}\mathbb{Z}, \quad a \bmod n \mapsto (a \bmod p_i^{k_i})_i.$$

The map is evidently a ring homomorphism; injectivity comes from CRT ($a \equiv 0 \pmod{p_i^{k_i}}$ for all i means $a \equiv 0 \pmod{n}$); surjectivity comes from $\gcd(p_i^{k_i}, p_j^{k_j}) = 1$ ($i \neq j$) and the solvability part of the Chinese remainder theorem.

Step 2. Taking projective limits over n (in the divisibility order) on both sides, the limit commutes with finite direct products:

$$\varprojlim_n \mathbb{Z}/n\mathbb{Z} \cong \varprojlim_n \prod_p \mathbb{Z}/p^{v_p(n)}\mathbb{Z} \cong \prod_p \varprojlim_k \mathbb{Z}/p^k\mathbb{Z} = \prod_p \mathbb{Z}_p.$$

Here the second step uses the interchange of (finite) direct products with projective limits (compatibility of categorical limits with products): for the primes in a fixed finite set S, $\varprojlim_n \prod_{p\in S}\mathbb{Z}/p^{v_p(n)}\mathbb{Z} = \prod_{p\in S}\mathbb{Z}_p$; then taking the topology (product topology) over all primes gives the full isomorphism.

Step 3. The map $\Phi : \widehat{\mathbb{Z}} \to \prod_p \mathbb{Z}_p$ is defined by: $(a_n)_{n\geq 1} \mapsto (a_{p^k})_{k\geq 1}\big|_{\text{each } p}$, i.e., for each prime p, extract the p-power components. This is a continuous ring isomorphism; the topological isomorphism follows because both sides are compact groups and the map is bijective and continuous. □

Local decomposition of the BC system. Combining the Euler product $\zeta(\beta) = \prod_p(1-p^{-\beta})^{-1}$ of §1 and the tensor product $\ell^2(\mathbb{N}^*) \cong \bigotimes_p \ell^2(\mathbb{Z}_{\geq 0})$, the BC system decomposes into local factors, one for each prime p:

$$\text{BC system} = \prod_p (\text{local system at } p).$$

Each local system corresponds to the structure of the full p-adic ring $\mathbb{Z}_p$.

§2.2 The Full Group

The local decomposition above reduces the BC system to independent factors at each prime p. Fix p; the local system at p is built on $\mathbb{Z}_p$. Taking the K-digit truncation of $\mathbb{Z}_p$, define the **full group**

$$G_p^{(K)} = \mathbb{Z}/p^K\mathbb{Z} \quad (\text{as a ring, with } p^K \text{ elements}),$$

i.e., the natural truncation $\pi_K : \mathbb{Z}_p \twoheadrightarrow \mathbb{Z}/p^K\mathbb{Z}$. Note that what is taken here is the full ring $\mathbb{Z}/p^K\mathbb{Z}$, not the unit group $(\mathbb{Z}/p^K\mathbb{Z})^*$. The reason is as follows.

The phase operator $e(a/p^K)$ of the BC algebra is defined for all $\alpha \in \mathbb{Z}/p^K\mathbb{Z}$, acting as:

$$e(a/p^K)\,|\alpha\rangle = e^{2\pi i a\alpha/p^K}\,|\alpha\rangle, \qquad \alpha \in \mathbb{Z}/p^K\mathbb{Z}.$$

This is equally well defined for $\alpha = 0, p, 2p, \ldots, p^K - p$ (i.e., the multiples of p), giving special phases:

$$e(a/p^K)\,|mp\rangle = e^{2\pi i am/p^{K-1}}\,|mp\rangle, \quad m = 0, 1, \ldots, p^{K-1}-1.$$

If only the unit group $\alpha \in (\mathbb{Z}/p^K\mathbb{Z})^*$ were taken (excluding the multiples of p), the crossed relations would break. A complete argument follows.

Step 1. The covariance relation $e(r)\,\mu_n = \mu_n\,e(nr)$ requires: for each $r = a/p^K$ and each $n \in \mathbb{N}^*$, μ_n maps $|\alpha\rangle$ (α a unit) to some $|\alpha'\rangle$ (α' should also lie in the state space).

Step 2. Under the natural action on $\mathbb{Z}/p^K\mathbb{Z}$, μ_p maps α to $p\alpha \bmod p^K$. Take $\alpha = 1$: $\mu_p\,|1\rangle = |p\rangle$. But $p\cdot 1 = p \notin (\mathbb{Z}/p^K\mathbb{Z})^*$ (p is not a unit).

Step 3. If the state space contained only $\{|\alpha\rangle : \gcd(\alpha, 3) = 1\}$, take $n = 3$, $r = 1/3^5$. Acting with both sides of the covariance relation on $|1\rangle$:

$$e(1/3^5)\,\mu_3\,|1\rangle = e(1/3^5)\,|3\rangle, \qquad \mu_3\,e(3/3^5)\,|1\rangle = e^{2\pi i/3^4}\,\mu_3\,|1\rangle = e^{2\pi i/3^4}\,|3\rangle,$$

both sides require $\mu_3|1\rangle = |3\rangle$ to exist in the state space– but $3 \equiv 0 \pmod 3$, i.e., 3 is a multiple of 3 in $\mathbb{Z}/3^K\mathbb{Z}$ and does not belong to the unit group. If $|3\rangle$ is not in the state space, the covariance relation $e(r)\,\mu_3 = \mu_3\, e(3r)$ cannot hold at $|1\rangle$ (the resulting vector lies outside the state space), signaling a break.

Hence the complete representation of the BC algebra requires the state space to contain the multiples of p, and the full group must be the full ring $\mathbb{Z}/p^K\mathbb{Z}$ rather than the unit group $(\mathbb{Z}/p^K\mathbb{Z})^*$. $\square$

§2.3 Determination of $K_{\max}$

The full-ring truncation $\mathbb{Z}/p^k\mathbb{Z}$ is defined for every $k \geq 1$, and the truncations are connected by the natural projections $\mathbb{Z}/p^{k+1}\mathbb{Z} \twoheadrightarrow \mathbb{Z}/p^k\mathbb{Z}$. The value of K should be fixed by the algebraic structure of the system itself, not by an external choice. The natural condition determining its maximal value is: the full ring should completely accommodate the smallest transformation group of $\mathbb{F}_p$ involving both addition and multiplication–the affine group

$$\mathrm{Aff}(\mathbb{F}_p) = \mathbb{F}_p \rtimes \mathbb{F}_p^* = \{x \mapsto ax + b : a \in \mathbb{F}_p^*,\, b \in \mathbb{F}_p\}, \quad |\mathrm{Aff}(\mathbb{F}_p)| = p(p-1).$$

Consider the augmentation map $\varepsilon : \mathbb{F}_p[\mathrm{Aff}(\mathbb{F}_p)] \to \mathbb{F}_p$ of its group algebra $\mathbb{F}_p[\mathrm{Aff}(\mathbb{F}_p)]$, $\varepsilon(\sum a_g g) = \sum a_g$, with augmentation ideal $I = \ker(\varepsilon)$. As an $\mathbb{F}_p$-vector space, $\mathbb{F}_p[\mathrm{Aff}(\mathbb{F}_p)]$ has dimension $p(p-1)$; ε is a rank-1 map, so $\dim(I) = p(p-1) - 1$. Set

$$\boxed{K_{\max} = \dim_{\mathbb{F}_p}(I) = p(p-1) - 1.} \tag{7}$$

p	$\lvert\mathrm{Aff}(\mathbb{F}_p)\rvert$	$K_{\max} = p(p-1) - 1$	$\dim \mathcal{H}_p = p^{K_{\max}}$
2	2	1	2
3	6	5	243
5	20	19	$\approx 1.9 \times 10^{13}$
7	42	41	$\approx 4.5 \times 10^{34}$
11	110	109	$\approx 10^{113}$

Affine completeness determines the maximal value of K; the state-space construction of this paper takes $K \equiv K_{\max}$. In §10, different physical structures take different values of k according to their own conditions.

§2.4 The Unit Group and the Teichmüller Decomposition

The p^K elements of the full ring $\mathbb{Z}/p^K\mathbb{Z}$ fall into two classes according to their p-adic valuation: the units $(\mathbb{Z}/p^K\mathbb{Z})^*$ (coprime to p, $\varphi(p^K) = p^{K-1}(p-1)$ in total, possessing multiplicative inverses) and the non-units (the multiples of p, $\{0, p, 2p, \ldots, p^K - p\}$, p^{K-1} in total, without multiplicative inverses). Classified modulo p, the full group forms p "chains": $(p-1)$ unit chains plus one boundary chain.

The unit group $(\mathbb{Z}/p^K\mathbb{Z})^*$ (p an odd prime) admits the canonical Teichmüller splitting[27]:

$$(\mathbb{Z}/p^K\mathbb{Z})^* \cong \mu_{p-1} \times (1 + p\mathbb{Z}/p^K\mathbb{Z}), \tag{8}$$

where $\mu_{p-1} \cong (\mathbb{Z}/p\mathbb{Z})^* \cong \mathbb{Z}/(p-1)\mathbb{Z}$ (the Teichmüller part) and $1 + p\mathbb{Z}/p^K\mathbb{Z} \cong \mathbb{Z}/p^{K-1}\mathbb{Z}$ (the group of 1-units). Each $\alpha \in (\mathbb{Z}/p^K\mathbb{Z})^*$ decomposes uniquely as $\alpha = \omega \cdot (1 + c_1 p + \cdots + c_{K-1}p^{K-1})$.

The parameter α of a KMS state is a specific unit in $\mathbb{Z}_p^* \subset \mathbb{Z}_p$. But the state space is built on the full ring $\mathbb{Z}/p^K\mathbb{Z}$ (§3), not just the unit group. The Teichmüller part μ_{p-1} will be further decomposed via CRT in §3.

§3 Algebraic Structure of the State Space

§2.2 determined that the underlying set of the BC algebra at the prime p is the full ring $\mathbb{Z}/p^K\mathbb{Z}$ (the crossed relations require the multiples of p to be included), and §2.3 determined $K_{\max} = p(p-1) - 1$ (the state space takes $K \equiv K_{\max}$). To build a state space on the full ring, an inner-product structure is needed to define orthogonality. The function space on the full ring, equipped with the standard inner product, gives the **state space**:

$$\boxed{\mathcal{H}_p = L^2(\mathbb{Z}/p^K\mathbb{Z},\ \mathbb{C}), \quad \dim \mathcal{H}_p = p^K.} \tag{9}$$

Inner product: for $f, g \in \mathcal{H}_p$,

$$\langle f, g \rangle = \frac{1}{p^K} \sum_{\alpha \in \mathbb{Z}/p^K\mathbb{Z}} \overline{f(\alpha)}\, g(\alpha).$$

$\mathcal{H}_p$ is a p^K-dimensional Hilbert space. Finite dimensionality guarantees that all operators are bounded and all spectra are completely discrete– the entire spectral structure of the state space can be described by finite matrices.

The additive Fourier basis

$$\chi_j^{(\mathrm{add})}(\alpha) = e^{2\pi i j\alpha/p^K}, \quad j = 0, 1, \ldots, p^K - 1,$$

forms an orthonormal basis of $\mathcal{H}_p$: $\langle \chi_j^{(\mathrm{add})}, \chi_k^{(\mathrm{add})} \rangle = \delta_{jk}$. Besides the additive Fourier basis, the state space carries a second family of basis functions–the multiplicative characters, defined on the unit group $(\mathbb{Z}/p\mathbb{Z})^*$.

This section establishes the algebraic structure of the state space: the multiplicative characters and their properties, the spectral structure, symmetries, and metrics.

§3.1 Multiplicative Characters

A **multiplicative character** of $(\mathbb{Z}/p\mathbb{Z})^*$ is a group homomorphism $\chi : (\mathbb{Z}/p\mathbb{Z})^* \to \mathbb{C}^*$, $\chi(ab) = \chi(a)\chi(b)$. Since $(\mathbb{Z}/p\mathbb{Z})^*$ is a cyclic group of order $p-1$ (with generator g), all multiplicative characters are given by $\chi_k(g) = e^{2\pi i k/(p-1)}$ ($k = 0, \ldots, p-2$), $p-1$ in total. The subsequent subsections describe the properties of these $p-1$ characters in turn: classification by order, the subgroup lattice, channel types, and CRT coordinates.

The multiplicative character χ_k is embedded into $\mathcal{H}_p$ as follows:

$$\tilde{\chi}_k(\alpha) = \begin{cases} \chi_k(\alpha \bmod p), & \text{if } \gcd(\alpha, p) = 1, \\ 0, & \text{if } p \mid \alpha. \end{cases} \tag{10}$$

That is, the character value is taken on the units, and 0 on the multiples of p. Comparison of the two kinds of characters:

	Additive character	Multiplicative character
Domain	$\mathbb{Z}/p^K\mathbb{Z}$ (full group)	$(\mathbb{Z}/p\mathbb{Z})^*$ (unit group)
Homomorphism property	$\chi(a+b) = \chi(a)\chi(b)$	$\chi(ab) = \chi(a)\chi(b)$
Number	p^K	$p-1$

The additive characters form a complete orthogonal basis of L^2[28]; the multiplicative characters are embedded into L^2 via (10) and carry the multiplicative structure of the full group.

3.1.1 Classification by Order

The **order** of a multiplicative character χ_k is the smallest positive integer d with $\chi_k^d = \chi_0$, namely $d = (p-1)/\gcd(k, p-1)$. The $p-1$ multiplicative characters are classified by order $d \mid (p-1)$:

- there are exactly $\varphi(d)$ characters of order d (φ the Euler function);
- check: $\sum_{d|(p-1)} \varphi(d) = p-1$ (Euler's identity).

Example: $p = 13$, $p-1 = 12 = 2^2 \times 3$. The divisors of 12 are $1, 2, 3, 4, 6, 12$. The characters classified by order:

Order d	$\varphi(d)$	Number of characters	Real/complex
1	1	1	real (trivial)
2	1	1	real ($\chi_6 = \bar{\chi}_6$)
3	2	2	complex (conjugate pair)
4	2	2	complex (conjugate pair)
6	2	2	complex (conjugate pair)
12	4	4	complex (two conjugate pairs)

In total $1+1+2+2+2+4 = 12 = p-1$ characters, falling into $|\{1, 2, 3, 4, 6, 12\}| = 6$ distinct orders.

3.1.2 The Subgroup Lattice

The subgroup lattice $\mathrm{Lat}(\mathbb{Z}/(p-1)\mathbb{Z})$ of $\mathbb{Z}/(p-1)\mathbb{Z}$ is a partially ordered set: for each $d \mid (p-1)$ there is a unique subgroup $\langle d\rangle \cong \mathbb{Z}/((p-1)/d)\mathbb{Z}$, with inclusion given by divisibility: $\langle d_1\rangle \subseteq \langle d_2\rangle \iff d_2 \mid d_1$.

3.1.3 Channel Topology

For a nontrivial character χ_k ($k \neq 0$), its square is $\chi_k^2 = \chi_{2k \bmod (p-1)}$. According to whether the square lands on the trivial character, two channel types are defined:

- **Type I channel** (closed): $\chi_k^2 \neq \chi_0$ (the square is still a nontrivial character);
- **Type II channel** (open): $\chi_k^2 = \chi_0$ (the square lands on the trivial character).

The channel type is completely determined by the order. For $k \neq 0$, $\chi_k^2 = \chi_0$ if and only if $2k \equiv 0 \pmod{p-1}$, i.e., $k = (p-1)/2$. In that case the order of χ_k is $(p-1)/\gcd(k, p-1) = (p-1)/((p-1)/2) = 2$. Hence among the nontrivial characters, only the one of order 2 is Type II; all others are Type I. $\square$

For example, take $p = 13$ ($p-1 = 12$):

χ_k	Order	χ_k^2	Channel
χ_1	12	χ_2 (order 6)	Type I
χ_2	6	χ_4 (order 3)	Type I
χ_3	4	χ_6 (order 2)	Type I
χ_4	3	χ_8 (order 3)	Type I
χ_6	2	χ_0	**Type II**

Only χ_6 (the unique character of order 2, $k = (p-1)/2 = 6$) is a Type II channel; the unlisted $\chi_5, \chi_7, \ldots, \chi_{11}$ are conjugate to the listed directions and have the same channel types.

3.1.4 CRT Decomposition

Let $p - 1 = q_1^{b_1} \cdots q_s^{b_s}$. Then the character group decomposes as

$$\mathbb{Z}/(p-1)\mathbb{Z} \cong \mathbb{Z}/q_1^{b_1}\mathbb{Z} \times \cdots \times \mathbb{Z}/q_s^{b_s}\mathbb{Z}, \tag{11}$$

inducing a tensor product of the mode space: $L^2(\mu_{p-1}) \cong \bigotimes_{i=1}^{s} L^2(\mathbb{Z}/q_i^{b_i}\mathbb{Z})$.

The argument proceeds in three steps.

Step 1. The q_i are pairwise coprime; the Chinese remainder theorem gives a group isomorphism: $\mathbb{Z}/(p-1)\mathbb{Z} \cong \mathbb{Z}/q_1^{b_1}\mathbb{Z} \times \cdots \times \mathbb{Z}/q_s^{b_s}\mathbb{Z}$, with map $a \mapsto (a \bmod q_1^{b_1}, \ldots, a \bmod q_s^{b_s})$.

Step 2. For a product group $G = \prod_i G_i$ ($G_i = \mathbb{Z}/q_i^{b_i}\mathbb{Z}$), Pontryagin duality gives $\widehat{G} \cong \prod_i \widehat{G_i}$. Each character decomposes uniquely as $\chi = \chi_1 \otimes \cdots \otimes \chi_s$.

Step 3. The orthogonal basis $\{\chi\}_{\chi\in\widehat{G}}$ of $L^2(G)$ corresponds bijectively via CRT to the basis $\{\chi_1 \otimes \cdots \otimes \chi_s\}$ of $\bigotimes_i L^2(G_i)$. The inner product is preserved:

$$\langle \chi_1 \otimes \cdots \otimes \chi_s,\ \chi_1' \otimes \cdots \otimes \chi_s' \rangle = \prod_i \delta_{\chi_i, \chi_i'}.$$

Hence $L^2(G) \cong \bigotimes_i L^2(G_i)$ is an isomorphism of Hilbert spaces. □

The mode space thus decomposes into $s = \omega(p-1)$ independent directions.

§3.2 Spectral Structure

3.2.1 The Convolution Operator

For $f \in \mathcal{H}_p$, the **convolution operator** K is defined as the KMS-weighted sum of scaling operators:

$$(Kf)(\alpha) = \sum_{\substack{n\geq 1 \\ p\nmid n}} n^{-\beta} f(n\alpha), \qquad \alpha \in \mathbb{Z}/p^K\mathbb{Z}, \quad \beta > 1. \tag{12}$$

$\beta > 1$ guarantees absolute convergence of the series, so K is bounded; $p \nmid n$ guarantees that $\alpha \mapsto n\alpha$ is invertible; the weights $n^{-\beta}$ come from the KMS state, and the scaling action comes from the covariance relation $e(r)\mu_n = \mu_n e(nr)$. K is a bounded linear operator on $\mathcal{H}_p$.

3.2.2 Diagonalization Theorem

The convolution operator K is diagonalized in the multiplicative character basis $\{\tilde{\chi}_k\}_{k=0}^{p-2}$ ((10)):

$$K\tilde{\chi}_k = \hat{K}(\chi_k) \cdot \tilde{\chi}_k, \tag{13}$$

with eigenvalues the Dirichlet L-functions:

$$\boxed{\hat{K}(\chi_k) = L(\beta, \chi_k) = \sum_{\substack{n\geq 1 \\ p\nmid n}} \chi_k(n)\, n^{-\beta}.} \tag{14}$$

In particular, $\hat{K}(\chi_0) = \zeta(\beta) \cdot (1 - p^{-\beta})$ (the ζ function with the Euler factor at p removed).

Argument. $\tilde{\chi}_k(\alpha) = \chi_k(\alpha \bmod p)$ takes the character value on units and vanishes on non-units. For $p \nmid n$, α and $n\alpha$ are both units or both non-units, and $\tilde{\chi}_k(n\alpha) = \chi_k(n)\, \tilde{\chi}_k(\alpha)$. Substituting into (12):

$$(K\tilde{\chi}_k)(\alpha) = \sum_{p \nmid n} n^{-\beta} \chi_k(n)\, \tilde{\chi}_k(\alpha) = L(\beta, \chi_k)\, \tilde{\chi}_k(\alpha). \qquad \square$$

3.2.3 Structure of the Eigenvalue Spectrum

For each prime p, K has $p-1$ eigenvalues in the multiplicative directions, $\{\lambda_k = L(\beta, \chi_k) : k = 0, \dots, p-2\}$:

(i) $\lambda_0 = L(\beta, \chi_0) \propto \zeta(\beta)$ (trivial character, proportional to the partition function);

(ii) for real characters ($\chi_k = \bar{\chi}_k$): $\lambda_k \in \mathbb{R}$;

(iii) for complex characters ($\chi_k \neq \bar{\chi}_k$): $\lambda_k \in \mathbb{C}$, with $\lambda_{\bar{k}} = \overline{\lambda_k}$ (the eigenvalues of conjugate characters are complex conjugates of each other).

§3.3 Symmetries

Two natural symmetry operators exist on the state space $\mathcal{H}_p$: the involution ι and the Galois action σ.

3.3.1 The Involution ι

The **involution** ι on $\mathcal{H}_p$ is defined by:

$$(\iota f)(\alpha) = f(p^K - \alpha), \quad \alpha \in \mathbb{Z}/p^K\mathbb{Z}.$$

ι is a self-adjoint unitary operator on $\mathcal{H}_p$: $\iota^2 = \mathrm{id}$, $\iota^* = \iota$.

Restricted to $\mu_{p-1} \cong (\mathbb{Z}/p\mathbb{Z})^*$, ι acts as $\iota(a) = p - a \bmod p$. The orbits of ι on μ_{p-1} are the **doublets** $\{a, p-a\}$ (2-element orbits when $a \neq p-a$). For odd primes p, $a \neq p-a$ holds for all $a \in \{1, \dots, p-1\}$ (since $2a \neq p$ in $\mathbb{Z}$, p being odd). Hence the number of orbits = the number of doublets = $(p-1)/2$.

The **tension** of the doublet $\{a, p-a\}$ is defined as

$$T(a) = |2a - p|.$$

$T(a)$ measures the "distance" between the two elements within the doublet. For odd primes p, the tension takes the values $1, 3, 5, \dots, p-2$ (the sequence of odd numbers).

Let g be a generator of $(\mathbb{Z}/p\mathbb{Z})^*$; by Euler's criterion:

$$g^{(p-1)/2} \equiv -1 \pmod{p}.$$

This means the $(p-1)/2$-th power of the generator amounts to negation (i.e., ι). On the antisymmetric state of a doublet, $g^{(p-1)/2}$ acts as -1: $g^{(p-1)/2}|\psi_-\rangle = -|\psi_-\rangle$ (sign flip at the $(p-1)/2$-th power), $g^{p-1}|\psi_-\rangle = +|\psi_-\rangle$ (a full cycle restores the sign).

3.3.2 The Galois Action σ

The multiplicative characters take values in the $(p-1)$-th roots of unity, lying in the cyclotomic field $\mathbb{Q}(\zeta_{p-1})$. The Galois group acting on the character values is $\mathrm{Gal}(\mathbb{Q}(\zeta_{p-1})/\mathbb{Q}) \cong (\mathbb{Z}/(p-1)\mathbb{Z})^*$, acting by

$$\sigma_a(\chi_k) = \chi_k^a = \chi_{ak \bmod (p-1)} \quad (a \in (\mathbb{Z}/(p-1)\mathbb{Z})^*).$$

$\gcd(a, p-1) = 1$ guarantees that $k \mapsto ak$ is a bijection of $\mathbb{Z}/(p-1)\mathbb{Z}$ preserving the order of each element, so this is a group action; a full Galois orbit consists of characters of the same order, of size $\varphi(d)$ (d the order of χ_k).

Key special case: -1 is always coprime to $p-1$, and $\sigma_{-1}(\chi_k) = \chi_{-k} = \bar{\chi}_k$ is complex conjugation. The orbits of the subgroup $\{1, \sigma_{-1}\}$ are called **conjugate orbits**, of size

$$G(\chi_k) = \begin{cases} 2, & \chi_k \neq \bar{\chi}_k \text{ (complex character, orbit } \{\chi_k, \bar{\chi}_k\}), \\ 1, & \chi_k = \bar{\chi}_k \text{ (real character, fixed point)}; \end{cases}$$

equivalently, $G(\chi_k) = [\mathbb{Q}(\chi_k) : \mathbb{Q}(\chi_k) \cap \mathbb{R}]$ (the degree of the character value field over its maximal real subfield). The exponents in the spectral weights later use the conjugate-orbit size G, not the full Galois-orbit size $\varphi(d)$.

3.3.3 The Joint Structure (σ, ι)

σ acts on **mode space** (changing "which character"), ι acts on **position space** (changing "which group element"). The two act on different spaces and are mutually independent (they commute).

For $a \in (\mathbb{Z}/p\mathbb{Z})^*$, define the ι-orbit dimension:

$$C(a) = \left|\{a,\, a^{-1}\}\right| = \begin{cases} 1, & a^2 \equiv 1 \pmod{p}, \\ 2, & a^2 \not\equiv 1 \pmod{p}. \end{cases}$$

Under the joint action of (σ, ι), the joint orbit dimension of mode χ_k at position a is

$$\dim \mathcal{O}(\chi_k, a) = G(\chi_k) \times C(a).$$

The exponent structure of the spectral weights is given in §6.4 by the covering structure of transition paths; the joint orbit dimension is its single-station prototype.

§3.4 Metric Structure

The eigenvalues of the Hamiltonian $H = \log n$ embed the state space $\mathbb{N}^*$ into the real line: $n \mapsto \log n$. This embedding defines the metric $d(m, n) = |\log m - \log n|$ on $\mathbb{N}^*$; the same eigenvalues $E_n = \log n$ give the energy spectrum; the time evolution $\sigma_t = e^{itH}$ is generated by the same H. Distance, energy, and time–all three structures are uniquely determined by H, sharing the same spacing $\log(1 + 1/n)$. This section establishes their properties one by one.

3.4.1 Spatial Metric

Define a metric on $\mathbb{N}^*$:

$$d(m,n) = |\log m - \log n| = \left|\log \frac{m}{n}\right|.$$

$(\mathbb{N}^*, d)$ is a metric space; verified item by item:

(i) Positive definiteness. $d(m,n) \geq 0$. log is strictly increasing on $\mathbb{N}^*$, so $m \neq n \Rightarrow \log m \neq \log n \Rightarrow |\log m - \log n| > 0$; for $m = n$ it is clearly 0.

(ii) Symmetry. $d(m,n) = d(n,m)$. $|\log m - \log n| = |\log n - \log m|$, by the symmetry of the absolute value.

(iii) Triangle inequality. $d(m,n) \leq d(m,k) + d(k,n)$, $\forall\, k \in \mathbb{N}^*$. By the triangle inequality for the absolute value on the reals:

$$|\log m - \log n| = |(\log m - \log k) + (\log k - \log n)| \leq |\log m - \log k| + |\log k - \log n|.$$

□

The spacing structure of the metric d is determined by the adjacent distances:

$$d(n, n+1) = \log\left(1 + \frac{1}{n}\right).$$

The following properties are verified item by item:

(a) Maximal spacing. $d(1,2) = \log 2$ (the maximal adjacent distance attained on $\mathbb{N}^*$). For $n = 1$, $\log(1+1) = \log 2$; for $n \geq 2$, $\log(1 + 1/n) < \log 2$ ($\log(1+x)$ is strictly increasing for $x > 0$).

(b) Monotone decrease. $f(n) = \log(1 + 1/n)$: as the composition of the strictly decreasing $1/n$ with the strictly increasing log, f is strictly decreasing.

(c) Asymptotic behavior. Taylor expansion: $\log(1 + 1/n) = 1/n - 1/(2n^2) + O(1/n^3) \sim 1/n$. □

The monotone decrease of the spacings means that $\mathbb{N}^*$ is a non-uniform discrete space under this metric: adjacent spacings can be arbitrarily small ($\inf_n d(n, n{+}1) = 0$), yet the distance between any two points is strictly positive.

(i) Equal distance intervals correspond to equal-ratio integer intervals. $d(n, kn) = |\log n - \log(kn)| = \log k$ is constant (independent of n). □

On the finite state space $(\mathbb{Z}/p^K\mathbb{Z})^*$, the spacings take finitely many values. The minimal spacing is

$$d_{\min} = d(p^K - 2,\, p^K - 1) = \log \frac{p^K - 1}{p^K - 2}.$$

Argument: $(\mathbb{Z}/p^K\mathbb{Z})^*$ is a finite set whose elements form a finite ordered sequence in the log coordinate. The adjacent spacing $d(n, n+1) = \log(1 + 1/n)$ is strictly decreasing in n, so the minimum is attained at the largest adjacent pair: $d_{\min} = d(p^K - 2,\, p^K - 1)$. □

The ratio of the maximal adjacent distance $d(1,2) = \log 2$ to the minimal adjacent distance $d_{\min} \approx 1/p^K$ is $p^K \log 2$, growing exponentially with K.

The intrinsic constant of the metric is given by the per-mode symmetric sum of squared discrete gradients. The squared gradient of mode n counts the squared spacings in its two neighboring directions (the standard pointwise convention of lattice field theory; $n = 1$ has only a right neighbor):

$$|\nabla H(n)|^2 = d(n, n{+}1)^2 + d(n{-}1, n)^2 \quad (n \geq 2), \qquad |\nabla H(1)|^2 = d(1, 2)^2,$$

and in the sum each adjacent link is counted once from each of its two ends:

$$\boxed{C^* = \sum_{n=1}^{\infty} |\nabla H(n)|^2 = 2\sum_{n=1}^{\infty} \left[\log\left(1 + \frac{1}{n}\right)\right]^2 = 1.9543783\ldots} \tag{15}$$

The series converges: $\log(1{+}1/n) \sim 1/n$, and after squaring $\sim 1/n^2$. The reason for the per-mode (rather than per-link) convention: the function of C^* is to normalize the energy scale $E(W) = \sqrt{W \log^2 W / C^*}$ (§5.3); the numerator $W \log^2 W$ is the collective fluctuation counted per mode, and the metric constant in the denominator must adopt the same convention to be on the same scale.

3.4.2 Energy Metric

The spectrum of the BC Hamiltonian H on the basis states $|n\rangle$ is

$$\mathrm{Spec}(H) = \{E_n = \log n : n \in \mathbb{N}^*\} = \{0, \log 2, \log 3, \log 4, \ldots\}.$$

The properties of $\mathrm{Spec}(H)$, verified item by item:

(i) Ground-state energy. $E_1 = \log 1 = 0$.

(ii) Minimal excitation energy. $\Delta E_{\min} = E_2 - E_1 = \log 2$. For $n \geq 2$, $E_{n+1} - E_n = \log(1 + 1/n) \leq \log 2$.

(iii) Additivity. $E_{mn} = \log(mn) = \log m + \log n = E_m + E_n$ (energy is additive with respect to multiplication).

(iv) Prime decomposition. $E_n = \sum_{p|n} v_p(n) \cdot E_p$ (superposition of prime excitations). By substituting the unique factorization $n = \prod_p p^{v_p(n)}$ into log.

(v) Finite-dimensional truncation. On the local state space $\mathcal{H}_p^{(K)}$, $E_{\max} < K \log p$. $\alpha \leq p^K - 1 < p^K$, so $\log \alpha < K \log p$.

(vi) Classical limit. $\Delta E_{n \to n+1} = \log(1 + 1/n) \to 0$ $(n \to \infty)$, and the spectrum tends to a continuum. □

Adjacent level spacing:

$$\Delta E(n) = E_{n+1} - E_n = \log\left(1 + \frac{1}{n}\right) = \frac{1}{n} - \frac{1}{2n^2} + \frac{1}{3n^3} - \cdots$$

In the low-energy region the spacings are large (quantum effects prominent); in the high-energy region they are small (classical limit). Energy quantization comes from the discreteness of $\mathbb{N}^*$.

Since $H = \log n$, the level spacing $\Delta E(n) = \log(1 + 1/n)$ coincides with the spatial spacing $d(n, n{+}1)$ of §3.4.2. Hence the $C^* = \sum_n |\nabla H(n)|^2 = 2\sum[\log(1 + 1/n)]^2$ defined in §3.4.2 is simultaneously the per-level sum of squared two-sided gaps: the spatial metric constant and the energy metric constant are one and the same quantity.

3.4.3 Time Metric

The time evolution is the one-parameter unitary group $\sigma_t = e^{itH}$ (§1(v)), $\sigma_t|n\rangle = n^{it}|n\rangle$, with continuous parameter t; the state space and the energy spectrum are discrete, so the resolvable time difference has a lower bound (see below).

The intrinsic period of the state $|n\rangle$ under time evolution is

$$T_n = \frac{2\pi}{\log n} \quad (n \geq 2).$$

The ground state $|1\rangle$ has no intrinsic period ($E_1 = 0$; it does not evolve).

Argument: $\sigma_t|n\rangle = e^{it\log n}|n\rangle$. The condition $e^{iT_n \log n} = 1$ has smallest positive solution $T_n = 2\pi/\log n$. $|1\rangle$: $\sigma_t|1\rangle = e^{it\cdot 0}|1\rangle = |1\rangle$ holds identically. □

The intrinsic periods of the prime states $|p\rangle$, $T_p = 2\pi/\log p$:

$$T_2 \approx 9.06, \quad T_3 \approx 5.72, \quad T_5 \approx 3.90, \quad T_7 \approx 3.23.$$

Small primes have long periods (low-energy long-wavelength modes); large primes have short periods (high-energy short-wavelength modes).

The shortest time required to distinguish adjacent states $|n\rangle, |n+1\rangle$:

$$\Delta t \geq \frac{1}{\Delta E(n)}.$$

The minimal time resolution (taking the maximal spacing $\Delta E_{\max} = \log 2$):

$$\Delta t_{\min} = \frac{1}{\log 2} \approx 1.443.$$

Argument: the phase difference is $\Delta\phi = \Delta t \cdot \Delta E(n)$; the resolution criterion takes the convention $\Delta\phi \geq 1$, giving $\Delta t \geq 1/\Delta E(n)$. Taking the infimum: $\Delta E_{\max} = \log 2$ gives $\Delta t_{\min} = 1/\log 2$. □

The resolution bound constrains observation, not evolution: the flow parameter t is continuous while resolvability is discrete– the two share the same origin $H = \log n$, residing respectively in the evolution layer and the observation layer.

From the resolution definition $\Delta t(n) = 1/\Delta E(n)$ one obtains directly

$$S(n) = \Delta E(n) \cdot \Delta t(n) = 1, \quad \forall\, n : \tag{16}$$

the product of resolution and energy spacing is identically 1, a direct consequence of the definition.

3.4.4 Unity of the Three Metrics

Let $f : \mathbb{N}^* \to \mathbb{R}_{\geq 0}$ satisfy: (a) complete additivity $f(mn) = f(m) + f(n)$; (b) nonnegativity $f(n) \geq 0$; (c) asymptotic flatness $f(n+1) - f(n) \to 0$. Then $f = c \cdot \log$ (the Erdős–Kátai theorem).

Three physical quantities are uniquely determined by the same function:

$$\text{Space:} \quad d(m,n) = |\log m - \log n|, \tag{17}$$

$$\text{Energy:} \quad E_n = \log n, \tag{18}$$

$$\text{Time:} \quad T_n = 2\pi/\log n. \tag{19}$$

The relation $d(1,n) = E_n = 2\pi/T_n$ is a corollary of the uniqueness theorem– no other function can simultaneously satisfy (a)(b)(c) while giving a different metric structure. $\square$

Minimal resolutions of the state space:

Metric	Minimal value	Expression
Minimal spatial distance	$d_{\min} \approx 1/p^K$	$\log \frac{p^K-1}{p^K-2}$
Minimal energy spacing	$\Delta E_{\min} \approx 1/p^K$	$\log \frac{p^K-1}{p^K-2}$
Time resolution	$\Delta t \approx p^K$	$1/\Delta E_{\min}$

They satisfy: $d_{\min} = \Delta E_{\min}, \quad d_{\min} \cdot \Delta t = 1$. For $K = K_{\max} = p(p-1) - 1$ (determined in §2.3), the three quantities are uniquely determined by p.

§4 Quantum Structure of the State Space

§3 established the complete algebraic structure of the state space $\mathcal{H}_p$: the multiplicative characters as the mode basis, the eigenvalues of the convolution operator as L-functions, symmetries given by the involution and the Galois action, and metrics given by $H = \log n$. Do these structures constitute a legitimate quantum mechanical system? This section verifies the five axioms of von Neumann quantum mechanics– Hilbert space, self-adjoint observables, tensor product, unitary evolution, and the Born rule[34]– each determined by a specific structure of §3.

§4.1 Hilbert Space

von Neumann (1932) requires the pure states of a physical system to form rays in a Hilbert space $\mathcal{H}$, guaranteeing the superposition principle, the probability interpretation, and spectral decomposition. For the BC system, $\mathcal{H}_p$ is constructed as follows.

The argument proceeds in four steps.

Step 1. The C^*-algebra axioms give $\phi(a^*a) \geq 0$ for all $a \in \mathcal{A}_{\mathrm{BC}}$. Define $\langle a, b\rangle_\phi = \phi(a^*b)$. This map automatically satisfies:

- conjugate symmetry: $\langle a, b\rangle = \overline{\langle b, a\rangle}$ (by the $*$-preservation of ϕ);
- linearity: $\langle a, \alpha b + \beta c\rangle = \alpha\langle a, b\rangle + \beta\langle a, c\rangle$ (by the linearity of ϕ);
- positive definiteness: $\langle a, a\rangle = \phi(a^*a) \geq 0$, with equality if and only if $a \in \mathcal{N} = \{a : \phi(a^*a) = 0\}$.

Step 2. Take the quotient $\mathcal{A}/\mathcal{N}$ by the null space $\mathcal{N}$; on it $\langle\cdot,\cdot\rangle$ becomes a strictly positive-definite inner product. Completing with respect to this inner product gives the Hilbert space $\mathcal{H}_\phi$.

Step 3. For the BC system, for $\beta > 1$ the KMS state gives $\phi_\beta(|n\rangle\langle n|) = n^{-\beta}/\zeta(\beta) > 0$ for all $n \in \mathbb{N}^*$, so $\mathcal{N} = \{0\}$ (no null vector is quotiented away). The GNS space $\mathcal{H}_\phi$[14, 15, 17] is of Hilbert–Schmidt type (basis $\{e_{nm}\Omega\}$; the modular structure in Step 6 of §4.5 acts precisely on this space), and its diagonal slice gives the original representation space $\ell^2(\mathbb{N}^*)$. After p-adic reduction (§2) one obtains the finite-dimensional local space $\mathcal{H}_p = L^2(\mathbb{Z}/p^K\mathbb{Z})$.

Step 4. $\dim \mathcal{H}_p = p^K < \infty$. A finite-dimensional normed space is automatically complete (no separate verification of the convergence of Cauchy sequences is needed), so $\mathcal{H}_p$ is a Hilbert space. $\square$

§4.2 Self-Adjoint Observables

With the Hilbert space established, the next question is the structure of observables. von Neumann requires observables to be represented by self-adjoint operators $A = A^*$, guaranteeing real measurement results. In the BC framework, self-adjointness is given by conjugation invariance.

Let O be an operator on $\mathcal{H}_p$ with the χ_k as eigenvectors and eigenvalues λ_k. If O is a physical observable, the effective eigenvalues are $|\lambda_k|^2 \in \mathbb{R}$–$O$ is equivalent to a self-adjoint operator.

The argument proceeds in four steps.

Step 1. The Galois group $\mathrm{Gal}(\mathbb{Q}(\zeta_{p-1})/\mathbb{Q}) \cong (\mathbb{Z}/(p-1)\mathbb{Z})^*$ of §3.3 acts on the character labels: $\sigma_a(\chi_k) = \chi_{ak \bmod (p-1)}$. If an operator O satisfies $O\chi_k = \lambda_k \chi_k$, this action induces a label permutation of the eigenvalues $\lambda_k \mapsto \lambda_{ak}$ (a rearrangement of the spectrum). Note that σ_a permutes the labels; it does not move λ_k around as an algebraic number– the L-function values themselves do not belong to any cyclotomic field.

Step 2. Physically observable values must be invariant under the label action of the conjugation subgroup $\{1, \sigma_{-1}\}$ (§3.3): σ_{-1} exchanges $\chi_k \leftrightarrow \bar{\chi}_k$, corresponding to $\lambda_k \mapsto \lambda_{-k} = \overline{\lambda_k}$ (the complex-conjugation relation of L-functions). For complex characters ($\chi_k \neq \bar{\chi}_k$), $\lambda_k \neq \overline{\lambda_k}$, so λ_k itself is not conjugation-invariant–it cannot serve directly as an observable.

Step 3. The norm of λ_k with respect to $\{1, \sigma_{-1}\}$ is $N(\lambda_k) = \lambda_k \cdot \sigma_{-1}(\lambda_k) = \lambda_k \cdot \bar{\lambda}_k = |\lambda_k|^2 \in \mathbb{R}$– the minimal invariant combination on the conjugate orbit.

Step 4. The effective observable values of O are $|\lambda_k|^2 \in \mathbb{R}_{\geq 0}$. Restricted to the conjugation-invariant subspace, O is equivalent to an operator whose eigenvalues are all real–that is, a self-adjoint operator. □

§4.3 Tensor Product

The state space of a composite system is required to have a tensor product structure $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$. For the BC system, this structure is given by the Chinese remainder theorem.

Let $p-1 = q_1^{b_1} \cdots q_s^{b_s}$ be the standard factorization of $p-1$. Then the Teichmüller part of the state space satisfies the unique tensor product decomposition:

$$L^2(\mu_{p-1}) \cong \bigotimes_{i=1}^{s} L^2(\mathbb{Z}/q_i^{b_i}\mathbb{Z}). \tag{20}$$

The isomorphism itself is the CRT decomposition of §3.1.4 (CRT group isomorphism → Pontryagin duality → bijection of orthogonal bases); the argument is not repeated. What is added here is the uniqueness required by the composite structure: the prime factorization is unique, hence the CRT decomposition is unique, hence the tensor product decomposition is unique– there is no freedom of choice in "how to split the system into subsystems". □

At the global level: $\ell^2(\mathbb{N}^*) \cong \widehat{\bigotimes}_p \ell^2(\mathbb{Z}_{\geq 0})$ (unique factorization theorem); at the local level: the CRT decomposition (20). For $p = 13$: $L^2(\mathbb{Z}/12\mathbb{Z}) \cong L^2(\mathbb{Z}/4\mathbb{Z}) \otimes L^2(\mathbb{Z}/3\mathbb{Z})$.

§4.4 Unitary Time Evolution

With the state space, observables, and composite structure determined, time evolution is still needed. Time evolution must preserve probability, i.e., form a unitary group. The unitary evolution σ_t ($t \in \mathbb{R}$) of the BC system is defined by:

$$(\sigma_t f)(\alpha) = e^{it\cdot v_p(\alpha)} f(\alpha),$$

where v_p is the p-adic valuation. σ_t preserves the inner product, satisfies the group law, and has H as its generator.

The argument proceeds in four steps.

Step 1. (Unitarity)

$$\begin{aligned}\langle \sigma_t f, \sigma_t g\rangle &= \frac{1}{p^K} \sum_{\alpha \in \mathbb{Z}/p^K\mathbb{Z}} \overline{e^{it\, v_p(\alpha)} f(\alpha)} \cdot e^{it\, v_p(\alpha)} g(\alpha) \\ &= \frac{1}{p^K} \sum_{\alpha} \overline{f(\alpha)}\, g(\alpha) = \langle f, g\rangle.\end{aligned}$$

Key point: $|e^{it\, v_p(\alpha)}| = 1$ ($v_p(\alpha) \in \mathbb{Z}_{\geq 0}$, a pure phase), and the phase factors cancel.

Step 2. (Group law)

$$(\sigma_s \circ \sigma_t f)(\alpha) = e^{is\, v_p(\alpha)} \cdot e^{it\, v_p(\alpha)} f(\alpha) = e^{i(s+t)\, v_p(\alpha)} f(\alpha) = (\sigma_{s+t} f)(\alpha).$$

Together with $\sigma_0 = \mathrm{id}$, $\{\sigma_t\}_{t\in\mathbb{R}}$ forms a one-parameter unitary group.

Step 3. (Generator) $\sigma_t = e^{itH}$, $H(\alpha) = v_p(\alpha)$. H is self-adjoint (v_p is real-valued), so σ_t is unitary.

Step 4. (Uniqueness) Let σ'_t be another one-parameter group of $*$-automorphisms of $\mathcal{A}_{\mathrm{BC}}$. Restricted to the phase subalgebra $C^*(\mathbb{Q}/\mathbb{Z})$, σ'_t gives a continuous homomorphism $\mathbb{R} \to \hat{\mathbb{Z}}^*$; $\hat{\mathbb{Z}}^*$ is totally disconnected, so the continuous image must be trivial, and σ'_t fixes each $e(r)$ pointwise. The BC algebra relations then force $\sigma'_t(\mu_n) = \lambda_n(t)\mu_n$; complete multiplicativity gives $\lambda_{mn}(t) = \lambda_m(t)\lambda_n(t)$, unitarity gives $|\lambda_n(t)| = 1$, and the group law gives the Cauchy functional equation $\lambda_n(s+t) = \lambda_n(s)\lambda_n(t)$, so $\lambda_n(t) = n^{ic_n t}$; the KMS condition unifies the phases at all primes to a single scale: $\lambda_n(t) = n^{ict}$. The scale c is absorbed into the time unit; take $c = 1$ (consistent with the scale convention of §1(iii)). Hence σ_t is the unique time evolution. □

§4.5 The Born Rule

The standard statement of the Born (1926) rule[33]: a quantum system is in the state $|\psi\rangle$; measuring an observable A (eigenvalues $\{a_n\}$, eigenstates $\{|a_n\rangle\}$) yields the result a_n with probability

$$P(a_n) = |\langle a_n|\psi\rangle|^2 = \langle\psi|P_n|\psi\rangle,$$

where $P_n = |a_n\rangle\langle a_n|$ is the projection onto the eigendirection. For a mixed state ρ: $P(a_n) = \mathrm{Tr}(\rho\, P_n)$.

For the BC system, the state is the KMS state φ_β, the observable directions are the characters χ_k, with corresponding character component operators P_{χ_k} (Step 1 below). The C^* positive functional φ_β provides nonnegativity ($\varphi_\beta(P^*P) \geq 0$) and additivity; normalization is completed by the explicit normalization in Step 5. The specific form of $\varphi_\beta(P_{\chi_k})$ is determined below.

The argument proceeds in six steps.

Step 1. Define the **component operator** (Fourier coefficient operator) of the character direction χ_k:

$$P_{\chi_k} = \frac{1}{p-1}\sum_{a=1}^{p-1} \overline{\chi_k(a)}\, e(a/p),$$

where $e(a/p)$ are the additive generators (phase operators) of the BC algebra. P_{χ_k} extracts the χ_k-Fourier component but is not an idempotent projection ($P^2 \neq P$)– its eigenvalues on the basis $|\alpha\rangle$ have modulus $\sqrt{p}/(p-1) \notin \{0,1\}$.

Step 2. The Gibbs form $\phi_\beta = \mathrm{Tr}(e^{-\beta H}\cdot)/\zeta(\beta)$ is evaluated directly on the phase operators: for $\gcd(a,p)=1$,

$$\phi_\beta(e(a/p)) = \frac{1}{\zeta(\beta)}\sum_{n\geq 1} n^{-\beta}\, e^{2\pi i n a/p} = \frac{1}{\zeta(\beta)}\sum_{\substack{n\geq 1\\ p\nmid n}} n^{-\beta}\, e^{2\pi i n a/p} \;+\; p^{-\beta},$$

where the last term comes from the modes with $p \mid n$ (on which $e^{2\pi i n a/p}=1$, the sum being $p^{-\beta}\zeta(\beta)$).

Step 3. Substitute Step 2 into the definition of P_{χ_k}. The contribution of the $p \mid n$ terms is weighted by $\sum_a \overline{\chi_k(a)}$: for nontrivial χ_k it is annihilated by character orthogonality $\sum_a \overline{\chi_k(a)} = 0$, leaving the constant term $p^{-\beta}$ only in the χ_0 direction. For the nontrivial directions:

$$\phi_\beta(P_{\chi_k}) = \frac{1}{(p-1)\zeta(\beta)}\sum_{\substack{n\geq 1\\ p\nmid n}} n^{-\beta}\underbrace{\sum_{a=1}^{p-1}\overline{\chi_k(a)}\, e^{2\pi i n a/p}}_{S_k(n)}. \tag{21}$$

Step 4. For $\gcd(n,p)=1$, make the substitution $a = n^{-1}b \pmod p$ (n^{-1} exists since $\gcd(n,p)=1$):

$$S_k(n) = \sum_{b=1}^{p-1}\overline{\chi_k(n^{-1}b)}\, e^{2\pi i b/p} = \chi_k(n)\sum_{b=1}^{p-1}\overline{\chi_k(b)}\, e^{2\pi i b/p} = \chi_k(n)\cdot\chi_k(-1)\,\overline{\tau(\chi_k)},$$

where $\tau(\chi_k) = \sum_{b=1}^{p-1}\chi_k(b)\, e^{2\pi i b/p}$ is the Gauss sum, $\tau(\bar\chi_k) = \chi_k(-1)\overline{\tau(\chi_k)}$. Substituting back:

$$\phi_\beta(P_{\chi_k}) = \frac{\chi_k(-1)\,\overline{\tau(\chi_k)}}{(p-1)\zeta(\beta)}\cdot L(\beta,\chi_k), \tag{22}$$

where the sign factor $\chi_k(-1)=\pm 1$ has no effect after taking the modulus squared. Here $L(\beta,\chi_k) = \sum_{n\geq 1,\, p\nmid n}\chi_k(n)\, n^{-\beta}$.

The Dirichlet L-function appears as the evaluation of the KMS state on the Fourier component.

Step 5. By §4.2, physical observables must be Galois invariant. $\phi_\beta(P_{\chi_k}) \propto L(\beta,\chi_k) \in \mathbb{C}$ (for complex characters), so the measurement probability (a real number) must be the modulus squared. Define the response (later called the resonance ratio) and the measurement probability:

$$R_k(\beta) = \frac{L(\beta,\chi_k)}{\zeta(\beta)}, \qquad P(\chi_k) = \frac{|R_k(\beta)|^2}{\sum_j |R_j(\beta)|^2}.$$

The explicit normalization guarantees $\sum_k P(\chi_k) = 1$ for any β– normalization itself does not select the evaluation point. The evaluation point is determined by the modular structure of the state space; this is the content of Step 6.

Step 6 (evaluation point: modular structure). For $\beta > 1$, ϕ_β is a Gibbs state, its GNS representation is in standard form, and the Tomita–Takesaki theorem[16] gives the polar decomposition of the antilinear Tomita operator

$$S = J\Delta^{1/2}, \qquad S\,a\Omega = a^*\Omega,$$

where J is the modular conjugation (antiunitary) and Δ the modular operator (positive), both uniquely determined by the uniqueness of the polar decomposition. On the basis $\{e_{nm}\Omega\}$, the weights of $\Delta^{1/2}$ are $(n/m)^{-\beta/2}$; the weight relative to the $m = 1$ direction is

$$n^{-\beta/2}.$$

The amplitude-level ($\Delta^{1/2}$-layer) counterpart of the probability-level weight $n^{-\beta}$ is $n^{-\beta/2}$– the half exponent is not a choice; it is the unique positive part of the polar decomposition.

The vertex response is the modular matrix element connecting a mode with its conjugate mode, i.e., the evaluation at the $\Delta^{1/2}$ layer. The evaluation point is uniquely determined by three requirements: complete multiplicativity and measurability of the weight give $w(n) = n^{-s}$ for some s; lying at the $\Delta^{1/2}$ layer forces $s = \beta/2$ (uniqueness of the polar decomposition); compatibility with the KMS structure at $\beta = 1$ takes $\beta = 1$–hence $s = 1/2$ is unique.

The structure above holds pointwise for $\beta > 1$; $\beta = 1$ is the critical point of the phase transition ($\zeta(1)$ diverges), and the limit $\beta \to 1^+$ is examined character direction by character direction: the nontrivial χ components $\sum_n \chi(n)\, n^{-\beta/2} \to L(1/2, \chi)$ (the Dirichlet series of non-principal characters converges for $\operatorname{Re} s > 0$), so the limit is well defined; only the χ_0 component diverges. The amplitude-level divergence of χ_0 is precisely the signal of emergent-state condensation (§6, §13): elementary vertex responses exist only in the nontrivial directions; the χ_0 direction has no elementary vertex, only collective emergence.

1/2 is not another temperature–there is no KMS state at $\beta = 1/2$. It is the midpoint of the KMS analytic strip $[0, i\beta]$ of the $\beta = 1$ state, i.e., the location where the modular conjugation J acts; probability is the modulus squared of amplitude, and probability level $\beta = 1$ corresponds to amplitude level $\beta = 1/2$.

Final result:

$$\boxed{R_k = \frac{L(1/2, \chi_k)}{\zeta(1/2)}, \qquad P(\chi_k) = \frac{|R_k|^2}{\sum_j |R_j|^2}.} \tag{23}$$

In standard quantum mechanics the Born rule is regarded as an independent postulate– about the probabilities of measurement outcomes. The derivation above shows: probability is not a statement about "what happens upon measurement". It is the overlap between two states–how much of the KMS state φ_β is visible from the perspective of the character direction χ_k. This overlap is uniquely determined by C^* positivity, Galois invariance, and the modular structure to be the formula above. Probability is an objective relation between two states– not the output of a measurement, not the belief of an observer, not an intrinsic property of a particle.

Each of the five axioms corresponds to a specific structure of the state space $\mathcal{H}_p$:

Axiom	State-space structure	Determining mechanism
Hilbert space	C^* positivity	GNS construction
Self-adjoint observables	Galois invariance	$\sigma_{-1}(\lambda_k) = \bar{\lambda}_k \Rightarrow$ readout takes the conjugation-invariant combination $\|\lambda_k\|^2$
Tensor product	unique factorization	CRT decomposition
Unitary evolution	$H = \log n$	Cauchy equation + monotonicity
Born rule	KMS state + modular structure	$\Delta^{1/2}$-layer evaluation $\Rightarrow P \propto \|L(1/2, \chi)/\zeta(1/2)\|^2$

§5 Physical Spacetime

§4 verified that the state space $\mathcal{H}_p$ possesses all the structure required by the five axioms of quantum mechanics. A natural further question is: is physical spacetime also determined by the state space?

Physical spacetime is the arena in which particle motion and observation take place – a finite-dimensional real space, with a metric, with causal structure. This section starts from the algebraic structure of the state space $\mathcal{H}_p$ and derives the dimension and metric signature of spacetime (§5.1), the time direction and the arrow of time (§5.2), and the energy scales that the state space naturally distinguishes (§5.3).

§5.1 Dimension and Metric

Spacetime coordinates label the positions of observation events. Observation is carried out relative to the ground state – the ground state is the reference state in which no mode is excited, the starting point of every physical process. Hence the structure of spacetime is determined by the observable space at the ground state.

In the BC system, the multiplicative unit $n = 1$ of $\mathbb{N}^*$ is the natural ground-state candidate: in the unique factorization $n = \prod p_i^{a_i}$, $n = 1$ corresponds to the empty product – no prime factor participates, that is, no excitation at all. What this subsection derives is the geometric type of spacetime (dimension $= 4$, signature $(1, 3)$, causal structure); the physical space itself remains the discrete $\mathbb{N}^*$ (minimal spacing $\log 2$, §3.4), and there are no spacetime coordinates taking continuous real values.

Verification: $H|1\rangle = \log 1 = 0$ (zero energy), $\sigma_t|1\rangle = e^{it\cdot 0}|1\rangle = |1\rangle$ (time invariant).

Since $H = \log n$ is strictly positive for $n \geq 2$, the **ground state** $|1\rangle$ is unique.

Once the ground state is fixed, the dimension of spacetime is determined by how many independent operations can be performed at the ground state. The involution ι of the state space (§3.3) pairs the group elements two by two, and the ι-orbit containing the ground state delimits the dimension of the effective operation space at the ground state. The following six steps derive the dimension, the metric, and the time direction of spacetime.

Step 1. The involution $\iota(\alpha) = p^K - \alpha$ of §3.3 is defined on the full ring; reduced mod p it descends to the species layer $(\mathbb{Z}/p\mathbb{Z})^*$, acting as $a \mapsto p - a$ (the full-ring orbit $\{1, p^K - 1\}$ of $|1\rangle$ reduces mod p to $\{1, p-1\}$; the two layers agree). The ι-orbit of the ground state $|1\rangle$ at the species layer is

$$D_1 = \{1,\ \iota(1)\} = \{1,\ p-1\}.$$

Since $p \geq 3$ is an odd prime, $1 \neq p - 1$, hence $|D_1| = 2$.

Step 2. The species layer $L^2((\mathbb{Z}/p\mathbb{Z})^*)$ decomposes along ι-orbits as $\mathcal{H}_{D_1} \oplus \mathcal{H}_{D_2} \oplus \cdots \oplus \mathcal{H}_{D_{(p-1)/2}}$. Among these only $\mathcal{H}_{D_1}$ contains the ground state $|1\rangle$. The involution ι binds $|1\rangle$ and $|p-1\rangle$ into an indivisible orbit; any operation on the ground state must act simultaneously on both components of D_1. The effective operation space at the ground state is

$$\mathcal{H}_{D_1} = \operatorname{span}\{|1\rangle, |p-1\rangle\} = \mathbb{C}^2.$$

The remaining subspaces $\mathcal{H}_{D_j}$ ($j \geq 2$) do not contain the ground state and correspond to the internal degrees of freedom of matter (§6).

Step 3. The self-adjoint operators on $\mathbb{C}^2$ constitute $\mathrm{Herm}_2(\mathbb{C})$. Self-adjointness guarantees real eigenvalues (§4.2), and Galois invariance is automatically satisfied. Hence the observable space at the ground state is $\mathrm{Herm}_2(\mathbb{C})$.

As a real vector space:

$$\dim_{\mathbb{R}} \mathrm{Herm}_2(\mathbb{C}) = 2^2 = 4.$$

The standard basis $\{I, \sigma_1, \sigma_2, \sigma_3\}$ (identity matrix + three Pauli matrices) gives the expansion

$$A = tI + x\sigma_1 + y\sigma_2 + z\sigma_3.$$

The four real parameter directions (t, x, y, z) give the dimension of spacetime: 4. $\mathrm{Herm}_2(\mathbb{C})$ is the observable space at the ground state; it determines the geometric type (dimension and signature) of the discrete physical space $\mathbb{N}^*$, rather than endowing the physical space with continuous coordinates.

Step 4. On $\mathrm{Herm}_2(\mathbb{C})$, $\mathrm{SL}(2, \mathbb{C})$ acts by $A \mapsto gAg^\dagger$. The unique (up to scalar multiple) invariant quadratic form under this action is the determinant:

$$\det \begin{pmatrix} t+z & x-iy \\ x+iy & t-z \end{pmatrix} = t^2 - x^2 - y^2 - z^2.$$

This is precisely the Minkowski metric $\eta = \mathrm{diag}(+1, -1, -1, -1)$ of signature $(1, 3)$.

Verification of $\mathrm{SL}(2, \mathbb{C})$ *invariance*:

$$\det(gAg^\dagger) = (\det g)(\det A)(\det g^\dagger) = (\det g)^2 \det A = \det A$$

(since $\det g = 1$ for $g \in \mathrm{SL}(2, \mathbb{C})$).

Step 5. If $|D_1| = a$, the amplitude space is $\mathbb{C}^a$ and the observable space is $\mathrm{Herm}_a(\mathbb{C})$ (dimension a^2), whose natural "metric" is det – a form of degree a.

- $a = 1$: det is a form of degree one (a linear function) and defines no metric.
- $a = 2$: det is a quadratic form and defines the metric η. ✓
- $a \geq 3$: det is a form of degree a (cubic, quartic, ...), not quadratic, and defines no metric.

Hence only for $|D_1| = 2$ is the determinant det a quadratic form, capable of defining a spacetime metric – this explains why physical spacetime is precisely a 4-dimensional Lorentz space.

Step 6. §4.4 confirmed $\sigma_t = e^{itH}$ as the unique dynamical time evolution. In the coordinates $\{t, x, y, z\}$ of $\mathrm{Herm}_2(\mathbb{C})$, the generator direction of σ_t (the I direction) corresponds to the time direction in the signature $(1, 3)$. Verification: $\det(tI) = t^2 > 0$, $\det(x\sigma_i) = -x^2 < 0$, so among the four basis directions I is the only one with $\det > 0$, i.e. the time coordinate corresponding to the "1" in the signature. The three Pauli directions are the spatial coordinates corresponding to the "3".

Step 7. The signature $(1, 3)$ determines the causal structure. The sign of the metric $\det(A - B)$ divides pairs of events into timelike ($\det > 0$, causally connectable), lightlike ($\det = 0$, the boundary of light signals), and spacelike ($\det < 0$, not causally connectable). The light cone $\mathcal{C}_B = \{A : \det(A - B) = 0\}$ defines the propagation boundary of light, and the speed of light $c = 1$ is a metric normalization convention. Combined with the time direction of Step 6, the future light cone is marked by $\mathrm{tr}(A - B) > 0$.

The seven steps above show that physical spacetime is given by the structure of the state space. The ground-state orbit of the involution ι determines the dimension, $\mathrm{SL}(2, \mathbb{C})$ invariance determines the metric signature $(1, 3)$, the generator of σ_t marks the time direction, and the sign of det determines the causal structure. Physical spacetime is a 4-dimensional Lorentz space. $\square$

§5.2 KMS Phase Transition and the Arrow of Time

§5.1 determined the dimension and metric of spacetime. This section derives the time direction from the analytic properties of the partition function.

The partition function $\zeta(\beta)$ of the BC system has a unique pole at $\beta = 1$, corresponding to the unique structural phase transition: the system passes irreversibly from disorder (Galois symmetry intact) into order (spontaneous symmetry breaking). This irreversibility determines the arrow of time.

The KMS inverse temperature $\beta > 0$ labels which global equilibrium state φ_β the system occupies (the state satisfying $\varphi_\beta(A\sigma_{i\beta}(B)) = \varphi_\beta(BA)$). $\beta = 1$ is the unique phase-transition point:

(i) $\beta \leq 1$: the KMS state is unique (high-temperature disordered phase, Galois symmetry intact);

(ii) $\beta > 1$: the KMS states are parametrized by $\widehat{\mathbb{Z}}^*$ (low-temperature ordered phase, spontaneous symmetry breaking).

The derivation proceeds in four steps.

Step 1. The partition function of the BC system is

$$Z(\beta) = \mathrm{Tr}(e^{-\beta H}) = \sum_{n=1}^{\infty} n^{-\beta} = \zeta(\beta).$$

The Riemann ζ function[20] has a unique simple pole at $\beta = 1$:

$$\zeta(\beta) \sim \frac{1}{\beta - 1} \quad (\beta \to 1^+).$$

Divergence of the partition function signals the failure of the Gibbs form – the structure of the equilibrium state changes here (for $\beta \leq 1$ equilibrium states still exist, but they are no longer of Gibbs type; see Step 2). Hence $\beta = 1$ is the unique phase-transition point (the unique pole of $\zeta(\beta)$ in $\mathrm{Re}(\beta) > 0$ is at $\beta = 1$).

Step 2. This is one of the core theorems of Bost–Connes (1995)[1, 3]: for $0 < \beta \leq 1$, the KMS_β state on $\mathcal{A}_{\mathrm{BC}}$ is unique.

In this regime the symmetry of the Galois group $\widehat{\mathbb{Z}}^*$ is unbroken.

Step 3. For $\beta > 1$, the KMS_β states form a Choquet simplex, whose extremal states are parametrized by $\widehat{\mathbb{Z}}^*$:

$$\{\varphi_{\beta,\alpha}\}_{\alpha \in \widehat{\mathbb{Z}}^*}.$$

Distinct extremal states are related by the Galois action: $\varphi_{\beta,\gamma\alpha} = \varphi_{\beta,\alpha} \circ \theta_\gamma$. This is spontaneous symmetry breaking: in the low-temperature phase the $\widehat{\mathbb{Z}}^*$ symmetry is broken down to the selection of a single extremal state.

Step 4. The pole of $\zeta(\beta)$ lies at $\beta = 1$ (on the positive real axis).

- β increasing = temperature decreasing = the direction of cooling;

- crossing $\beta = 1$ triggers the phase transition (spontaneous symmetry breaking);
- the substance of the irreversibility is the choice of extremal state: the low-temperature phase has infinitely many extremal states ($|\widehat{\mathbb{Z}}^*| = \infty$); once the system has condensed onto some $\alpha \in \widehat{\mathbb{Z}}^*$, the information of that choice cannot be undone by any thermodynamic process within the low-temperature phase – the selection of the breaking direction is irreversible.

Irreversibility defines the arrow of time: the system can only evolve from disorder to order, never back.

In the cosmological context, the phase-transition point $\beta = 1$ corresponds to the Big Bang: the irreversible passage of the system from $\beta < 1$ (disorder) into $\beta > 1$ (order). The arrow of time is fixed by the pole of $\zeta(\beta)$ at $\beta = 1$. □

5.2.1 Dynamical Evolution and the KMS Condition

The dynamics on the state space is given by the automorphisms σ_t:

$$\sigma_t(A) = e^{itH} A e^{-itH}, \quad A \in \mathcal{A}_{\mathrm{BC}}.$$

The KMS condition links the evolution σ_t to the equilibrium state φ_β:

$$\varphi_\beta(A\sigma_{i\beta}(B)) = \varphi_\beta(BA).$$

A state satisfying this condition is a β-KMS state[11, 12, 13, 18].

Step 6 of §5.1 already identified the generator direction of σ_t (the I direction) as the time direction in the signature $(1, 3)$. Combined with the irreversibility of the phase transition in this section, the structure of time is fully determined: σ_t provides the evolution, and the $\beta = 1$ phase transition provides the direction.

§5.3 Energy Scales

Every physical process involves a set of modes, and the collective excitation of these modes determines the characteristic energy scale of the process.

Step 1 (spectral counting function). The spectrum of $H = \log n$ is the discrete point set $\{\log n : n \in \mathbb{N}^*\}$. Define the counting function

$$\mathcal{N}(\omega) = \#\{n \in \mathbb{N}^* : \log n \le \omega\} = \lfloor e^\omega \rfloor.$$

There are exactly $\lfloor e^\omega \rfloor$ modes below the energy level ω. The BC spectrum is of exponential (Hagedorn[42]) type: the number of modes grows exponentially with energy. Read in reverse: the mode set participating with spectral weight W exactly fills the energy interval $[0, \log W]$, with **boundary energy level**

$$\omega(W) = \log W$$

– the inversion of the counting function.

Step 2 (the energy-scale formula). For a physical process participating with spectral weight W, the characteristic energy scale is defined by

$$\boxed{E(W)^2\, C^* = W \cdot (\log W)^2, \qquad E(W) = \sqrt{\frac{W \log^2 W}{C^*}},} \tag{24}$$

where $C^* = \sum_{n=1}^{\infty} |\nabla H(n)|^2 = 1.9543783$ is the intrinsic metric constant of §3.4. Physical reading: W modes are coherently excited at the collective frequency $\omega(W) = \log W$; each mode carries a fluctuation ω^2, giving total fluctuation $W\omega^2$, normalized by C^*. This formula is the definition of a collective excitation, not an incoherent pile-up of the individual mode frequencies (the trace-type sum $\sum_{n \le N} \log^2 n = N \log^2 N - 2N \log N + \cdots$ differs from it appreciably at physically relevant N).

The participating spectral weight W is taken in two ways: for an integer cutoff, $W = N$ (mode counting); for a thermal state, $W = Z(\beta) = \sum_n n^{-\beta}$ (Boltzmann-weighted counting of discrete modes, §6.5). The same formula (24) thereby covers three energy scales: two from mode counting—taking the unit group and the full group respectively (§10.1, §10.2)— and one from thermal weighting (§13).

Step 3 (two mode sets and the scale hierarchy). The ring structure of the state space $\mathcal{H}_p = L^2(\mathbb{Z}/p^K\mathbb{Z})$ distinguishes two mode sets:

- the full ring $S_{\text{tot}} = \mathbb{Z}/p^K\mathbb{Z}$, $W = p^K$ (including multiples of p, p^{K-1} non-unit elements in all);
- the unit group $S_{\text{unit}} = (\mathbb{Z}/p^K\mathbb{Z})^*$, $W = \varphi(p^K) = p^{K-1}(p-1)$ (coprime to p; the characters are defined on it).

Substituting into (24) gives two characteristic scales $E_{\text{tot}} = E(p^K)$ and $E_{\text{unit}} = E(\varphi(p^K))$, with hierarchy ratio

$$\frac{E_{\text{tot}}}{E_{\text{unit}}} = \sqrt{\frac{p^K \log^2 p^K}{\varphi(p^K) \log^2 \varphi(p^K)}} \xrightarrow{K \gg 1} \sqrt{\frac{p^K}{\varphi(p^K)}} = \sqrt{\frac{p}{p-1}}.$$

The hierarchy ratio depends only on p, not on the level K. $\square$

Remark (common origin with the phase transition): the exponential counting $\mathcal{N}(\omega) = \lfloor e^{\omega} \rfloor$ is at the same time the root of the phase transition of §5.2: $Z(\beta) = \sum_n e^{-\beta \log n}$ diverges for $\beta \le 1$ precisely because the exponential growth of the mode count overwhelms the Boltzmann suppression at $\beta = 1$. The phase transition (§5.2), the energy-scale formula (this section), and the metric-force constant $G = 1/E^2$ (§8) all stem from this exponential counting.

§5.1–§5.3 derived spacetime, the arrow of time, and the energy hierarchy from the algebraic structure of the state space. Comparing these three sets of results shows: the dimension and metric of spacetime are independent of p: $|D_1| = 2$ holds for every odd prime $p \ge 3$, and the pole position of $\zeta(\beta)$ is likewise independent of p. The state space $\mathcal{H}_p$ of any odd prime p yields a 4-dimensional Lorentz spacetime with signature $(1, 3)$. But the energy hierarchy ratio $\sqrt{p/(p-1)}$ depends on p – different state spaces correspond to the same spacetime structure, but to different physical content.

§6 Stable States

§5 has shown that the state space $\mathcal{H}_p = L^2(\mathbb{Z}/p^K\mathbb{Z})$ determines physical spacetime and the energy scale. When the state space as a whole is excited, the excitation is distributed over different directions. A generic excitation is a superposition over several directions; its interference pattern changes continuously in time, with no fixed shape. But if the excitation pattern along certain directions can remain permanently unchanged under time evolution– that is, if it forms a stable standing-wave mode– then those directions correspond to elementary matter: excitations whose shape is preserved at every moment. Therefore the problem of elementary matter is equivalent to: *which directions of the state space can form stable excitation patterns?*

This is precisely the mathematical definition of a **stable state**: an excitation mode whose shape is invariant under the propagation dynamics–acquiring only an overall amplitude and phase factor. This section will show that the existence of stable states is determined by the spectral structure of the state space, and that every stable state carries complete geometric coordinates (position), topological quantum numbers (shape), and a definite mass.

§6.1 Definition and Classification

The propagation of excitations in the state space $\mathcal{H}_p$ is given by the convolution operator K induced by the KMS state (§3.2), and a general state is a superposition of several eigenmodes of K. After m propagation steps, each component is multiplied by the m-th power of its eigenvalue: if $\psi = \sum_j c_j\psi_j$ (where $K\psi_j = \lambda_j\psi_j$), then

$$K^m\psi = \sum_j c_j\,\lambda_j^m\,\psi_j, \quad m = 1, 2, \ldots \tag{25}$$

When the λ_j are pairwise distinct, the relative amplitudes $|\lambda_i/\lambda_j|^m$ and relative phases $e^{im\arg(\lambda_i/\lambda_j)}$ of the components change continuously with the propagation step m, so the distribution of $|\psi|^2$ keeps changing–the shape of the state is distorted by propagation. Such a state cannot represent matter–the shape of matter must not change under propagation.

An excitation corresponding to matter must be shape-invariant under propagation, i.e. ψ must be an eigenvector of K:

$$K\psi = \lambda\,\psi, \quad \lambda \in \mathbb{C}. \tag{26}$$

Physical meaning: propagation changes only the overall amplitude and phase, while the shape of the spatial distribution $|\psi(\alpha)|^2$ of the wave function remains unchanged. A nonzero $\psi \in \mathcal{H}_p$ satisfying (26) is called a **stable state**.

Stable states fall into two classes: those defined on the unit group $(\mathbb{Z}/p^K\mathbb{Z})^*$ are called **elementary states**, and those defined on the full ring $\mathbb{Z}/p^K\mathbb{Z}$ are called **emergent states**. We now establish their existence and stability mechanisms in turn.

6.1.1 Elementary states

The first class of stable states in the state space are the eigenmodes of K– the multiplicative characters $\tilde{\chi}_k$ on the species-level unit group $(\mathbb{Z}/p\mathbb{Z})^*$.

By the diagonalization theorem of §3.2, $\tilde{\chi}_k$ is an eigenvector of K with eigenvalue $\lambda_k = L(\beta, \chi_k)$:

$$K\tilde{\chi}_j = \lambda_j \tilde{\chi}_j,$$

which is exactly the form of the stability condition (26). Conversely, any $\psi \neq 0$ satisfying (26) is by definition an eigenvector of K, and must lie in some eigenspace of K: the elementary states are precisely the eigenvectors of K.

$(\mathbb{Z}/p\mathbb{Z})^*$ is a cyclic group of order $p-1$, and its dual group has exactly $p-1$ characters. For $k \neq k'$, the difference $L(\beta, \chi_k) - L(\beta, \chi_{k'}) = \sum_n [\chi_k(n) - \chi_{k'}(n)]\, n^{-\beta}$ is a Dirichlet series with not-all-zero coefficients (the coefficient at the first n with $\chi_k(n) \neq \chi_{k'}(n)$ is nonzero), hence an analytic function on $\beta > 1$ that does not vanish identically, whose zeros form at most a discrete set. Therefore, except for an at most discrete set of exceptional values of β, the $p-1$ eigenvalues are pairwise distinct–the spectrum of K is simple; at the exceptional points the character basis is still an eigenbasis, only the simplicity degenerates. Nondegeneracy at the physically realized value of β can be verified numerically. Every eigenspace of K is one-dimensional: $\ker(K - \lambda_k I) = \mathbb{C} \cdot \tilde{\chi}_k$. Here χ_0 is the trivial character (identically 1), discussed separately in §6.1.2. The elementary states are the $p-2$ nontrivial character directions $\{\mathbb{C} \cdot \tilde{\chi}_k\}_{k=1}^{p-2}$, with stability protected by the simplicity of the spectrum.

Besides these $p-2$ elementary states, the state space also contains a second class of stable states, driven by the pole of the partition function.

6.1.2 Emergent states

The state space also contains a **second class of stable states**– emergent states protected not by a single eigenvalue of K, but driven by the pole of the partition function $Z(\beta) = \sum_{n=1}^{p^K} n^{-\beta}$ as $\beta \to 1$.

In the thermodynamic limit ($p^K \to \infty$), $Z(\beta) \to \zeta(\beta)$. $\zeta(\beta)$ has a simple pole at $\beta = 1$:

$$\zeta(\beta) = \frac{1}{\beta - 1} + \gamma + O(\beta - 1), \quad \beta \to 1^+.$$

The pole means that as $\beta \to 1$, the KMS state φ_β assigns infinite spectral weight to certain directions– in the thermodynamic limit, these directions "condense" into an emergent state that cannot be described as a superposition of finitely many eigenmodes.

For each character χ_k, the spectral weight of the KMS state in direction k is controlled by $L(\beta, \chi_k)$.

- For $k = 0$ (the trivial character $\chi_0 \equiv 1$): $L(\beta, \chi_0) = \zeta(\beta)$ has a pole at $\beta = 1$.
- For $k \neq 0$ (nontrivial characters): $L(1, \chi_k)$ is finite and nonzero (Dirichlet's theorem [21]).

Hence condensation occurs only in the χ_0 direction– this is determined by the uniqueness of the pole structure of the ζ function among all Dirichlet L-functions.

The stability of elementary states comes from the simplicity of the spectrum of K. The stability of the emergent state comes from the pole of $\zeta(\beta)$ at $\beta = 1$. The pole is a topological invariant of the ζ function: as a meromorphic function, the pole set of ζ in $\mathrm{Re}(s) > 0$ (only $\{s = 1\}$) cannot be removed by continuous deformation of parameters. The stability of elementary states can be broken by spectral degeneracy;

the stability of the emergent state is guaranteed by the topological structure of an analytic function.

§6.2 Structure of Stable States

The existence of stable states has been established. This section determines the position of a stable state within the state space, and the shape of its excitation at that position.

A stable excitation pattern carries two sets of physical properties: it occupies a definite position in the state space, and it oscillates in a definite way at that position. The algebraic structure of the state space describes these properties: the CRT decomposition describes the coordinates of the stable state on a multidimensional torus, and the involution ι describes its generational membership within a doublet.

Elementary states possess rich CRT coordinates and doublet structure, whereas the emergent state sits at the origin of the torus, with no fine structure but participating in all lattice sites.

6.2.1 Structure of elementary states

The two algebraic structures of the state space–the prime factorization of $p-1$ (CRT) and the involution ι of the multiplicative group– determine two sets of position coordinates for every stable state.

The unique factorization of $p-1$ is

$$p-1 = q_1^{b_1} \cdot q_2^{b_2} \cdots q_s^{b_s}, \tag{27}$$

where $q_1 < q_2 < \cdots < q_s$ are the $s = \omega(p-1)$ distinct prime factors of $p-1$, and $b_i \geq 1$ are their exponents. The Chinese Remainder Theorem (CRT) gives the ring isomorphism

$$\mathbb{Z}/(p-1)\mathbb{Z} \;\cong\; \mathbb{Z}/q_1^{b_1}\mathbb{Z} \;\times\; \mathbb{Z}/q_2^{b_2}\mathbb{Z} \;\times\; \cdots \;\times\; \mathbb{Z}/q_s^{b_s}\mathbb{Z}. \tag{28}$$

This isomorphism decomposes $\mathbb{Z}/(p-1)\mathbb{Z}$ (the labeling group of the character group $\widehat{(\mathbb{Z}/p\mathbb{Z})^*}$) into a direct product of s independent factors.

The physical meaning of the isomorphism (28) is that the character group has the structure of an s-dimensional discrete torus. The label $k \in \mathbb{Z}/(p-1)\mathbb{Z}$ of each stable state χ_k decomposes uniquely under CRT as

$$k \;\longleftrightarrow\; (k_1,\, k_2,\, \ldots,\, k_s), \quad k_i \in \mathbb{Z}/q_i^{b_i}\mathbb{Z}. \tag{29}$$

The component k_i is the "frequency coordinate" of χ_k along the i-th direction of the torus. The direct-product structure of CRT guarantees that the coordinates k_i and k_j $(i \neq j)$ along different directions are completely independent– changing k_i does not affect k_j; the directions do not interfere with one another.

The vector $(k_1, k_2, \ldots, k_s)$ is called the **mode vector** of the stable state χ_k. It gives the lattice position of the stable state on the s-dimensional torus. Each direction admits $q_i^{b_i}$ possible coordinate values, for a total of $\prod_i q_i^{b_i} = p-1$ lattice sites. The origin $(0, 0, \ldots, 0)$ corresponds to χ_0 (the emergent-state direction), and the remaining $p-2$ lattice sites correspond to the $p-2$ elementary states.

The unit group $(\mathbb{Z}/p\mathbb{Z})^*$ carries an involution (order-two automorphism) $\iota : a \mapsto -a \pmod p$. Since p is an odd prime, consider whether ι has fixed points: if $\iota(a) = a$, then $-a \equiv a \pmod p$, i.e. $2a \equiv 0 \pmod p$. But $a \in (\mathbb{Z}/p\mathbb{Z})^*$ means $a \not\equiv 0$, and p being an odd prime means $2 \not\equiv 0 \pmod p$, so $2a \equiv 0$ has no solution in $(\mathbb{Z}/p\mathbb{Z})^*$. Hence ι is fixed-point-free: for every $a \in (\mathbb{Z}/p\mathbb{Z})^*$ we have $-a \neq a$.

A fixed-point-free involution partitions the $p-1$ group elements exactly into

$$\frac{p-1}{2} \text{ doublets (dual pairs) } \{a,\ -a\}, \quad a = 1, 2, \ldots, \tfrac{p-1}{2}. \tag{30}$$

Each doublet contains two group elements, exchanged by ι. These are $(p-1)/2$ inequivalent positions– the doublets are the elements of the ι-orbit space $(\mathbb{Z}/p\mathbb{Z})^*/\langle\iota\rangle$.

The stable state χ_k is defined on all of $(\mathbb{Z}/p\mathbb{Z})^*$: let g be a primitive root and write the discrete logarithm $t = \mathrm{ind}_g(a)$ (i.e. $a = g^t$); then for each doublet $\{a, -a\}$,

$$\chi_k(a) = e^{2\pi i k\,\mathrm{ind}_g(a)/(p-1)}, \qquad \chi_k(-a) = \chi_k(-1)\,\chi_k(a), \qquad \chi_k(-1) = (-1)^k \tag{31}$$

(the last equality holds because $-1 = g^{(p-1)/2}$, so $\chi_k(-1) = e^{\pi i k} = (-1)^k$). Fixing the frequency direction χ_k and varying the doublet position a: all quantum numbers determined by k (the order d, the CRT components k_i, the Galois properties) remain unchanged– because they depend only on the label k, not on the evaluation point a. But χ_k takes different phase values $\chi_k(a)$ on different doublets, which makes the lattice projection ρ_j in the mass formula vary with the doublet (§6.4.7). Consequence: particles in the same frequency direction have different masses at different doublet positions.

To characterize the internal structure of a doublet, define the **tension** of each doublet:

$$T(a) = |2a - p|, \quad a = 1, 2, \ldots, \tfrac{p-1}{2}. \tag{32}$$

$T(a)$ is the separation between a and $-a$ $(= p - a)$ within $\{1, \ldots, p-1\}$. As a increases from 1 to $(p-1)/2$, $T(a)$ takes the values $p-2$, $p-4$, …, 3, 1– the tensions of the $(p-1)/2$ doublets form an arithmetic progression with common difference 2.

The same frequency direction χ_k at different doublet positions a–i.e. sharing all quantum numbers but differing in mass– is defined as different generations of the same family. One frequency direction has $(p-1)/2$ generations. Combining the CRT coordinates and the doublet position: the complete position of an elementary state is given by the mode vector plus the fundamental-domain position:

$$\text{position} = \Big(\ \underbrace{k_1,\, k_2,\, \ldots,\, k_s}_{\text{CRT torus coordinates}}\ ,\qquad \underbrace{a}_{\text{fundamental-domain position (generation)}}\ \Big). \tag{33}$$

The first s coordinates fix the lattice site of the particle on the torus (which frequency), and the last coordinate fixes its specific generation within that frequency direction. Together the $s+1$ coordinates uniquely determine the position of a physical particle. The independence of the two sets of coordinates is precisely the joint structure (σ, ι) of §3.3: the CRT coordinates run along the mode-space direction (the Galois action σ changes the character label), while the doublet position runs along the position-space direction (the involution ι changes the evaluation point); σ and ι act on different spaces, so species and generation can be labeled independently.

With the position fixed, the next question is what shape the excitation takes at that position.

We now examine the excitation shape of a stable state at its position– the spatial period, winding direction, and nodal structure of the wave function on the group. These three shape quantities are derived one by one from three algebraic properties of the state space (the homomorphic image, Galois symmetry, and the first isomorphism theorem).

The stable state $\chi_k : (\mathbb{Z}/p\mathbb{Z})^* \to S^1$ is a group homomorphism–it maps the multiplicative group into the unit circle. The image of a group homomorphism is a subgroup of the target group S^1. Since the domain $(\mathbb{Z}/p\mathbb{Z})^*$ is a finite group (of order $p-1$), the image is also finite. A finite subgroup of S^1 must be a group of roots of unity:

$$\mathrm{Im}(\chi_k) \leq S^1 \text{ finite} \implies \mathrm{Im}(\chi_k) = \mu_d \text{ (the group of } d\text{-th roots of unity).} \tag{34}$$

The integer d is the **order** of χ_k. Explicit computation: let g be a generator (primitive root) of $(\mathbb{Z}/p\mathbb{Z})^*$; then

$$\chi_k(g) = e^{2\pi i k/(p-1)}. \tag{35}$$

The order of χ_k is the smallest positive integer d with $\chi_k(g^d) = 1$, i.e. the smallest d with $e^{2\pi i k d/(p-1)} = 1$. This requires $(p-1) \mid kd$, i.e. $d = (p-1)/\gcd(k, p-1)$. Therefore:

$$d = \mathrm{ord}(\chi_k) = \frac{p-1}{\gcd(k,\, p-1)}. \tag{36}$$

The physical meaning of the order d is the spatial period: along the generator direction, the value of χ_k returns to 1 after d steps (completing one full phase cycle $0 \to 2\pi$). The larger d is, the faster the phase of the wave function varies–high-frequency oscillation; the smaller d, the slower the variation–low-frequency oscillation.

In CRT language, each component of the mode vector $(k_1, \ldots, k_s)$ has its own local order:

$$o_i = \mathrm{ord}_{q_i^{b_i}}(k_i) = \frac{q_i^{b_i}}{\gcd(k_i,\, q_i^{b_i})}, \quad i = 1, \ldots, s. \tag{37}$$

The total order d is related to the local orders by

$$d = \mathrm{lcm}(o_1,\, o_2,\, \ldots,\, o_s). \tag{38}$$

This is because CRT decomposes χ_k into a tensor product of s component characters χ_{k_i}, each with period o_i; returning to 1 overall requires all components to return to 1 simultaneously, i.e. taking the least common multiple of the local periods. In torus language: d is the number of steps needed to first return to the starting lattice site when walking from it in the direction $(k_1, \ldots, k_s)$–the length of the closed orbit on the torus.

By the structure of homomorphisms into S^1, every stable state has a definite period d. d can only take values dividing $p-1$, giving $\tau(p-1)$ possible periods (τ is the divisor-counting function). For each $d \mid (p-1)$, exactly $\varphi(d)$ characters have order d (φ is the Euler function–the number of elements of μ_d of order exactly d).

Checking the total count: $\sum_{d|(p-1)} \varphi(d) = p-1$, exactly the number of characters. Here $d=1$ corresponds to χ_0 (the emergent-state direction), and the $p-2$ characters with $d \geq 2$ are the elementary states.

The state space carries the Galois automorphism σ_{-1}, which maps each character to its complex conjugate:

$$\sigma_{-1}(\chi_k) = \overline{\chi_k} = \chi_{-k}. \tag{39}$$

Explicitly: $\chi_k(g) = e^{2\pi ik/(p-1)}$ becomes $e^{-2\pi ik/(p-1)} = \chi_{-k}(g)$ under complex conjugation. The phase of χ_k rotates in the positive direction (counterclockwise), while $\bar{\chi}_k = \chi_{-k}$ rotates at the same rate in the negative direction (clockwise). The two describe the two winding directions of the same standing wave–positive and negative winding.

$\chi_k = \bar{\chi}_k$ if and only if χ_k is real-valued, i.e. $\mathrm{Im}(\chi_k) \subseteq \{+1, -1\} = \mu_2$. This requires $d = \mathrm{ord}(\chi_k) \mid 2$, i.e. $d=1$ or $d=2$.

- $d=1$: χ_0 is the trivial character (identically 1), $\bar{\chi}_0 = \chi_0$, with no winding distinction.
- $d=2$: $\chi_{(p-1)/2}$ is the unique character of order two (the Legendre symbol), taking values ± 1, $\bar{\chi}_{(p-1)/2} = \chi_{(p-1)/2}$, self-conjugate, no winding.
- $d>2$: $\chi_k \neq \bar{\chi}_k$, and the two form a **chirality pair** $\{\chi_k, \bar{\chi}_k\}$. The two stable states in a chirality pair share:
  - the same order d (since $\mathrm{ord}(\bar{\chi}_k) = \mathrm{ord}(\chi_k)$);
  - the same eigenvalue modulus $|\lambda_k| = |L(\beta, \chi_k)| = |L(\beta, \bar{\chi}_k)|$ (the L-function satisfies $L(s, \bar{\chi}) = \overline{L(\bar{s}, \chi)}$, so the moduli agree at real β);
  - the same CRT local orders o_i (since $-k$ has the same order as k in each component).

  But their phases are opposite: $\arg \lambda_k = -\arg \lambda_{-k}$.

The $p-3$ stable states with $d>2$ (excluding one each with $d=1$ and $d=2$) split exactly into $(p-3)/2$ chirality pairs. By the existence of the Galois automorphism σ_{-1}, stable states with $d>2$ come in pairs– every positive-winding particle has a negative-winding partner.

Define χ_k ($k < (p-1)/2$) as positive-winding and χ_{-k} as negative-winding; stable states with $d \leq 2$ are achiral (self-conjugate). In the formulas, the winding is encoded by the parameter G:

$$G = \begin{cases} 2, & \chi_k \neq \bar{\chi}_k \text{ (chirality pair, } d>2), \\ 1, & \chi_k = \bar{\chi}_k \text{ (self-conjugate, } d \leq 2). \end{cases} \tag{40}$$

The first isomorphism theorem (a fundamental theorem of group theory) gives: for the group homomorphism $\chi_k : (\mathbb{Z}/p\mathbb{Z})^* \to \mu_d$,

$$(\mathbb{Z}/p\mathbb{Z})^* / \ker \chi_k \;\cong\; \mathrm{Im}(\chi_k) = \mu_d. \tag{41}$$

Hence:

$$|\ker \chi_k| = \frac{|(\mathbb{Z}/p\mathbb{Z})^*|}{|\mu_d|} = \frac{p-1}{d}. \tag{42}$$

The kernel $\ker \chi_k = \{a \in (\mathbb{Z}/p\mathbb{Z})^* : \chi_k(a) = 1\}$ is the set of group elements on the nodal surface where the wave function is identically 1. For any two elements $a, b \in \ker \chi_k$ on the nodal surface, $\chi_k(a) = \chi_k(b) = 1$– the wave function χ_k cannot distinguish them. More generally, χ_k partitions $(\mathbb{Z}/p\mathbb{Z})^*$ into d equivalence classes (the d cosets of $\ker \chi_k$), and elements in the same class are indistinguishable to χ_k.

The size of the nodal surface $|\ker \chi_k| = (p-1)/d$ reflects the resolving power of the excitation:

- When d is small, $|\ker| = (p-1)/d$ is large–the nodal surface contains most group elements, the wave function partitions the group into only a few equivalence classes, and the resolving power is weak.
- When d is large, $|\ker| = (p-1)/d$ is small–the nodal surface contains only a few group elements, the wave function partitions the group into many equivalence classes, and the resolving power is strong.
- In the extreme case $d = p-1$, $|\ker| = 1$ (the kernel contains only the identity), and the wave function takes $p-1$ distinct values on the $p-1$ elements–complete resolution, each group element corresponding to a unique phase value.
- At the other extreme $d = 1$, $|\ker| = p-1$ (the kernel is the whole group), and the wave function $\chi_0 \equiv 1$ takes the same value on all elements–no resolution at all.

The smaller the kernel, the finer the microscopic structure of the excitation. $|\ker \chi_k|$ and d are in one-to-one correspondence (for given p, d determines $|\ker| = (p-1)/d$), but they characterize oscillation frequency and resolving power respectively.

Combining period, winding, and nodal surface: the excitation shape of an elementary state at its position is completely characterized by three quantities:

$$\text{shape} = \big(\underbrace{d}_{\text{period}},\ \underbrace{\pm \text{ or } 0}_{\text{winding}},\ \underbrace{|\ker| = (p-1)/d}_{\text{nodal surface}}\big). \tag{43}$$

These three quantities are not independent–d and $|\ker|$ are functions of each other–but each conveys a different physical intuition. All the information of the shape is encoded in the order d and the winding sign $\pm$. Together with the position coordinates (33), the complete description of a stable state is:

$$\text{complete description} = \big(\underbrace{k_1, \ldots, k_s,\ a}_{\text{position}},\ \underbrace{d,\ \pm}_{\text{shape}}\big). \tag{44}$$

This completes the geometric description of elementary states. The emergent state has different geometric properties.

6.2.2 Structure of the emergent state

The discussion of position and shape above concerned elementary states. We now derive the position and shape of the emergent state–which is collective and full-ring.

The emergent state sits in the χ_0 direction. In the language of the CRT torus, $k = 0$ corresponds to the mode vector $(k_1, k_2, \ldots, k_s) = (0, 0, \ldots, 0)$– the origin of the torus. Physical meaning: the emergent state has zero frequency in every direction of the CRT torus– it oscillates in no direction, has no spatial structure, and is a pure “zero mode”. Elementary states are defined on the species-level group $(\mathbb{Z}/p\mathbb{Z})^*$ (§6.1.1); after lifting to the full space along the reduction $\mathbb{Z}/p^K\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$, they involve only the $\varphi(p^K) = p^{K-1}(p-1)$ elements coprime to p. The multiples of p ($p, 2p, 3p, \ldots, p^{K-1} \cdot p$) are not in the unit group, and elementary states do not involve these elements.

The participation range of the emergent state is different. The partition function $Z(\beta) = \sum_{n=1}^{p^K} n^{-\beta}$ sums over all p^K lattice sites– including the multiples of p. The pole-driven condensation involves the full lattice set, without distinguishing $\gcd(n,p) = 1$ from $p \mid n$. The participation range of the emergent state is all p^K elements of the full ring $\mathbb{Z}/p^K\mathbb{Z}$– larger than the $\varphi(p^K)$ of the elementary states by a factor $p/(p-1)$.

An elementary state $\tilde{\chi}_k$ is a single eigenvector of K– under propagation it is merely multiplied by the fixed factor λ_k, orthogonal to and independent of the other elementary states. The emergent state corresponds to no single eigenvalue of K. It arises in the limit $\beta \to 1$, where the pole of $Z(\beta)$ makes the χ_0-direction component of the KMS state condense cooperatively out of all lattice sites into an indecomposable emergent state.

Elementary states have $(p-1)/2$ doublet positions (generations) and s-dimensional CRT coordinates. The emergent state has none of this fine structure: $\chi_0 \equiv 1$ takes the value 1 on all group elements– there is no phase distinction within a doublet ($\chi_0(a) = \chi_0(-a) = 1$), and no fine structure within the CRT lattice (the mode vector is zero). The emergent state has only one direction and one generation.

	Elementary state	**Emergent state**
CRT coordinates	$(k_1, \ldots, k_s)$, at least one $k_i \neq 0$	$(0, 0, \ldots, 0)$
Doublet position	$(p-1)/2$ generations	none (unique)
Participating lattice sites	$\varphi(p^K)$	p^K
Spatial structure	nodal surfaces, fine oscillation	no nodal surface, globally coherent

This determines the position and shape of both classes of stable states.

§6.3 Formation of Stable States

This section examines the formation process of stable states– how energy in the KMS heat bath reaches a given direction and builds up a stable excitation pattern.

The formation mechanisms of the two classes of stable states differ in nature:

- **Elementary states**: energy propagates step by step through the finite Dirac operator D_F, starting from χ_0 and proceeding station by station along the CRT torus to the nontrivial character directions;
- **Emergent states**: energy condenses collectively at the pole of $\zeta(\beta)$, with all p^K lattice sites participating in concert, condensing at once.

The two are described in turn below.

6.3.1 Formation of elementary states

The process by which the $p-2$ nontrivial elementary states draw energy from the heat bath is driven by the finite Dirac operator D_F, starting from the trivial character

χ_0 and reaching the nontrivial directions step by step. The formation process uses the geometric labels of the full description (44).

D_F encodes the noncommutativity of addition and multiplication in the BC algebra: it is the effective propagation operator of the additive translation T_a in the space of multiplicative characters. The CRT decomposition $V = V_1 \otimes V_2 \otimes \cdots \otimes V_s$ writes the mode space of elementary states as a tensor product of s independent factors. Under this decomposition D_F advances one step along a single factor direction at a time, leaving the state of the remaining factors unchanged.

Starting from $\chi_0 = (0, 0, \ldots, 0)$, each step advances along a single CRT factor direction–the admissible steps correspond to single-factor characters (exactly one nonzero coordinate, §6.4.3). The number of steps required to reach the lattice site labeled by the mode vector $(k_1, \ldots, k_s)$ equals the number of nonzero coordinates. The more steps, the longer the propagation needed to establish the standing wave, and each additional step attenuates the amplitude further– the difference in step count thereby produces the mass hierarchy.

Every step and every node of the propagation path carries a definite weight. The step strength is determined by the spectrum of the convolution operator K; the node weight is determined by the Galois symmetry at the site, the CRT factor dimension d_i, and the normalization of the KMS state. The unifying principle: indistinguishable degrees of freedom are normalized at amplitude level, distinguishable degrees of freedom are counted at probability level. Upon arrival at a site, the pairing with the Galois-conjugate direction $\bar{\chi}_k$ is fixed by the winding direction– propagation pairing the positive and negative windings takes the direct channel, propagation across doublets takes the indirect channel. The precise propagation strengths and node factors are defined in §6.4.

Different generations within the same doublet correspond to different projection positions on the same transition path. The involution ι pairs characters into doublets; the generations occupy different relative positions within the doublet, leading to different accumulated attenuation along the propagation path–whence the generational mass hierarchy.

6.3.2 Formation of emergent states

Emergent states do not undergo the stepwise propagation of D_F. Their formation mechanism is **collective condensation**: as $\beta \to 1^+$, the pole of the partition function $Z(\beta) = \zeta(\beta)$ makes the spectral weight of the KMS state in the χ_0 direction diverge, and all p^K lattice sites inject energy into the χ_0 direction simultaneously and in concert–a one-shot global condensation.

In contrast:

- Elementary states: energy propagates step by step along D_F paths, attenuating at each step; more steps yield smaller amplitude–this produces the mass hierarchy.
- Emergent states: no stepwise propagation; the pole drives a global condensation in which all lattice sites participate simultaneously–this produces an extremely large spectral weight.

§6.4 will use these formation structures–the step counts, per-step strengths, and per-station normalizations of elementary states, and the pole-condensation mechanism of emergent states–to compute the masses of the two classes of stable states.

§6.4 Elementary-State Masses

The physical meaning of mass is: how much spectral weight a stable excitation occupies in the thermal-equilibrium background. The larger the weight, the larger the inertia of that mode in the heat bath, corresponding to a larger mass. The KMS state φ_β is precisely the thermal-equilibrium state of the state space at temperature $1/\beta$, and it assigns a definite spectral weight to each direction. Mass is therefore defined as:

$$m = v \cdot \varphi_\beta(P_\psi),$$

where v is the energy-scale parameter and P_ψ is the complete geometric projection operator of the stable state ψ. The two classes of stable states differ in the specific form of the projection P_ψ and in the outcome of the KMS evaluation: when the evaluation is finite, the mass is read off linearly; when the evaluation diverges, the mass is read off from the energy-scale formula (24) (§6.5).

Elementary states have fine geometric structure (Galois, doublet, CRT lattice), the projection P_f is correspondingly fine, and the KMS evaluation gives finite values of L-functions. The emergent state has none of this fine structure; its projection P_{em} is merely the full-ring projection in the χ_0 direction, and the KMS evaluation gives the pole of $\zeta(\beta)$. We compute these in turn.

6.4.1 The mass operator

The starting point of the mass computation is the definition of the entanglement seed–it measures the coupling strength of the internal Dirac operator in each character direction.

We first give the formal definition of the internal structure. The internal degree-of-freedom space is taken to be the group algebra of the finite group $G_F \simeq \mathbb{Z}/p\mathbb{Z}$, $\mathcal{H}_F = \mathbb{C}[\mathbb{Z}/p\mathbb{Z}]$; T_a ($a \in \mathbb{Z}/p\mathbb{Z}$) is the translation operator $(T_a\psi)(x) = \psi(x+a)$ on $\mathcal{H}_F$; the **internal Dirac operator** D_F is a self-adjoint operator on $\mathcal{H}_F$, given in the position basis by a kernel $D(x-y)$ whose Fourier coefficients are denoted $\hat{D}(m)$. D_F does not commute with translations–the commutator $[D_F, T_a]$ encodes the noncommutativity of the additive and multiplicative structures (§6.3), and it drives the mass transfer between directions. The triple $(\mathcal{A}_F, \mathcal{H}_F, D_F)$ (with $\mathcal{A}_F = \mathbb{C}[\mathbb{Z}/p\mathbb{Z}]$ acting on itself) constitutes a finite spectral triple.

The eigenbasis of translations on $\mathcal{H}_F$ consists of the additive characters $e_j(x) = \omega^{jx}$ ($j \in \mathbb{Z}/p\mathbb{Z}$, $\omega = e^{2\pi i/p}$): $T_a e_j = \omega^{ja} e_j$. The matrix elements of $[D_F, T_a]$ in this basis are

$$\langle e_j|[D_F, T_a]|e_l\rangle = \hat{D}(j-l)\big(\omega^{la} - \omega^{ja}\big).$$

The diagonal matrix elements vanish identically (for $j = l$ the two phase factors cancel): a single additive mode by itself carries no coupling; the nonzero coupling comes entirely from transfer between modes– in particular the cross coupling between conjugate mode pairs $\{e_j, e_{-j}\}$, which carries the winding structure of the chirality pairs (§6.4.2). In what follows we take the flat normalization of the kernel $|\hat{D}(m)| = 1$ ($m \neq 0$): the overall strength of D_F is absorbed into the energy-scale parameter v, and the direction dependence is given entirely by the character structure.

The elementary-state directions are the multiplicative characters χ_k (defined on $(\mathbb{Z}/p\mathbb{Z})^*$, extended by $\chi_k(0) = 0$), while the translation eigenbasis consists of the additive characters e_j–the two dualities are bridged by Gauss sums. The Fourier coefficients of χ_k in the additive basis are

$$\hat{\chi}_k(j) = \sum_x \chi_k(x)\,\omega^{-jx} = \tau(\chi_k)\,\overline{\chi_k}(-j), \qquad \tau(\chi_k) = \sum_x \chi_k(x)\,\omega^x,$$

where $\tau(\chi_k)$ is the Gauss sum. For nontrivial χ_k, the standard computation (substituting $x = ty$ and using character orthogonality) gives $|\tau(\chi_k)|^2 = p$; moreover $\hat{\chi}_k(0) = \sum_x \chi_k(x) = 0$ (orthogonality). Therefore: every nontrivial multiplicative direction spreads uniformly in the additive dual, occupying all $p-1$ nonzero additive modes, with amplitude modulus exactly $|\hat{\chi}_k(j)| = \sqrt{p}$ on each mode.

For a nontrivial multiplicative direction χ_k, define the **entanglement seed** $\mu_j(a)$ as the effective amplitude deposited by the translation T_a into the χ_k direction via the additive mode e_j: its modulus is the expansion amplitude $|\hat{\chi}_k(j)| = \sqrt{p}$ on that mode, and its phase is the eigenphase ω^{ja} of T_a on that mode,

$$\mu_j(a) \;=\; \sqrt{p}\,\omega^{ja}, \qquad j \neq 0. \tag{45}$$

The modulus $\sqrt{p} = |\tau(\chi_k)|$ is independent of the direction label k: $|\mu_j(a)| = \sqrt{p}$ holds for all nontrivial directions, all nonzero modes, and all a– **local equal strength**: the entanglement seed does not distinguish directions, guaranteed by the uniform-modulus property of Gauss sums. The global mass is given by the modulation of the entanglement seed by L-function values: local equal strength provides the same base for all directions, and the absolute values of the L-function values distinguish the directions, giving different masses.

6.4.2 Direct/indirect channels

With the entanglement seed established, the symmetrization of chirality pairs divides the characters into direct and indirect channels. Among the multiplicative characters, χ_k and $\chi_{p-1-k} = \bar{\chi}_k$ are complex conjugates of each other, forming the chirality pair $\{\chi_k, \bar{\chi}_k\}$ (§6.2); in the additive dual, complex conjugation maps the occupied mode j to $-j$, with conjugate coefficients: $\overline{\hat{\chi}_k(j)} = \widehat{\bar{\chi}_k}(-j)$. For each chirality pair define the real combinations

$$\chi_k^{(\pm)} \;=\; \frac{1}{\sqrt{2}}\big(\chi_k \pm \bar{\chi}_k\big).$$

The trivial character χ_0 is real (self-conjugate). For odd prime p there also exists a unique real nontrivial character of order 2, $\chi_{(p-1)/2}$ (the Legendre symbol, taking values ± 1, satisfying $\bar{\chi}_{(p-1)/2} = \chi_{(p-1)/2}$). These two real characters require separate treatment.

Define the **symmetrized seed** as the antisymmetric coherent combination of the seeds of the two windings of a chirality pair on the conjugate mode pair $\{j, -j\}$:

$$\mu_j^{\mathrm{sym}}(a) \;=\; \mu_j(a) - \bar{\mu}_j(a) \;=\; 2i\,\mathrm{Im}\,\mu_j(a),$$

which measures whether the two winding components can produce a net coupling. In the symmetrized basis of chirality pairs:

1. **Direct channel**: if χ_k is a complex character ($\mathrm{ord}(\chi_k) > 2$), the two members of the chirality pair carry independent winding components; from $\mu_j(a) = \sqrt{p}\,\omega^{ja}$ and $\bar{\mu}_j(a) = \sqrt{p}\,\omega^{-ja}$,

$$\mu_j^{\mathrm{sym}}(a) = 2i\sqrt{p}\,\sin\frac{2\pi ja}{p} \neq 0$$

($j \neq 0$ and $a \not\equiv 0 \pmod p$ guarantee $ja \not\equiv 0 \pmod p$), corresponding to a massive elementary-state channel;

2. **Indirect channel**: if the character is real ($\mathrm{ord}(\chi_k) \leq 2$, i.e. χ_0 or $\chi_{(p-1)/2}$), the chirality pair degenerates to a single element– the two winding components coincide identically, and the antisymmetric carrier $\chi_k^{(-)}$ vanishes identically:

$$\mu_j^{\mathrm{sym}} = 0,$$

the symmetrized seed cancels exactly, and mass must be acquired through indirect (higher-order) transitions. For χ_0, moreover, the seed itself vanishes identically (a nonprimitive direction: the Gauss sum degenerates, $|\hat{\chi}_0(j)| = 1$, and the trivial representation commutes with D_F).

This criterion is precisely the winding classification of §6.2: chirality pairs with winding $\pm$ have nonzero symmetrized seed and take the direct channel; real characters with winding 0 have exactly cancelling symmetrized seed and take the indirect channel. Particles in the indirect channel–corresponding to real character directions–acquire mass only through higher-order transitions. Their mass must detour through the off-diagonal couplings of complex character directions and be transferred back, a longer path further diluted through χ_0, several orders of magnitude smaller than direct-channel particles. That the symmetrized seed vanishes exactly is determined by the self-conjugacy of real characters.

6.4.3 CRT transition rules

With the channel type determined, the CRT structure gives the concrete transition paths from χ_0 to each nontrivial character direction. In the internal space on which D_F acts, transitions take place in the multiplicative sector $\mathbb{C}[(\mathbb{Z}/p\mathbb{Z})^*] \subset \mathcal{H}_F$ (spanned by all multiplicative characters, of dimension $p-1$; the complementary direction δ_0 is the extension zero point and does not participate in transitions). By the CRT decomposition (28) of the label group, the multiplicative sector is written as $V_1 \otimes V_2 \otimes \cdots \otimes V_s$, with each factor V_i corresponding to the i-th direction of the CRT decomposition.

Each transition step changes the state of only one factor V_i, leaving the other factors unchanged. Formally, on the multiplicative sector

$$D_F = \sum_{i=1}^{s} \mathbb{1}_{V_1} \otimes \cdots \otimes D^{(i)} \otimes \cdots \otimes \mathbb{1}_{V_s},$$

where $D^{(i)}$ is the local Dirac operator on the i-th factor.

By the direct-sum form of D_F, each transition step proceeds along a single CRT factor direction: the admissible transition generators are the **single-factor**

characters– characters with exactly one nonzero CRT coordinate; characters with two nonzero coordinates correspond to diagonal moves, which are not elementary steps. The coordinates $(k_1, \ldots, k_s)$ of the target residence χ_f give its generator decomposition: each nonzero coordinate k_i corresponds to a unique single-factor generator $\ell_i = (0, \ldots, k_i, \ldots, 0)$,

$$\chi_f = \prod_{i:\, k_i \neq 0} \ell_i,$$

the decomposition being uniquely determined by coordinate arithmetic. A transition path is a sequence starting from χ_0 and passing through the stations in order; the station set is one of the following two:

- **Open path**: the stations are the generators, and the product of the stations equals the residence– the residence is reached by the generator composition and does not appear as a station;
- **Closed path**: the stations are the generators plus the residence itself, and the product of the stations equals χ_0– the path closes through the residence station.

For the Galois pair $\{\chi, \bar{\chi}\}$ of a residence with two nonzero coordinates, the generator sets of the two members are conjugates of each other, giving an open path and a closed path respectively; which member takes which path belongs to the identification data of §9. The order of the stations is fixed by the background coupling: the KMS background is σ-symmetric, while the natural component of an odd character ($\chi(-1) = -1$) is the antisymmetric combination (§6.4.7), whose overlap with the background is zero: the entry station must be an even character; the exit station is the residence (closed path) or the remaining generator (open path). Each station contributes one L-function factor, and the product gives the total coupling strength.

6.4.4 Decomposition of the evaluation

With the transition path determined, the KMS evaluation $m_f = v\,\varphi_\beta(P_f)$ of §6.4.1 expands along that path into a product of matrix elements; the factors attached to the individual links of the path are obtained one by one as the expansion proceeds.

The expansion proceeds in three steps, and the evaluation level of each factor is fixed afterwards. Step 1, decomposition of the projection: the generator decomposition of the residence (§6.4.3) writes P_f as a station-by-station transfer along the path; inserting the completeness decomposition of the character basis between adjacent stations, the KMS evaluation becomes a chain of matrix elements

$$m_f = v \left| \langle \mathrm{out} \,|\, \mathcal{V}(s_n) \cdots \mathcal{V}(s_1) \,|\, \Omega \rangle \right|^c, \tag{46}$$

where $|\Omega\rangle$ is the cyclic vector of the KMS background. Step 2, structure of the single-station operator:

$$\mathcal{V}(s) = P_s\, \hat{K}^{1/2}\, D(s), \tag{47}$$

where $\hat{K}^{1/2}$ is the amplitude-level evaluation of the convolution operator at the vertex evaluation point $\beta = \frac{1}{2}$ (the $\Delta^{1/2}$ layer of §4.5), diagonal in the character basis with diagonal elements the resonance ratios $R_s = L(\frac{1}{2}, \chi_s)/\zeta(\frac{1}{2})$; P_s is the endpoint projection determined by the CRT coordinates of the station (§6.4.6); and $D(s)$ is the virtual-process correction factor of the station (§6.4.7; stations without insertion processes take 1). Step 3, final state and powers: for a closed path the product of the stations returns to χ_0, the final state is $|\Omega\rangle$, and the matrix element is taken in squared modulus as a whole; an open path ends at the residence, and the evaluation is the amplitude-level projection $\langle\chi_f|$ onto the residence direction; the generational evaluation is given by the response vector $\langle r_g|$ (§6.4.7). When the path breaks into decoherent segments, each segment is taken in squared modulus independently and the results multiplied.

The evaluation level of each factor in the matrix element (Born-level versus corrected-level, §11–§12) is determined by the temporal structure of the process, in two parts.

First, inputs internal to a virtual process take Born-level values. The insertion sequence of a dwell segment is generated by the insertion operator $\hat{V}$ (§12.1): n insertions are indistinguishable, the count divides by $n!$, and the series sums to $\exp(i\alpha^{(0)}\hat{V})$. This summation is performed only once: if an intermediate state inside a closure were evaluated at the corrected level, the insertion series would appear a second time in the expansion, and the second-order insertion contribution would be accounted for twice, once among the outer insertions and once in the inner summation, contradicting the indistinguishable counting.

Second, real processes take corrected-level values. The phase of a real process accumulates in units of the dwell period (the frequency shift of §12.1), and the reading is the time-averaged matrix element:

$$\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T} e^{-it\alpha^{(0)}\hat{V}}\,A\,e^{it\alpha^{(0)}\hat{V}}\,dt \;=\; \sum_{\pm} P_{\pm}\,A\,P_{\pm}, \tag{48}$$

where $P_{\pm}$ are the eigenprojections of $\hat{V}$ (the dimension is finite, so the limit always exists). The average diagonalizes the reading in the eigenbasis of $\hat{V}$, which is the mass-state basis (the symmetrized combinations, §12.1); the eigenphase accumulates over dwell periods, and the reading carries the frequency-shift factor $(1-\alpha_{\text{eff}})$. A virtual process is truncated by the projections at its two ends within a single propagation step; no time average is available, and the reading stays at the Born level. This average is the same operation as the time average of the modular flow in §4.5: the distinction between the corrected level and the Born level is whether the reading has passed through the time average of the insertion flow.

The remaining two assignments fail: a real process at the Born level reads a non-stationary matrix element, which varies with the truncation instant and does not constitute an observable; a virtual process at the corrected level contradicts the counting of the first part.

The following subsections compute the links in the order of the evaluation. No principle independent of this matrix element is introduced hereafter; the derivation, completeness, and uniqueness of the evaluation rule set are established in Appendix A.

6.4.5 L-factors and exponents

Once the transition path is determined, the contribution of each station to the mass is given by the L-function value at that station and its exponent. The exponent is determined by two degrees: the Galois degree counts the contribution of the complex-conjugate partner, and the covering number counts how many times the station is covered by independent KMS evaluations.

Each station s_k of the transition path corresponds to a character χ_{s_k}, which determines a Dirichlet L-function $L(s, \chi_{s_k})$– namely the eigenvalue of K in that direction. The mass contribution is taken to be $|L(\frac{1}{2}, \chi_{s_k})|$– the absolute value of the L-value on the critical line (the measurement point $\beta = 1/2$ is fixed by the modular structure of §4: the amplitude-level weight of the $\Delta^{1/2}$ layer).

The conjugate-connection structure of the vertex can be checked numerically on its own: replacing it by the diagonal evaluation $\langle\chi^{(+)}|K|\chi^{(+)}\rangle = \mathrm{Re}\, L$ turns the L-factor of the spectral weight from a power of the modulus into a power of the real part, and the two choices differ substantially; the comparison with observation is given in §12.1, where the diagonal choice is excluded.

The exponent is determined by two independent degrees:

For the character χ_{s_k} at transition station s_k, define:

- **Galois degree** G: $G = 2$ if χ_{s_k} is a complex character (its conjugate orbit $\{\chi, \bar{\chi}\}$ has size 2), $G = 1$ if χ_{s_k} is a real character;
- **Covering number** C: the number of times the station is covered by independent KMS evaluations. Each probability-level closure (KMS evaluation taking the modulus squared) contributes 2 to the stations it covers, and an amplitude-level readout (the vertex response of the $\Delta^{1/2}$ layer, §4.5) contributes 1.

The covering structure is uniquely determined by the segment topology of the path:

- Closed path: the whole path forms a single probability closure, and all stations have $C = 2$;
- Open path: the residence is reached by the generator composition and does not appear as a station (§6.4.3), no probability closure can be made at the residence, and the readout takes place at the vertex response of the final generator: the final generator has $C = 1$, the remaining stations $C = 2$;
- Decoherent segments: each segment closes independently, the shared endpoint station is covered once by each of the two segments, $C = 1 + 1 = 2$;
- Multiple passes: each pass counts one closure, and covering numbers add.

The effective exponent of the station is

$$\mathrm{power}(s_k) \;=\; G \times C. \tag{49}$$

For a single-closure path, a complex-character station with $G = C = 2$ gives power $= 4$, a real-character station with $G = 1$, $C = 2$ gives power $= 2$; a real-character station at the end of an open path with $G = C = 1$ gives power $= 1$. The covering number equals, place by place, the background-normalization exponent of the node factor (§6.4.6): both arise from the same segment decomposition. Combining all stations along the path, the total contribution of the L-factors is

$$\mathcal{L}_f \;=\; \prod_{k=1}^{n} \left|L(\tfrac{1}{2},\, \chi_{s_k})\right|^{G_k \cdot C_k}, \tag{50}$$

where n is the number of transition steps and G_k, C_k are the Galois degree and covering number of the k-th station.

6.4.6 Node normalization factors

Besides the L-factors, the endpoint stations of a transition path also carry **node factors** $\mathcal{T}_{s_k}$, which convert the algebraic normalization of the signal into a physically observable probability amplitude.

Node factors arise only at the entry and exit stations. The convolution operator K is diagonal in the character basis (§3.2); the projection at an intermediate station acts as the identity on the eigendirections of K, merely selecting the path without producing numerical factors; numerical factors can only come from the state overlaps at the two ends– the σ-symmetric background coupling at the entry and the projection at the exit, i.e. the evaluation of the endpoint projection P_s in (47). For a single-station path the start and end coincide, and the factor is counted once.

The value of the projection is uniquely determined by the distinguishability on the CRT factor. For $a_i \neq 0$ the selection in that direction has already been made by the generator multiplication; the projection is the identity and the factor is 1. For $a_i = 0$ there are two cases: when the degrees of freedom in that direction are inseparable (χ and $\bar{\chi}$ belong to the same conjugate orbit), the physical state is the symmetric combination in the invariant subspace, the endpoint evaluation is the overlap of the unit vector $u = (1/\sqrt{q_i^{b_i}}) \sum_x |x\rangle$ with the entry state, and the factor is the projection norm $1/\sqrt{q_i^{b_i}}$; when the degrees of freedom are distinguishable, there is no observable connecting different sectors (superselection), the cross terms vanish, and the evaluation is the trace $\mathrm{Tr}\,\mathbf{1}_{V_i} = q_i^{b_i}$ over the trivial representation, normalized by the background amplitude $|\zeta(\frac{1}{2})|$ at the vertex evaluation point. The factor at an endpoint station is the product of these evaluations over the CRT factors:

For a station s_k, let its CRT coordinates be $(a_1, a_2, \ldots, a_s)$. Each CRT factor V_i contributes a node factor $\mathcal{N}_i(a_i)$:

$$\mathcal{T}_{s_k} = \prod_{i=1}^{s} \mathcal{N}_i(a_i), \tag{51}$$

where the form of each factor is determined by the symmetry in that direction:

$$\mathcal{N}_i(a_i) = \begin{cases} q_i^{b_i}/|\zeta(\frac{1}{2})|, & a_i = 0,\ \text{distinguishable}, \\ 1/\sqrt{q_i^{b_i}}, & a_i = 0,\ \text{indistinguishable}, \\ 1, & a_i \neq 0, \end{cases} \tag{52}$$

where $q_i^{b_i} = \dim V_i$ is the dimension of the i-th CRT factor; for $a_i = 0$ the two cases are: degrees of freedom in that direction distinguishable (trivial representation, counting $q_i^{b_i}$ distinguishable states), or indistinguishable (χ and $\bar{\chi}$ inseparable, projection onto the invariant subspace). The counting in the distinguishable case is probability-level: there is no observable connecting different sectors (superselection), the cross terms vanish, and inclusive summation gives the factor $q_i^{b_i}$; the normalization is taken with respect to the background amplitude $|\zeta(\frac{1}{2})|$ at the vertex evaluation point $\beta = \frac{1}{2}$ (the full-ring spectral weight, one amplitude-level reference). This is the unified principle of §6.3.1: indistinguishable degrees of freedom are normalized at amplitude level ($1/\sqrt{d_i}$), distinguishable degrees of freedom are counted at probability level ($\times d_i$).

The entry projection and the counting of distinguishable labels are both attributes of the $\Delta^{1/2}$-layer vertex: the former is the amplitude-level projection onto the symmetric combination at the vertex, the latter is the inclusive counting at the vertex readout. Decoherent-segment paths contain no vertex (every segment is a probability-level closure, §6.4.5), so both attributes are exempted simultaneously: the probability-level closure sums inclusively over intermediate labels, the two Galois copies each carry $(1/\sqrt{d_1})^2$, and the sum $d_1 \times (1/\sqrt{d_1})^2 = 1$– the amplitude-level projection is exactly compensated after squaring and inclusive summation; each probability-level closure retains only one factor of background normalization $1/|\zeta(\frac{1}{2})|$. For a path decohered into n segments, the node factor is $(1/|\zeta(\frac{1}{2})|)^n$. χ_0 as a **relay station** (indirect channel) carries a dilution factor on each pass:

$$\mathcal{T}_0 \;=\; \frac{1}{N}, \tag{53}$$

where N is the effective number of modes of the unit group (for the value of the effective exponent see §10.1): the signal re-enters the KMS background ensemble and is shared equally among N modes. χ_0 as the starting point of a path does not carry this factor– the departure is already normalized by the σ-symmetric coupling at the entry. The total node-normalization factor of the full path is the product of the endpoint factors and the relay dilution factors.

6.4.7 Generational factors

The generations of an elementary state correspond to different lattice projections along the same transition path– that is, different values of the doublet position a (§6.2.1). The difference between generations comes from the projection weights of the L-function phase on the discrete lattice.

Let $m = (p-1)/2$ be the generation number. The generational evaluation is the inner product of two vectors. The first is the lattice configuration: at the j-th doublet site, the phases of the excitation on the three components $\{\chi_0, \chi, \bar{\chi}\}$ are

$$|g_j\rangle = |\chi_0\rangle + e^{i\theta_j}|\chi\rangle + e^{-i\theta_j}|\bar{\chi}\rangle, \qquad \theta_j = \delta + \frac{2\pi j}{m}\,: \tag{54}$$

the χ_0 component carries no doublet label and hence no phase; the two members of the conjugate orbit are the two winding directions of the same standing wave, with opposite phases. The second is the response vector: the σ-symmetric coupling at the entry station supplies the χ_0 component with unit weight, and the G members of the conjugate orbit are inseparable and normalized by $1/\sqrt{G}$,

$$\langle r| = \langle\chi_0| + \frac{1}{\sqrt{G}}\big(\langle\chi| + \langle\bar{\chi}|\big). \tag{55}$$

Expanding the inner product gives

$$f_j \;=\; \langle r|g_j\rangle \;=\; 1 + \sqrt{G}\cdot\cos\!\left(\delta + \frac{2\pi j}{m}\right), \qquad \delta = \arg L(\tfrac{1}{2},\, \chi_k), \tag{56}$$

where χ_k is the representative character of the particle type. The generational factor is the probability-level ratio

$$\rho_j \;=\; \left(\frac{f_j}{f_0}\right)^2. \tag{57}$$

The lattice projection ρ_j encodes how the L-function phase δ distributes weight over the m equally spaced lattice sites: when δ is close to 0 or π the generations degenerate, while a displaced δ produces a strong generational hierarchy. m is the generation number, i.e., the number of involution doublets $(p-1)/2$ (§6.2), arising from the cyclic structure among the doublets and independent of the dimension of the CRT factor (the two have different origins; in the channel identified in §9 their values happen to coincide).

The unit weight of the χ_0 component of the response vector can be checked numerically on its own: scanning w in $f_j = w + \sqrt{G}\cos\theta_j$, two independent observations are reproduced exactly at the same value $w = 1$ (§12.1).

The number of available positions for the lattice projection is determined by the path. Generational readout takes place in the m-dimensional doublet readout space; the natural component of each station's character on the doublets (even characters take the symmetric combination $\chi(a) + \chi(-a)$, odd characters the antisymmetric one) gives a readout vector. Letters located in different CRT factors have readout vectors belonging to different σ-orbits, which are always linearly independent: the rank of the path's system of vectors equals the number of stations. The dimensions occupied by the path can no longer be used for generational readout, so the number of free lattice sites is

$$\#\{\text{lattice positions}\} \;=\; m - (\text{number of stations} - 1), \tag{58}$$

and the generational steps exceeding this budget are carried by the virtual-process channels of the links (see (b) below). Free lattice sites and link channels are counted by the same rank, and their total always equals m: each family has exactly m generations, independent of residence.

When the generational mediator sits at a station on the path, the same phase degree of freedom is simultaneously required to maintain propagation coherence and to be projectively read out, and the two are incompatible: the propagation breaks into independent decoherent segments, the L-factor power at that station doubles, and the endpoint projection factors degenerate to the per-segment background normalization $1/|\zeta(\frac{1}{2})|$; when the mediator lies outside the path (lattice projection, virtual process), the path remains coherent and the nodes are unchanged.

Generational transitions are accompanied by virtual-process corrections. Their type is uniquely determined by the CRT topology of character self-multiplication.

(a) Self-multiplication classification. The virtual process is driven by the character self-multiplication $\chi_k^2 = \chi_{2k \bmod (p-1)}$, and its classification is exactly the channel topology of §3.1: Type I channels ($\chi_k^2 \neq \chi_0$, i.e., $\mathrm{ord}(\chi_k) > 2$, self-multiplication returning to its own subgroup) and Type II channels ($\chi_k^2 = \chi_0$, i.e., $\mathrm{ord}(\chi_k) = 2$, $k = (p-1)/2$, self-multiplication falling into the trivial direction). By §3.1, for every odd prime p there exists exactly one Type II character.

(b) The four-case classification and the unified principle. A generational transition requires one Galois flip ($\chi \mapsto \bar{\chi}$, the winding-direction flip of §6.2), and the generational factor of each step factorizes into the product of two factors:

$$\text{generational factor} \;=\; \text{flip readout} \times \text{dwell weight}, \tag{59}$$

where the flip readout is the squared modulus of the projection of the flip output onto the readout direction of the target generation, and the dwell weight is the delivery metering of the virtual-process channel supplying the flip. Flip sources fall into two classes (phase projection and self-loop inclusion), dwell falls into two classes (without and with self-loop), and the product gives four mutually exclusive cases. The self-loop factor is the spectral evaluation of the transfer operator: when the self-loop forms a circuit, the sum over per-cycle transfers is the resolvent series, giving $1/\alpha_c$; when the self-loop lands in the trivial direction and no circuit can form, the evaluation is a single matrix element, giving α_o. The case in which the g-th generation finds itself is uniquely determined by its readout dimension:

- **Free lattice site**: the flip is supplied by the lattice phase projection, with no self-loop involved; factor ρ_g (Eq. (57));
- **Locked dimension + decoherent segment**: the mediator station has already broken into decoherent segments; the flip is supplied by the readout of the locked-link phase, factor λ^2 (corrected-level, Eq. (61));
- **Locked dimension + coherent + closed self-loop** ($\chi_s^2 = \bar{\chi}_s$): the self-loop output is precisely the Galois conjugate, so the flip is contained within the virtual process; the mass state (the symmetrized combination, §6.4.2) is an eigenstate of the self-loop insertion operator, the flip output is collinear with the target readout direction, the readout projection is the identity, factor 1; the per-cycle delivery probability is α_c, and the coherent series over the undelivered survival weight $(1-\alpha_c)$, $\sum_{n\geq 0}(1-\alpha_c)^n = 1/\alpha_c$ (the same resummation mechanism as the Dyson iteration of §7.4; here the delivery lands on the symmetric eigenvector, adding constructively in phase cycle by cycle, so the series has uniform sign), gives dwell weight $1/\alpha_c$ and generational factor $1/\alpha_c$;
- **Locked dimension + coherent + open self-loop** ($\chi_s^2 = \chi_0$): the self-loop falls into the trivial direction and carries no flip; the flip is still supplied by the locked-phase readout (λ^2, corrected-level, Eq. (61)); once the emitted quantum falls into χ_0 the process terminates, leaving no surviving branch to sum, so the dwell weight is the single-emission probability α_o and the generational factor is $\alpha_o\lambda^2$.

The factors $1/\alpha_c$ and α_o are the values of one and the same coupling constant on the two kinds of processes, with and without branching: the waiting series of delivery gives the reciprocal, and the single emission event gives the direct value. Evaluation level: dwell and emission are real processes in the post-transition vacuum, so the coupling takes its corrected-level observed value (the same rule as the counting level of §14.3); the inputs internal to the correction layer (δ and the Dyson ratio r) take Born-level values (§11–§12). This level assignment is uniquely determined by the temporal structure of the process (§6.4.4). The exhaustiveness of the four cases depends on the order of the link generator: $\chi_s^2 = \bar{\chi}_s \iff \mathrm{ord}(\chi_s) = 3$, $\chi_s^2 = \chi_0 \iff \mathrm{ord}(\chi_s) = 2$; each CRT factor $\mathbb{Z}/q_i^{b_i}\mathbb{Z}$ contributes single-factor generators of order exactly $q_i^{b_i}$, so the orders of all single-factor generators are exhausted by these two values if and only if every $q_i^{b_i} \in \{2, 3\}$, that is, $p-1 \mid 6$; in that case the four cases are complete. In every other case there exist generators whose self-multiplication is neither the conjugate nor trivial: this occurs when $p-1$ contains a square factor (for $p = 13$, the generators of order 4 in the $\mathbb{Z}/4\mathbb{Z}$ factor square to the generator of order 2, neither the conjugate nor trivial), and likewise when $p-1$ contains a prime factor $q_i \geq 5$; in either case the classification would need to be extended. Furthermore, the closed self-loop requires a generator of order 3 and the open self-loop a generator of order 2, so both loop types are available and the classification is complete if and only if $p-1=6$, that is, $p=7$: the four-case structure itself constitutes one more independent characteristic quantity for the channel identification of §9. Here the Born-level amplitude of λ is the two-component projection chain of the locked-link phase: leaving the original lattice site (cosine component) and entering the flip channel (sine component), in series,

$$\lambda^{(0)} \;=\; |\sin\delta^{(0)}\,\cos\delta^{(0)}| \;=\; \tfrac{1}{2}|\sin 2\delta^{(0)}|, \tag{60}$$

where $\delta^{(0)}$ is the bare value of the link phase: Decoherent segments accumulate no insertion phase, so the phase entering the projection chain contains no correction term (§12.1). Eq. (60) is the Born-level amplitude of the link readout. The readout is a real transmission in the post-transition vacuum, so its modulus takes the corrected-level value (the same clause as the evaluation level of α_c and α_o). The lowest-order virtual process of the correction is the two-vertex composite straddling the readout: the two vertices sit on the two sides of the readout, single-sided dwell being excluded by the instantaneous projection, so there is exactly one copy of the composite and no series (the flip generator is a real character: no phase reversal, no Dyson iteration). The generator multiplication of the even-family insertion against $\{\chi,\bar{\chi}\}$ (the entry generator and its conjugate, both even characters) gives two routes, whose intermediate generators are the products of the flip generator with the insertion generator, landing in the two double-coordinate residence directions. The counting consists of three items: the phase coefficient 1 of the dwell insertion contains the coherent sum of the two conjugate routes, each route carrying a share of $\frac{1}{2}$; the segment has decohered, so the two routes are instead added at probability level; the two ends of the readout carry no $\Delta^{1/2}$ vertex and lock no distinguishable label (the decoherent-segment exemption of §6.4.6), so the virtual intermediate state is summed over the d_2 distinguishable states of the released label direction; the generation label is fixed by the link endpoints and does not participate in the sum. The probability-level correction totals $2\times d_2\times\frac{1}{2}=d_2$ units of insertion coupling; at amplitude level

$$\lambda \;=\; \lambda^{(0)}\Big(1+\tfrac{d_2}{2}\,\alpha^{(0)}\Big), \tag{61}$$

where $\alpha^{(0)}$ is the Born-level coupling of the even-family self-loop (d_2 is the dimension of the corresponding CRT factor; for its value see §12.1). In the corresponding insertion on a dwell segment, this label is conserved along the propagation line (the insertion generator carries no such label), and the intermediate state is a single term–the frequency-shift correction (§12.1) carries no d_2 factor.

(c) Generational structure of the indirect channel. The direct channel in the real-character direction is closed (§6.4.2), and no coherent path is available for phase projection: the generational structure is instead given by the arithmetic data of the doublets. The weight of the a-th doublet is the product of the spectral weights of its two chains (§2.4):

$$w_a \;=\; \zeta_a(m)\,\zeta_{p-a}(m), \qquad \zeta_a(s) = \sum_{n\equiv a \pmod p} n^{-s}, \tag{62}$$

with the exponent taken to be the generation number $m=(p-1)/2$. The doublet product $V_a=a(p-a)\equiv -a^2 \pmod p$ satisfies $V_a^m\equiv(-1)^m \pmod p$ (by Fermat's little theorem $a^{p-1} \equiv 1$; by Euler's criterion this is precisely the quadratic-residue sign of V_a): when $p \equiv 3 \pmod 4$, m is odd and $V_a^m \equiv -1$ holds for all doublets– the involution -1 is a nonresidue, and the lift in which particle and antiparticle coincide is arithmetically realizable; channels with $p \equiv 1 \pmod 4$ admit no such lift. The weights (62), combined with the circulant matrix $\mathcal{C} = \mathrm{circ}(f_0, f_{m-1}, \ldots, f_1)$, give the mass matrix $\mathcal{M} = \mathcal{C}\,\mathrm{diag}(w_1, \ldots, w_m)\,\mathcal{C}^T$, whose eigenvalues give the generational factors of the indirect channel. Let the reference generation be g_0 (the lattice site with $\rho_{g_0} = 1$). The generational factor of the g-th generation elementary state f is the product of the four-case factors along the readout steps from g_0 to g:

$$\mathrm{Gen}_f^{(g)} = \rho_g \times \prod_{\substack{c:\,\text{closed} \\ \text{self-loop channels}}} \frac{1}{\alpha_c} \times \prod_{\substack{o:\,\text{open} \\ \text{self-loop channels}}} \alpha_o \cdot \lambda^2 , \tag{63}$$

where the applicable case and value of each factor are determined by the classification of (b); generations on free lattice sites carry only ρ_g, while generations borne by channels carry the corresponding self-loop factors. The lattice projection ρ_g encodes the weight distribution of the L-function phase over the discrete lattice, providing the Born-level structure of the generational hierarchy ($\rho_0 : \rho_1 : \rho_2 \sim 1 : \epsilon : \epsilon^2$). The specific values of the virtual-process factors $1/\alpha_c$ and $\alpha_o\lambda^2$ come from the KMS-fluctuation correction to the effective number of paths ($N_k \to N_k^{\mathrm{eff}}$), whose systematic derivation is given in §7.4; here we only use its results: in Type I directions ($\chi_k^2 \neq \chi_0$) the correction is $N_k^{\mathrm{eff}} = N_k + V_0 - 2|R_k|^2$ (V_0 is the connected value of the background fluctuation, R_k the resonance ratio of §4; §7.4); in Type II directions ($\chi_k^2 = \chi_0$) the correction is determined by cross-sector redistribution; a closed cycle triggers an alternating geometric series (Dyson iteration), whose exact sum is $\sum_{n\geq 0}(-r)^n = 1/(1+r)$, $r \ll 1$. These factors, combined with the lattice projection, determine the complete numerical values of the generational hierarchy.

6.4.8 The unified mass formula

Combining the structures established in the preceding sections– the CRT transition path (§6.4.3), the L factors and powers at each station (§6.4.5), the node normalization (§6.4.6), and the generational factors (§6.4.7)– the mass of a stable state f is:

Let v be the energy-scale parameter and f a g-th generation elementary state whose shortest transition path from χ_0 passes through the stations $\{s_1, \ldots, s_n\}$. Then

$$\boxed{m_f = v \times \prod_{k=1}^{n} \left|L(\tfrac{1}{2}, \chi_{s_k})\right|^{G_k \cdot C_k} \times \prod_{k=0}^{n} \mathcal{T}_{s_k} \times \mathrm{Gen}_f^{(g)}} \tag{64}$$

where:

- G_k, C_k are the Galois degree and covering number of the k-th station (§6.4.5);
- $\mathcal{T}_{s_k}$ is the node-normalization factor (§6.4.6); for the starting point $s_0 = \chi_0$ we adopt the convention $\mathcal{T}_{s_0} = 1$ (the starting point of a path carries no factor), while a χ_0 passed as a relay counts as $\mathcal{T}_0 = 1/N$;
- $\mathrm{Gen}_f^{(g)}$ is the generational factor of the g-th generation (§6.4.7).

For indirect-channel particles (directions with $\mu_0 = 0$), the transition path must detour through extra intermediate stations; the path is longer and passes through the χ_0 dilution $\mathcal{T}_0 = 1/N$, so the mass is suppressed by $\mathcal{O}(1/N)$. Combined with the special values of the L-function at the real character, the indirect-channel elementary-state mass $m_\nu \sim v/N \cdot |L|^{\text{power}}$ lands at the sub-eV scale. Formula (64) is precisely the expansion of the mass definition $m = v \cdot \varphi_\beta(P_f)$: the complete projection P_f is refined step by step according to Galois, doublet, and CRT lattice, and the KMS evaluation $\varphi_\beta(P_f)$ factorizes along the transition path into the product of the L-values at each station (with powers $G \cdot C$), the node normalization $\mathcal{T}$, and the generational factor Gen; the compact form is $m_f = v \cdot |L|^{\text{pow}} \cdot \text{node} \cdot \text{Gen}$.

§6.5 Emergent-State Mass

Completely parallel to §6.4: the mass is defined as the evaluation of the KMS state on the complete projection of the emergent state.

The emergent state resides in the χ_0 direction and involves all p^K elements of the full ring. Its projection operator is the full-ring projection in the χ_0 direction:

$$P_{\text{em}} = P_{\chi_0}^{\text{full}},$$

where the superscript "full" emphasizes that this is the χ_0-direction projection on the full ring $\mathbb{Z}/p^K\mathbb{Z}$– not restricted to the unit group $(\mathbb{Z}/p^K\mathbb{Z})^*$.

Compare with the elementary-state projection $P_f = P_{\chi_k}^{(\sigma)} \cdot P_a^{(\iota)} \cdot P_j^{(3)}$ (on the unit group, refined step by step according to Galois, doublet, and CRT lattice): the emergent-state projection has no doublet distinction (the χ_0 direction has only one generation) and no CRT fine structure (the mode vector is zero): P_{em} projects only onto the full-ring subspace of zero frequency (χ_0).

The mass of the emergent state is given by the spectral weight of the collective projection. The spectral weight is defined as follows.

The (unnormalized) KMS density matrix is $\rho = e^{-\beta H}$, i.e., $\rho|n\rangle = n^{-\beta}|n\rangle$. For any character χ_k, define its spectral weight as the unnormalized sum of ρ in the χ_k direction:

$$N_{\chi_k} = \sum_{n=1}^{p^K} \chi_k(n) \cdot n^{-\beta}, \tag{65}$$

where the summation ranges over the p^K lattice sites of the finite system; everything below is read in the thermodynamic limit $K \to \infty$. This is the effective number of participating modes in the direction of the character χ_k– the total thermodynamic weight that ρ allocates to that direction.

For a nontrivial character ($k \neq 0$), χ_k vanishes on multiples of p, so the sum is restricted to the unit group:

$$N_{\chi_k} = \sum_{\substack{n \le p^K \\ \gcd(n,p)=1}} \chi_k(n)\, n^{-\beta} \xrightarrow{K\to\infty} L(\beta, \chi_k) \quad \text{(finite)},$$

namely the Dirichlet L-function.

For the trivial character $\chi_0 \equiv 1$ (the full-ring direction), χ_0 applies no filtering to the full ring, and the spectral weight is the partition function itself:

$$N_{\chi_0} = \sum_{n=1}^{p^K} n^{-\beta} = Z(\beta) \xrightarrow{K\to\infty} \zeta(\beta).$$

This is the precise meaning of the statement that the collective projection yields $\zeta(\beta)$: the χ_0 direction filters nothing, so the spectral weight equals the sum of the contributions of all lattice sites, i.e., the partition function.

The spectral weight N_{χ_k} is precisely the thermal-state prescription for the participation spectral weight W of §5.3: the participating modes are restricted to the χ_k direction, and each mode is counted with Boltzmann weight $n^{-\beta}$ (the second prescription $W = Z(\beta)$ written explicitly in §5.3). Taking $W = N_{\chi_0} = Z(\beta)$, the energy-scale formula (24) gives the mass of the emergent state:

$$\boxed{m_{\text{em}} = \sqrt{\frac{Z(\beta) \cdot \log^2 Z(\beta)}{C^*}},} \tag{66}$$

where C^* is the intrinsic metric constant of §3.4 and β is the energy-scale parameter at which the emergent state resides.

The derivation chain:

$$\text{collective projection} \xrightarrow{(65)} N_{\chi_0} = Z(\beta) \xrightarrow{\S5.3} m_{\text{em}} = \sqrt{Z \log^2 Z / C^*}.$$

Comparison with the elementary states: the KMS evaluation of an elementary state is finite, and the mass takes the linear readout $m = v \cdot \varphi_\beta(P_f)$; the KMS evaluation of the emergent state gives a divergent number of participating modes $Z(\beta)$, and the mass is obtained by the energy-scale formula converting the participation number into energy. Both kinds of mass are determined by KMS evaluation; the only difference is whether the evaluation result is finite (linear readout) or divergent (energy-scale readout).

The complete comparison of the two kinds of matter:

	Elementary states	**Emergent state**
Definition	nontrivial eigenmodes of K	collective condensation driven by the ζ pole
Number	$p-2$	1
Stability mechanism	spectral simplicity	the pole of $\zeta(\beta)$ at $\beta = 1$
Location	$p-2$ lattice sites on the CRT torus	the origin of the torus
Formation mechanism	D_F propagating stepwise along the CRT torus	one-shot global condensation at the ζ pole
Spectral weight	$N_{\chi_k} = \sum_n \chi_k(n) n^{-\beta} = L(\beta, \chi_k)$	$N_{\chi_0} = \sum_n n^{-\beta} = \zeta(\beta)$
Spectral-weight character	finite	pole (divergent as $\beta \to 1$)
Mass	$m_f = v \cdot \lvert L\rvert^{\text{pow}} \cdot \text{node} \cdot \text{Gen}$	$m_{\text{em}} = \sqrt{\zeta(\beta) \log^2 \zeta / C^*}$

§7 Resonance Forces

§6 identified matter in the state space – the stable states, divided into elementary states and emergent states. This chapter examines the interactions between stable states carried by the convolution operator K, called the **resonance force**. The underlying mechanism is uniformly frequency resonance; the distinction lies only in the statistics: between elementary states, equivalent paths are averaged over and the force is diluted, called the **elementary resonance force**; when emergent states participate, independent channels are summed over and the force is enhanced, called the **emergent resonance force**. The dichotomy of matter induces the dichotomy of resonance forces: the classification of forces is not a new input but a corollary of the classification of stable states. Beyond K, the metric structure of the Hamiltonian H gives another, universal force (the metric force, §8); this chapter is confined to the resonance forces carried by K. §7.1 gives the definition of the force and its selection rule, §7.2–7.3 derive the species of forces and their strength hierarchy, §7.4–7.5 compute the coupling constants of the two resonance forces, and §7.6 gives the system of dynamical equations.

§7.1 Definition of the Resonance Force

The diagonalization theorem of §3.2 states: the translation-invariant linear operator on the state space is the convolution operator K, diagonal in the character basis $\{\chi_k\}_{k=0}^{p-2}$, with eigenvalues given by Dirichlet L-function values:

$$K\chi_k = \lambda_k\,\chi_k, \qquad \lambda_k = L(\beta, \chi_k), \qquad k = 0, 1, \ldots, p-2. \tag{67}$$

K is the operator of the resonance force: $|\lambda_k|$ is the strength of the force in direction k, and $\arg\lambda_k$ is the phase deflection introduced by the force. The corresponding spectral projections $\{P_k\}_{k=0}^{p-2}$ decompose the unit-group part of the state space into $p-1$ orthogonal directions (§4).

On this basis we define resonance.

Resonance: two modes ψ_1, ψ_2 resonate if and only if there exists a spectral projection P_k with $P_k\psi_1 \neq 0$ and $P_k\psi_2 \neq 0$ – that is, they both have nonzero components in the same eigendirection of K. Resonating modes share the same eigenspace, oscillate at the same frequency under time evolution, and thus couple automatically.

Resonance strength: $\varphi_\beta(P_k\,K^\dagger K\,P_k)$ – the energy weight that the KMS state φ_β (§4.4) assigns to the shared direction. Large weight means strong resonance; zero weight means no force.

Selection rule: the orthogonality of characters $\langle\chi_k, \chi_{k'}\rangle = \delta_{kk'}$ gives

$$\langle\chi_{k'},\, K\chi_k\rangle = \lambda_k\,\langle\chi_{k'}, \chi_k\rangle = \lambda_k\,\delta_{kk'}, \tag{68}$$

that is, the action of K produces no $k \to k'$ transitions ($k \neq k'$): an excitation in direction k remains in direction k after propagation, with no component entering other directions. Whether two elementary states can communicate through the force K depends entirely on whether their frequency labels coincide – the selection rule is a direct corollary of the diagonal structure (67).

Two direct corollaries of the selection rule:

Conservation law: under character multiplication $\chi_i \cdot \chi_j = \chi_{i+j}$, the character label is conserved through the interaction, $\chi_{\text{in}} = \chi_{\text{out}}$ – the frequency label is the conserved quantum number of the interaction.

Time evolution: the time flow $\sigma_t = e^{itH}$ of §3.4 acting on direction k gives $A_k(t) = A_k(0)e^{i\omega_k t}$ – the resonance amplitude oscillates at the eigenfrequency; the propagation of the force does not disturb the frequency decomposition (with time-resolution lower bound $\Delta t_{\min} = 1/\log 2$, §3.4).

This definition applies uniformly to both resonance forces: the elementary resonance force operates in the nontrivial character directions ($k \neq 0$), with rich CRT subdivision and selection rules (§7.2–7.4); the emergent resonance force operates in the χ_0 direction, with no selection rule (frequencies match automatically) but with pole amplification (§7.5).

§7.2 Species of Resonance Forces

Under the CRT decomposition $k \leftrightarrow (k_1, k_2, \ldots, k_s)$ of §6.2 ($s = \omega(p-1)$), the character group decomposes into a direct product, and character multiplication proceeds coordinatewise:

$$\chi_k = \prod_{i=1}^{s} \chi^{(i)}_{k_i}, \qquad \chi_k \chi_{k'} \;\longleftrightarrow\; (k_1 + k'_1, \ldots, k_s + k'_s). \tag{69}$$

The selection rule for resonance (§7.1) therefore separates coordinatewise: each coordinate k_i is an independently conserved quantum number; the resonance channel labels of the i-th direction are nonzero only in the i-th coordinate and do not touch the others – resonances in the i-th and j-th directions are labeled by distinct conserved charges, constituting physically independent coupling mechanisms, and cross-factor mixing is forbidden by the selection rule. The amplitude itself does not factorize ($L(\beta, \chi_k)$ depends on the full k): the independence comes from the coordinatewise conservation of quantum numbers, not from a product decomposition of L-values.

Besides the resonances of the s individual CRT factors, there also exists a global resonance across all factors. The **factor-preserving unitary group** of the CRT tensor product $\mathbb{C}^{q_1^{b_1}} \otimes \cdots \otimes \mathbb{C}^{q_s^{b_s}}$ (the unitary transformations preserving each tensor factor) is the image of the tensor-product action

$$\mathcal{G} = \{A_1 \otimes \cdots \otimes A_s :\ A_i \in U(q_i^{b_i})\} \;\subset\; U(\mathbb{C}^{p-1}). \tag{70}$$

The surjection $U(1) \times \prod_i SU(q_i^{b_i}) \to \mathcal{G}$, $(u, A_1, \ldots, A_s) \mapsto u\, A_1 \otimes \cdots \otimes A_s$, has kernel characterized by $A_i = \alpha_i I$, $\alpha_i^{q_i^{b_i}} = 1$, and $u \prod_i \alpha_i = 1$, so the kernel is $\cong \prod_i \mathbb{Z}/q_i^{b_i}\mathbb{Z} \cong \mathbb{Z}/(p-1)\mathbb{Z}$ (CRT),

$$\boxed{\mathcal{G} \;\cong\; \big(U(1) \times SU(q_1^{b_1}) \times \cdots \times SU(q_s^{b_s})\big)/\mathbb{Z}_{p-1}.} \tag{71}$$

The central quotient $\mathbb{Z}_{p-1}$ is a direct imprint of the CRT decomposition. At the Lie-algebra level $\mathcal{G}$ decomposes into $s+1$ independent factors: s SU factors (one per CRT direction) plus 1 central $U(1)$. Each factor corresponds to an independent force:

- resonance purely in the i-th direction ($k_i \neq 0$, all other $k_j = 0$) $\longleftrightarrow$ the $SU(q_i^{b_i})$ factor;

- resonance across all directions (all $k_i \neq 0$) $\longleftrightarrow$ the central $U(1)$ factor.

The total number of force species is

$$\boxed{\text{force species} = \omega(p-1)+1.} \tag{72}$$

§7.3 Strength Hierarchy

The character order $d = \operatorname{ord}(\chi_k) = (p-1)/\gcd(k, p-1)$ of §6.2 determines the strength of the force: $|\lambda_k| = |L(\beta,\chi_k)|$ decreases monotonically as the order d increases – the force is strong in low-order character directions and weak in high-order character directions.

The key tool is the Euler product representation of the L-function:

$$L(\beta,\chi_k) = \prod_{\ell \text{ prime}} \frac{1}{1-\chi_k(\ell)\,\ell^{-\beta}}, \qquad |L(\beta,\chi_k)| = \prod_{\ell} \frac{1}{|1-\chi_k(\ell)\,\ell^{-\beta}|}. \tag{73}$$

Each prime ℓ contributes one local factor. Writing $\chi_k(\ell) = e^{2\pi i\theta_\ell}$,

$$|1-\chi_k(\ell)\ell^{-\beta}|^2 = 1 - 2\ell^{-\beta}\cos(2\pi\theta_\ell) + \ell^{-2\beta}. \tag{74}$$

For $d = 1$ (χ_0): $\theta_\ell = 0$ for all ℓ, and each local factor $(1-\ell^{-\beta})^{-1}$ attains its maximum. Since $|1 - z\,\ell^{-\beta}| \geq 1 - \ell^{-\beta}$ (for any $|z| = 1$), $|L(\beta,\chi_k)| \leq |L(\beta,\chi_0)|$ holds for all k, with strict inequality for $k \neq 0$ (there is always a prime class with $\chi_k(\ell) \neq 1$): the first link of the hierarchy is a theorem.

For $d > 1$: the image of χ_k is exactly the complete set μ_d of d-th roots of unity. The larger d is, the more dispersed $\chi_k(\ell)$ is on the unit circle, the more uniformly $\cos(2\pi\theta_\ell)$ ranges over $[-1, 1]$, and the more the positive contributions ($\cos > 0$) and negative contributions ($\cos < 0$) in the Euler product tend to cancel, reducing the total product $|L(\beta,\chi_k)|$. This is **destructive interference**: the larger d, the more dispersed the phases of the different prime channels, the more the contributions cancel in the product, and the weaker the net force; for $d = 1$ there is no cancellation and the force is strongest; for $d = p-1$ the phases are uniformly distributed over the $(p-1)$-th roots of unity, cancellation is maximal, and the force is weakest.

The strengths of the forces therefore form a hierarchy:

$$\boxed{|L(\beta,\chi_0)| > |L(\beta,\chi_{k_1})| > |L(\beta,\chi_{k_2})| > \cdots, \quad d_1 < d_2 < \cdots.} \tag{75}$$

The first link of the chain (χ_0 strongest) holds rigorously (the inequality above); the remaining ordering by order comes from the leading analysis of destructive interference, holds at order jumps for general $\beta > 1$, and has been verified numerically; the ordering among characters of equal order is determined by the specific p and β.

The strength of the force corresponds to the nodal structure of the stable states (§6.2): low-order modes have large kernels, more group elements participate in the propagation, and the net force is strong; high-order modes have small kernels, fewer group elements participate in the propagation, and the net force is weak.

§7.4 Coupling Constants of the Elementary Resonance Force

§7.2–7.3 established the qualitative properties of the forces. This section computes the coupling constant of the elementary resonance force in each direction. The computation parallels the mass computation of §6.4: construct the response operator, take the KMS expectation value, normalize.

7.4.1 Response Operator and Bare Coupling

To measure the response of the system after the force acts in direction k, define the **response operator**

$$\Pi_k \;=\; P_k\, K^\dagger K\, P_k, \tag{76}$$

where $P_k = |\chi_k\rangle\langle\chi_k|$ and $K^\dagger K$ measures the squared strength of the force's action. By the diagonal structure (67),

$$K^\dagger K\, \chi_k = |\lambda_k|^2\, \chi_k, \tag{77}$$

Π_k is a rank-one operator whose unique nonzero eigenvalue is $|L(\beta,\chi_k)|^2$ – the bare response strength of the force in direction k, prior to thermal-equilibrium processing.

Besides the bare eigenvalue, the KMS expectation value also involves averaging over the degeneracy of **equivalent paths** in direction k. The degeneracy has three sources. First, the kernel of the character: by the order formula of §6.2, $\ker\chi_k \cap (\mathbb{Z}/p\mathbb{Z})^*$ contains $|\ker\chi_k| = (p-1)/\mathrm{ord}(\chi_k) = \gcd(k, p-1)$ unit-group elements satisfying $\chi_k(n) = 1$ – they are completely indistinguishable in direction k, contribute the same phase, and constitute equivalent propagation paths to the same resonance direction. Second, the involution cosets: the involution ι partitions the unit group into $m = (p-1)/2$ doublets (§6.2), and realizations of the same path through any coset are equivalent in direction k. Third, self-conjugate doubling: for real characters ($\chi_k = \bar\chi_k$) the forward and backward propagations are indistinguishable, contributing the factor $c_k = 1 + [\chi_k = \bar\chi_k]$. The total number of equivalent paths is

$$N_k \;=\; |\ker\chi_k| \cdot m \cdot c_k \;=\; \frac{p-1}{\mathrm{ord}(\chi_k)} \cdot \frac{p-1}{2} \cdot c_k\,. \tag{78}$$

Choice of the direction k: the resonance channel is labeled by the coordinate difference (§7.2), whereas the response operator acts on the matter modes participating in the resonance – k is taken to be the residence coordinate of the matter family participating in the force, as distinguished from the channel's coordinate difference (values given in §11.2).

The KMS state is the maximally symmetric equilibrium state; it assigns no preference among indistinguishable choices, and therefore averages with equal weight over these N_k equivalent paths:

$$\varphi_\beta(\Pi_k) \;=\; \frac{|L(\beta,\chi_k)|^2}{N_k}. \tag{79}$$

The total response strength is diluted by the N_k equivalent paths – the same logic by which mass is diluted by degeneracy in §6.4.

The normalization baseline is taken to be $\zeta(\beta)^2$ – the strict upper-bound scale of the bare response: by the local-factor inequality $|1-\chi_k(\ell)\ell^{-\beta}| \geq 1-\ell^{-\beta}$ of §7.3, $|L(\beta,\chi_k)| \leq \zeta(\beta)$ holds for all k, so $\zeta(\beta)^2$ is the maximal scale of the bare response $|L|^2$; this baseline is consistent with the ζ normalization used in defining R_k in §4. Using the resonance ratio $R_k = L(\beta,\chi_k)/\zeta(\beta)$ introduced in §4, the **bare coupling constant** (tree level, virtual processes not yet included) is

$$\alpha_k^{(0)} \;=\; \frac{\varphi_\beta(\Pi_k)}{\zeta(\beta)^2} \;=\; \frac{|R_k|^2}{N_k}. \tag{80}$$

The numerator $|R_k|^2$ measures the squared relative resonance strength of direction k, and the denominator N_k is the number of equivalent paths: the stronger the resonance the stronger the force, the larger the degeneracy the weaker the force.

For the χ_0 direction, the resonance ratio can be computed explicitly. The sum defining $L(\beta,\chi_0)$ is restricted to $p \nmid n$ (the diagonalization theorem of §3.2), hence

$$L(\beta,\chi_0) = \sum_{p\nmid n} n^{-\beta} = \zeta(\beta)\,(1-p^{-\beta}), \qquad R_0(\beta) = 1-p^{-\beta}. \tag{81}$$

The deficit term $p^{-\beta}$ is the contribution of the p-local modes removed by the unit-group restriction ($p \nmid n$) – it measures the common deficit of the χ_0 direction in the KMS fluctuations, and will reappear in the virtual-process corrections.

7.4.2 Virtual-Process Corrections

The bare coupling (80) assumes all N_k paths are open. The virtual-process fluctuations of the KMS state correct the effective path count from N_k to N_k^{eff}, and the coupling constant becomes

$$\boxed{\alpha_k = \frac{|R_k|^2}{N_k^{\text{eff}}}, \qquad R_k = \frac{L(\beta,\chi_k)}{\zeta(\beta)}.} \tag{82}$$

The correction type is uniquely determined by the destination of the character self-loop χ_k^2:

- **Type I** ($\chi_k^2 \neq \chi_0$): the self-loop stays in a nontrivial sector; the virtual processes involve only the independent corrections from KMS fluctuations and self-energy.
- **Type II** ($\chi_k^2 = \chi_0$): the self-loop directly excites the χ_0 direction, from which recoupling to all sectors is possible; cross-sector redistribution must be tracked.

Type I correction formula ($\chi_k^2 \neq \chi_0$): two independent contributions superpose – background fluctuation $+V_0$, Born-propagation consumption $-2|R_k|^2$:

$$N_k^{\text{eff}} = N_k + V_0 - 2|R_k|^2, \tag{83}$$

where V_0 is the connected value of the background fluctuation: the bare value R_0 (Eq. (81)) contains a disconnected part; after subtraction, $V_0 = |R_{(p-1)/2}|^2$ – the self-loop probability of the unique Type II direction (numerical values in §11.3).

Type II correction formula ($\chi_k^2 = \chi_0$): after the self-loop reaches the χ_0 direction, each sector absorbs or returns propagation probability:

$$N_k^{\text{eff}} = N_k + \sum_{k'} \epsilon_{k'}\,|R_{k'}|^2 + c_k|R_k|^2, \tag{84}$$

where the sum runs over the sectors participating in the redistribution, the coefficients $\epsilon_{k'}$ (taking values ±1 or $\pm d_i$, with d_i the dimension of the corresponding CRT factor) are determined by each sector's absorption or return of propagation probability in the χ_0 direction, and c_k is the self-conjugacy factor of (78).

Dyson iteration (higher-order corrections): when $\chi_k^2 = \bar{\chi}_k$ (phase reversal), the self-loop couples again within the corrected propagation environment, producing an alternating geometric series. Let the first-order offset ratio be r (r taken at Born level; for the evaluation level see §6.4, §11); the Dyson iteration acts only on the virtual-process offset $\Delta N = N_k^{\text{eff}} - N_k$:

$$N_k^{\text{full}} = N_k + \frac{N_k^{\text{eff}} - N_k}{1+r}, \tag{85}$$

with final coupling constant $\alpha_k^{\text{full}} = |R_k|^2/N_k^{\text{full}}$.

§7.5 Coupling Constants of the Emergent Resonance Force

7.5.1 Coupling between Emergent States

The emergent resonance force has the same underlying mechanism as §7.1 (frequency matching by K), but operates in the χ_0 direction: all emergent states reside in the χ_0 direction, their frequencies match automatically (all zero frequency), and the resonance condition is natively satisfied.

Two key differences from resonance between elementary states:

- **Pole amplification**: propagation in the χ_0 direction is given by $\zeta(\beta)$, which has a pole at $\beta = 1$, far exceeding the propagators $L(\beta, \chi_k)$ ($k \neq 0$, finite) of the elementary directions.
- **Inclusive summation**: emergent states are blind to the character labels of the unit group (§6.2); the internal operations visible to them constitute the affine group of the residue field, $\text{Aff}(\mathbb{F}_p)$ ($|\text{Aff}(\mathbb{F}_p)| = p(p-1)$), providing $|\text{Aff}|_{\text{eff}}$ independent coupling channels (effective value in §11.5); the total cross-section is the superposition (sum) of all channels, not their average.

Contrast of statistics: in §7.4 the N_k equivalent paths of the elementary resonance force are redundancy (indistinguishable equivalent descriptions); the KMS state averages over redundancy, and the result is dilution ($1/N_k$ suppression); the independent channels between emergent states are distinguishable physical processes; the total effect is a sum, and the result is enhancement (by a factor $|\text{Aff}|_{\text{eff}}$). Redundancy $\Rightarrow$ averaging $\Rightarrow$ dilution; independent channels $\Rightarrow$ summation $\Rightarrow$ enhancement.

Bare amplitude ratio: emergent states reside on the full group; the condensation involves all lattice sites without distinguishing $\gcd(n, p)$ (§6.2); its propagator is the complete $\zeta(\beta)$, carrying no p-local deficit, and its bare amplitude ratio relative to the normalization baseline $\zeta(\beta)^2$ is 1. Multiplying by the channel count gives the coupling constant between emergent states:

$$\alpha_{\text{emg}} = 1 \cdot |\text{Aff}|_{\text{eff}}. \tag{86}$$

The coupling between emergent states has no independent experimental counterpart, so its numerical value is not listed separately; what has an experimental counterpart is the joint coupling of §11.5.

7.5.2 Joint Coupling between Emergent and Elementary States

The coupling between an emergent state (the χ_0 direction) and an elementary state (a χ_{k_1} direction, $k_1 \neq 0$) is composed of two factors: the elementary direction supplies the resonance amplitude – the squared resonance ratio $|R_{k_1}|^2$; the emergent direction supplies the channel amplification – the sum over $|\mathrm{Aff}|_{\mathrm{eff}}$ independent channels. The joint coupling constant is

$$\boxed{\alpha_{\mathrm{mix}}(k_1) = |\mathrm{Aff}|_{\mathrm{eff}} \cdot |R_{k_1}(\beta)|^2.} \tag{87}$$

α_{emg} does not run with the energy scale (the ζ pole structure and $|\mathrm{Aff}|_{\mathrm{eff}}$ are both algebraic constants), whereas α_{mix} runs with the energy scale through $|R_{k_1}(\beta)|^2$ – its evolution equation is given in §7.6.

§7.6 System of Dynamical Equations

§7.1–7.5 established the static properties of the resonance forces at fixed energy scale β. This section first gives the response equation for the action of the resonance force on matter, then the system of equations governing the variation of resonance strength and coupling constants with the frequency direction k and the energy scale β.

7.6.1 Matter Response Equation

The source term $\mathcal{J}_k$, the Green kernel, and the self-coupling (below) give one side of the resonance amplitude A_k; the reverse direction – how the resonance force changes the stable states participating in the resonance – is given directly by the diagonal structure of K. $A_k = R_k$ is fixed by the L-function; it is a measure of inter-mode coupling strength, not a field carrying degrees of freedom. For an arbitrary excitation ψ, its direction-k component after a single propagation becomes

$$P_k(K\psi) = \lambda_k\, P_k\psi = |\lambda_k|\, e^{i \arg \lambda_k}\, P_k\psi\ : \tag{88}$$

the entire action of the resonance force on matter is amplitude scaling ($|\lambda_k|$) and phase deflection ($\arg \lambda_k$) within direction k; the direction label k itself is unchanged – the force does not alter the identity of the stable state, which is another reading of the selection rule (68). Contrast with §8: the metric force shapes the equilibrium distribution of excitations over lattice sites (w_β, §8.5), while the resonance force shapes the amplitude and phase of excitations within a direction – the response equations of the two forces act respectively on positional and directional degrees of freedom.

7.6.2 Resonance Constraint Equation

By the Euler-product analysis of §7.3, the dependence of $\log|\lambda_k|$ on the direction k has leading term given by the density $1/d_k$ of aligned primes ($d_k = \mathrm{ord}(\chi_k)$): the prime classes satisfying $\chi_k(\ell) = 1$ have proportion $1/d_k$; their contributions add coherently, while the contributions of the remaining prime classes cancel. Define the **resonance ladder** $S(k) = \lambda_{k+1}/\lambda_k$, the **resonance deviation** $\Delta(k) = (\lambda_{k+2}\lambda_k - \lambda_{k+1}^2)/(\lambda_{k+1}\lambda_k) = S(k{+}1) - S(k)$, and the **order-jump source** $J(k) = 1/d_{k+1} - 1/d_k$. From the leading approximation $S(k) \approx 1 + \Phi(\beta)\, J(k)$,

$$\boxed{\Delta(k) = \Phi(\beta)\, \big[J(k+1) - J(k)\big],} \tag{89}$$

where the proportionality function $\Phi(\beta)$ is independent of k, given by the prime sum of the Euler product (the leading linear coefficient of $\log|\lambda_k|$ in $1/d_k$). The resonance deviation is the readout in the resonances of the second difference of $1/d_k$, uniquely determined by the variation of the order jumps: where the order is constant, J vanishes and the resonances vary geometrically (flat); at order jumps, J changes abruptly and the resonances bend.

7.6.3 Green-Kernel Evolution Equation

Taking the logarithm of the Euler product (73) and differentiating with respect to β, written as a Dirichlet series in the von Mangoldt function $\Lambda(n)$[26]:

$$\frac{\partial}{\partial\beta}\log L(\beta,\chi_k) = -\sum_{n=2}^{\infty}\frac{\Lambda(n)\,\chi_k(n)}{n^{\beta}}.$$

Subtracting the same identity for $\zeta(\beta)$ (the denominator of R_k; note that the L of the $k=0$ direction is $\zeta(\beta)(1-p^{-\beta})$, not ζ) gives the evolution of the resonance ratio R_k:

$$\boxed{\frac{\partial}{\partial\beta}\log R_k(\beta) = -\sum_{n=2}^{\infty}\frac{\Lambda(n)\,[\chi_k(n)-1]}{n^{\beta}}.} \tag{90}$$

The source term $\chi_k(n)-1$ measures the deviation of direction k from the χ_0 direction. As $\beta\to\infty$, $R_k\to 1$ (uniformization); as $\beta\to 1^+$, $R_k\to 0$ (asymptotic freedom[40, 41]).

7.6.4 Nonlinear Resonance Equation

Character multiplication $\chi_i\cdot\chi_j=\chi_{i+j}$ produces frequency mixing: the amplitude product of directions i and j injects a source into direction $k=i+j$. Define the structure constants $f_{ijk}=\delta_{i+j,k}$: the frequency mixing $\chi_i\chi_j=\chi_{i+j}$ is symmetric in (i,j), and so are the structure constants; the CRT direct product guarantees no cross-factor mixing (f_{ijk} is nonzero only within a single CRT factor). The diagonal terms $i=j$ are admissible: A_i^2 injects into direction $2i$, which is exactly the equation counterpart of the χ_i^2 self-loop in the virtual-process classification of §7.4. With the resonance amplitude $A_k=R_k$ as variable, the complete resonance equation is

$$\boxed{\frac{\partial A_k}{\partial\beta}+\nu_k(\beta)\,A_k+\sqrt{\alpha_k}\sum_{i+j\equiv k}f_{ijk}\,A_iA_j=\mathcal{J}_k(\beta),} \tag{91}$$

where the linear damping coefficient

$$\nu_k(\beta)=\sum_{n=2}^{\infty}\frac{\Lambda(n)\,[\chi_k(n)-1]}{n^{\beta}}$$

is such that the linear part on its own recovers exactly the Green-kernel evolution (90); $\sqrt{\alpha_k}\sum f_{ijk}A_iA_j$ is the nonlinear self-coupling (with coupling strength given by the α_k of §7.4); and $\mathcal{J}_k(\beta)$ is the matter source term – the excitation injection of elementary states in direction k.

7.6.5 Evolution Equation of the Emergent-State Coupling

The effective number of participating modes of the emergent state, $N_{\chi_0} = \zeta(\beta)$ (§6.4), varies with the energy scale according to the $k = 0$ limit logic of (90) ($\chi_0(n) \equiv 1$, the source term vanishes; the deficit factor of R_0 does not participate; what runs is $\zeta(\beta)$ itself):

$$\frac{\partial}{\partial\beta} \log\zeta(\beta) = -\sum_{n=2}^{\infty} \frac{\Lambda(n)}{n^\beta}. \tag{92}$$

As $\beta \to 1^+$, $\zeta(\beta) \to \infty$ (the pole) and the effective participation number of emergent states diverges – emergence occurs; as $\beta \to \infty$, $\zeta(\beta) \to 1$ – in the low-temperature limit the emergent effects weaken.

In the joint coupling $\alpha_{\text{mix}}(k_1) = |\text{Aff}|_{\text{eff}} \cdot |R_{k_1}|^2$, $|\text{Aff}|_{\text{eff}}$ is fixed by the affine group of the residue field together with the Loewy correction (§11.5, independent of β), so the β dependence comes entirely from the running of $|R_{k_1}|^2$ ((90) with $k = k_1$):

$$\boxed{\frac{\partial}{\partial\beta}\alpha_{\text{mix}} = |\text{Aff}|_{\text{eff}} \cdot \frac{\partial |R_{k_1}|^2}{\partial\beta} = |\text{Aff}|_{\text{eff}} \cdot 2\,\text{Re}\left[\bar{R}_{k_1} \frac{\partial R_{k_1}}{\partial\beta}\right].} \tag{93}$$

Structure:

- the running is driven entirely by the Green-kernel evolution of the χ_{k_1} direction; the higher the order of χ_{k_1}, the more dispersed the values of $\chi_{k_1}(n) - 1$ over the primes, and the stronger the driving;
- $|\text{Aff}|_{\text{eff}}$ is a constant amplification factor, making the absolute running rate of the joint coupling far exceed that of a single elementary resonance force;
- as $\beta \to 1^+$, $R_{k_1} \to 0$ (asymptotic freedom), and the joint coupling shuts off near the phase-transition point.

7.6.6 Summary of the Equations

Equation	Content	Physical meaning	Source
Sector classification	$K^{(i)} A_k^{(i)} = L_i \cdot A_k^{(i)}$	CRT factors independent	§7.2
Mixing forbidden	$A_{\text{mix}} = 0$	cross-factor resonance vanishes	§7.2
Conservation law	$\chi_{\text{in}} = \chi_{\text{out}}$	quantum-number conservation	§7.1
KMS time	$A_k(t) = A_k(0)e^{i\omega_k t}$	eigenfrequency oscillation	§7.1
Matter response	$P_k K\psi = \lvert\lambda_k\rvert e^{i \arg\lambda_k} P_k\psi$	amplitude scaling + phase deflection, identity unchanged	§7.6
Resonance constraint	$\Delta(k) = \Phi(\beta)J(k)$	frequency-space distribution constraint	§7.6
Green-kernel evolution	$\partial_\beta \log R_k = -\sum \frac{\Lambda(n)[\chi_k(n)-1]}{n^\beta}$	energy-scale running	§7.6
Nonlinear resonance	$\partial_\beta A_k + \nu_k A_k + \sqrt{\alpha_k}\sum f_{ijk} A_i A_j = \mathcal{J}_k$	self-coupling	§7.6
Joint-coupling evolution	$\partial_\beta \alpha_{\text{mix}} = \lvert\text{Aff}\rvert_{\text{eff}} \cdot \partial_\beta \lvert R_{k_1}\rvert^2$	emergent–elementary coupling running	§7.6

Here $A_k^{(i)}$ is the resonance amplitude within the i-th CRT factor, $K^{(i)}$ is the restriction of K to that factor, and L_i is the L-function value of the corresponding factor direction. The first five rows are the equation forms of the static properties; the last four are the evolution equations derived in this section. All evolution equations run along the energy scale β; the evolution along the time direction is free oscillation (the KMS-time row), and scattering-type processes are accounted for in the static form of the virtual-process corrections through N_k^{eff} (§7.4), with no separate interaction equation in the time direction.

The three couplings obey the unified formula

$$\alpha_X = |A_X|^2 \cdot \eta_X, \tag{94}$$

where $|A_X|^2$ is the bare amplitude squared and η_X is a statistical factor:

Coupling	$\lvert A_X\rvert^2$	η_X	**Statistics**
Elementary resonance force	$\lvert R_k\rvert^2$	$1/N_k^{\text{eff}}$	averaging (dilution)
Emergent resonance force (emergent–elementary joint)	$\lvert R_{k_1}\rvert^2$	$\lvert\text{Aff}\rvert_{\text{eff}}$	summation (enhancement)

The full range of force strengths is determined entirely by the range of η_X – from $O(1/N_k)$ (dilution) to $O(p(p-1))$ (enhancement), arising purely from the statistical dichotomy of averaging versus summation.

§8 The Metric Force

§7 established the resonance forces: the convolution operator K (§3.2) produces selective coupling in the character directions through frequency matching, divided into the elementary resonance force ($\alpha_k = |R_k|^2/N_k^{\mathrm{eff}}$, averaging over equivalent paths, §7.4) and the emergent resonance force (summation over independent channels, §7.5). The resonance forces are selective – there is no force between directions whose frequencies do not match.

Another force exists on the state space. It comes not from the spectral structure of K but from the metric structure of the Hamiltonian $H = \log n$ itself (§3.4): the non-uniformity of the metric exerts a systematic action on all excitations over the full ring $\mathbb{Z}/p^K\mathbb{Z}$. We call it the **metric force**. The resonance forces come from K, the metric force from H – two components of the same dynamical structure, giving two forces with entirely different modes of action.

§8.1 Definition of the Metric Force

The resonance forces come from the spectral structure of the operator K; the metric force comes from a more basic layer – the geometry of the state space itself. The metric induced by $H = \log n$ is unevenly distributed over the lattice sites, and a systematic difference in geometric environment is, in itself, a systematic action on the excitations residing within it.

Non-uniformity. By §3.4, $H = \log n$ induces the metric $d(m, n) = |\log m - \log n|$ on the lattice-site set. The step length between adjacent sites is

$$d(n, n+1) = \log(n+1) - \log n = \log\left(1 + \frac{1}{n}\right), \tag{95}$$

strictly decreasing in n: $\log 2 \approx 0.693$ at $n = 1$ (the maximal spacing, §3.4), and $\log(1 + 1/n) \sim 1/n \to 0$ at large n. The metric length of a unit lattice step is large at the small-n end ($\log 2$) and small at the large-n end ($\sim 1/n$) – equally spaced lattice sites are not equidistant in this metric.

Gradient and force field. The discrete gradient of H at site n is defined as the adjacent step length:

$$\nabla H(n) = H(n+1) - H(n) = \log\left(1 + \frac{1}{n}\right). \tag{96}$$

A uniform metric corresponds to constant ∇H; the gradient of $H = \log n$ is strictly decreasing in n, so the metric is non-uniform. The **metric force field** is defined as the negative of the metric gradient:

$$\boxed{F_{\mathrm{grav}}(n) = -\nabla H(n) = -\log\left(1 + \frac{1}{n}\right).} \tag{97}$$

The minus sign gives the direction of the force: toward decreasing n, i.e. toward increasing metric step length. With H as potential, the large-n region has high potential and the small-n region low potential; the metric force pulls excitations from high potential toward low potential.

The metric–energy identity. Metric and energy are given by the same structure: the metric of §3.4 is defined as

$$d(m,n) = |\log m - \log n| = |H(m) - H(n)| \tag{98}$$

that is, distance and energy difference are identical at the level of definition. Consequently the force field $F_{\text{grav}}(n) = -[H(n+1) - H(n)]$ has a double reading: the energy drop between adjacent sites and the metric spacing between adjacent sites are the same quantity; the spatial reading and the energy reading yield the same value.

This identity also locates the non-uniformity. In the metric coordinate $r = \log n$, the potential $H = r$ is linear in metric distance, with gradient identically 1; the non-uniformity appears in the comparison between lattice coordinate and metric coordinate: the metric length spanned by the n-th lattice step is $\log(1 + 1/n)$, decreasing in n, from $\log 2$ at $n = 1$ down to about $1/n$ at large n. What the metric force measures is precisely the mismatch rate of the two coordinates: the metric length contained in a unit lattice step, which is also the energy drop per unit lattice step.

Generating mechanism. The source of the force is the weight distribution of the KMS state in the non-uniform metric. By §4, the weight of the KMS state φ_β at site n is $w_\beta(n) = n^{-\beta}/Z(\beta)$, decreasing in n. In a uniform metric the thermal equilibrium state weights all sites equally and there is no preferred direction; in the non-uniform metric of $H = \log n$, the weight concentrates toward small n – all excitations are systematically pushed toward small n by the metric structure. This push does not depend on which character direction an excitation belongs to: it is given by the monotonicity of the weight function $w_\beta(n)$, and w_β depends only on the site label n.

Magnitude of the force. The values of $|F_{\text{grav}}(n)| = \log(1 + 1/n)$ at the sites:

n	1	2	3	5	10	100	p^K
$\lvert F_{\text{grav}}(n)\rvert$	$\log 2$	$\log(3/2)$	$\log(4/3)$	$\log(6/5)$	$\log(11/10)$	$\log(101/100)$	$\sim p^{-K}$

The force is strongest at $n = 1$ (≈ 0.693), decreases monotonically with n, and tends to zero at large n while remaining everywhere nonzero.

§8.2 Geometric Properties

The force-field formula (97) contains only H, not K – every property of the metric force is already written in the analytic behavior of $\log n$. Each of its differences from the resonance forces stems from one and the same difference: the positional nature of H versus the directional nature of K.

(1) Full-ring action. $H = \log n$ is defined on all p^K lattice sites of the full ring $\mathbb{Z}/p^K\mathbb{Z}$ (the state space is built on the full ring, §3.1), so $F_{\text{grav}}(n)$ is defined at every site. The resonance forces differ: the character analysis of K is carried out on the species-layer unit group $(\mathbb{Z}/p\mathbb{Z})^*$ (§3.2), and after lifting to the full space involves only the $\varphi(p^K)$ sites coprime to p (§6.2). The range of action of the metric force is strictly larger than that of the resonance forces – sites not belonging to the unit group feel the metric force all the same.

(2) Universality. $\nabla H(n) = \log(1+1/n)$ depends only on the site label n: not on the character direction k, not on the CRT coordinates $(k_1, \ldots, k_s)$, not on the order $\mathrm{ord}(\chi_k)$.

$$\begin{aligned} & H : \{1, \ldots, N\} \to \mathbb{R} \text{ depends only on } n \\ \Longrightarrow\ & F_{\mathrm{grav}}(n) = -\nabla H(n) \text{ depends only on } n \\ \Longrightarrow\ & \text{every excitation defined at site } n \text{ feels the same } F_{\mathrm{grav}}(n). \end{aligned} \tag{99}$$

The resonance forces are the opposite: the eigenvalues $\lambda_k = L(\beta, \chi_k)$ of K depend on the direction k, and K provides coupling only between frequency-matched modes (the selection rule of §7.1). The root of the difference: K is an operator diagonalized in the character basis (directional), H is a scalar function on the lattice sites (positional) – its gradient carries no character information. The trivial character χ_0 and the highest-order character feel exactly the same metric force at the same site n.

(3) Globality. $|F_{\mathrm{grav}}(n)| = \log(1 + 1/n) > 0$ for all $n \geq 1$ – the metric force is strictly nonzero at every site; no region of vanishing metric force exists. The variation of the magnitude with n is given directly by the difference:

$$|F_{\mathrm{grav}}(n)| - |F_{\mathrm{grav}}(n+1)| = \log \frac{1 + 1/n}{1 + 1/(n+1)} = \log \frac{(n+1)^2}{n(n+2)} = \log\left(1 + \frac{1}{n(n+2)}\right) > 0, \tag{100}$$

so $|F_{\mathrm{grav}}|$ is strictly decreasing in n: everywhere nonzero, yet arbitrarily weak. The asymptotic behavior at large n is

$$|F_{\mathrm{grav}}(n)| = \frac{1}{n} - \frac{1}{2n^2} + \frac{1}{3n^3} - \cdots \sim \frac{1}{n}. \tag{101}$$

In the metric coordinate $r = d(1, n) = \log n$ (with r ranging over the discrete set $\{\log n : n \geq 1\}$), $|F_{\mathrm{grav}}| = \log(1 + e^{-r}) \sim e^{-r}$ – on these discrete points the force decays exponentially with metric distance, its influence concentrated in the small-n region.

(4) Extremely weak coupling. The bare magnitude of F_{grav} is not small (≈ 0.693 at $n = 1$), but the state space $L^2(\mathbb{Z}/p^K\mathbb{Z})$ has p^K orthogonal modes in all, and the KMS state averages over all modes in thermal equilibrium. The metric force acts universally on all modes, making no selection, and its total effect is shared out uniformly among them all – each mode's share is $O(p^{-K})$. The elementary resonance force is not subject to this large-scale dilution: it acts only on frequency-matched modes, with dilution factor equal to the equivalent-path count $N_k \leq (p-1)^2/2$ (§7.4), far smaller than p^K ($K \geq 3$). The quantitative statement is given in §8.4.

§8.3 The Discrete Poisson Equation

The force field gives the push felt at each site; the push has another face – the convergence of force lines at each site. In the continuum theory this face is captured by the Laplacian of the potential; on discrete lattice sites it is carried by the second difference of the potential.

On a discrete space, a field is a lattice-site function – a quantity assigned to each point (site n) of physical space. The potential H and the force field F_{grav} are both fields in this sense.

Discrete Laplacian. For a lattice-site function f, define the second difference

$$(\Delta f)(n) = f(n+1) + f(n-1) - 2f(n), \qquad n \geq 2. \tag{102}$$

Poisson identity. Direct computation for the potential $f(n) = \log n$:

$$\begin{aligned}\Delta(\log n) &= \log(n+1) + \log(n-1) - 2\log n \\ &= \log\frac{(n+1)(n-1)}{n^2} = \log\frac{n^2-1}{n^2} \; < \; 0.\end{aligned} \tag{103}$$

Hence

$$\boxed{-\Delta(\log n) = R(n) = \log\frac{n^2}{n^2-1}, \qquad n \geq 2.} \tag{104}$$

$R(n)$ is everywhere positive – the metric force is attractive everywhere.

The direction is opposite to that of continuum field theory: the continuum theory solves for the potential from a given source distribution; here the potential $\log n$ is given by the definition of H, (104) is the Poisson-type identity it satisfies, and the source density $R(n)$ is a derived quantity. The metric structure is not determined by the distribution of excitations but given by H itself.

Asymptotic behavior.

$$R(n) = -\log\left(1 - \frac{1}{n^2}\right) = \frac{1}{n^2} + \frac{1}{2n^4} + \frac{1}{3n^6} + \cdots. \tag{105}$$

Three properties:

- **Everywhere positive**: $R(n) > 0$ for all $n \geq 2$ – attraction, with no repulsive region;
- **Flat at the far end**: $R(n) \to 0$ $(n \to \infty)$ – the source density tends to zero far from the origin;
- **Inverse-square dominance**: $R(n) = 1/n^2 + O(1/n^4)$ – the leading term is inverse-square; the first correction $1/(2n^4)$ is significant only at small n (near the minimal-spacing scale).

§8.4 The Metric-Force Constant

That the metric force is universal and weak is a qualitative conclusion of its geometric properties. For the qualitative to become a comparable statement, one needs a number on the same scale as the resonance couplings α_k. The construction exactly parallels §7.4: response operator, KMS expectation value, normalization.

Response operator. The bare squared strength of the metric force at site n is the symmetric discrete gradient squared (the per-mode convention of §3.4) $|\nabla H(n)|^2 = d(n, n+1)^2 + d(n-1, n)^2$ ($n \geq 2$; $n = 1$ has only a right neighbor). Define the **metric-force response operator**

$$\Pi_{\text{grav}} = \sum_{n=1}^{N-1} |\nabla H(n)|^2 \, |n\rangle\langle n|, \tag{106}$$

where $|n\rangle$ is the lattice-site basis vector. Compare with the response operator $\Pi_k = P_k K^\dagger K P_k$ of §7.4: Π_k is a rank-one operator in a character direction (directional), Π_{grav} is a diagonal operator in the lattice-site basis (positional) – the difference again stems from the directional nature of K versus the positional nature of H.

KMS expectation value.

$$\varphi_\beta(\Pi_{\text{grav}}) = \frac{1}{Z(\beta)} \sum_{n=1}^{N-1} |\nabla H(n)|^2 \, n^{-\beta} = \frac{C^*_\beta}{Z(\beta)} + O(N^{-\beta}), \tag{107}$$

where

$$C^*_\beta = \sum_{n=1}^{\infty} |\nabla H(n)|^2 \, n^{-\beta}. \tag{108}$$

The summand is $\sim n^{-\beta-2}$, so the series converges for all $\beta \geq 0$. At $\beta = 0$ it reduces to the metric constant of §3.4: $C^*_0 = C^* = 1.9543783$. C^*_β is the Dirichlet coupling of the geometric content of the metric (squared gradient) with the thermodynamic content (Boltzmann weight $n^{-\beta}$), completely determined by the structure of H and the temperature β, independent of p and K. Numerically $C^*_2 \approx 0.685$ (dominated by $n = 1, 2$; convergence is very fast), and as $\beta \to \infty$, $C^*_\beta \to (\log 2)^2 \approx 0.480$ (only the right-neighbor term of $n = 1$ survives).

Dilution mechanism. The effective action of the metric force on a single mode requires division by the total mode count p^K:

$$G_{\text{mode}} = \frac{\varphi_\beta(\Pi_{\text{grav}})}{p^K} = \frac{C^*_\beta}{p^K \, Z(\beta)} \approx \frac{C^*_\beta}{p^K \, \zeta(\beta)}. \tag{109}$$

The numerator C^*_β is an $O(1)$ constant; the denominator grows linearly with the size of the state space: the metric-force constant is diluted linearly in p^K. The ratio to the resonance coupling (§7.4) is

$$\frac{G_{\text{mode}}}{\alpha_k} \sim \frac{C^*_\beta \, N_k}{p^K \, |R_k|^2} \lesssim \frac{O(1)}{p^{K-2}}. \tag{110}$$

The metric force is additionally suppressed relative to the elementary resonance force by at least a factor p^{K-2} (for typical directions $N_k = O(p)$, suppression by p^{K-1}); the larger K, the stronger the suppression. The root of the suppression is precisely universality: the metric force acts on all p^K modes, and KMS equilibrium spreads the total effect over them all; the elementary resonance force, being selective (acting only on matched directions, $N_k \leq p - 1$), escapes this dilution.

Calibration. The formal metric-force constant is given by the energy-scale formula (24) of §5.3. For a system participating with spectral weight N, $E(N) = \sqrt{N \log^2 N / C^*}$; define

$$\boxed{G = \frac{1}{E(N)^2} = \frac{C^*}{N \log^2 N}, \qquad N = p^K.} \tag{111}$$

Consistency of the dilution mechanism with the calibration: $G_{\text{mode}} \sim C^*_\beta/(p^K \zeta(\beta))$ and $G = C^*/(N \log^2 N)$ are both $O(p^{-K})$ at leading order, differing only in the logarithmic factor and the ζ factor – (109) gives the mechanism of the force's weakness (per-mode share), (111) gives the quantitative constant consistent with the energy-scale system.

§8.5 Dynamical Equations

The static characterization is now complete. What remains is change – the behavior of the force field under time evolution, and its running with the temperature β.

Staticity. $F_{\text{grav}}(n) = -\log(1 + 1/n)$ is given directly by the definition of H, while the time evolution is the time flow $\sigma_t = e^{itH}$ (§3.4), whose generator is H itself; H commutes with itself, so the force field defined by H is invariant under time evolution:

$$\sigma_t(F_{\text{grav}}) = F_{\text{grav}}, \qquad t \in \mathbb{R}. \tag{112}$$

The metric force is a static background.

Universality equation. By property (2) of §8.2:

$$\frac{\partial F_{\text{grav}}}{\partial k} = 0. \tag{113}$$

Equilibrium distribution. The response of excitations to the metric potential is given by the KMS state. The KMS weight of §4 is precisely the Boltzmann distribution in the potential H:

$$w_\beta(n) = \frac{e^{-\beta H(n)}}{Z(\beta)} = \frac{n^{-\beta}}{Z(\beta)}. \tag{114}$$

This is the equilibrium form of the equation of motion of the metric force: the distribution concentrates toward low potential (small n), and the lower the temperature (the larger β) the stronger the concentration.

A non-equilibrium trajectory equation does not exist in this framework, for a structural reason: the unitary evolution σ_t changes only phases; for any excitation $\psi = \sum_n c_n |n\rangle$, the magnitudes $|c_n|$ are preserved under σ_t, so no individual excitation moves along the lattice toward small n. The metric force does not drive trajectories; it shapes the equilibrium distribution: it is a statistical force, consistent with its origin in the metric structure rather than in a propagation operator. The resonance forces have evolution equations (§7.6); the metric force has only a distribution equation – this difference is precisely the expression of the difference in level between the two kinds of force.

Energy-scale evolution. The derivative of C^*_β with respect to β is

$$\frac{\partial C^*_\beta}{\partial \beta} = -\sum_{n=2}^{\infty} |\nabla H(n)|^2 \, n^{-\beta} \log n \; < \; 0 \tag{115}$$

(the $n = 1$ term does not contribute since $\log 1 = 0$; all remaining terms are strictly negative). C^*_β is strictly decreasing in β and bounded:

$$(\log 2)^2 \; < \; C^*_\beta \; \leq \; C^*, \qquad \beta \geq 0. \tag{116}$$

As the temperature falls (β increases), the Boltzmann weight contracts toward small n, and the effective strength of the metric fluctuations decreases monotonically from C^*, tending to the single-site contribution $(\log 2)^2$ of $n = 1$.

Summary of the equations.

Equation	Content	Meaning	Source
(112)	$\sigma_t(F_{\text{grav}}) = F_{\text{grav}}$	static background	§8.5
(113)	$\partial F_{\text{grav}}/\partial k = 0$	universality	§8.2
(114)	$w_\beta(n) = e^{-\beta H(n)}/Z(\beta)$	equilibrium distribution (response)	§8.5
(104)	$-\Delta(\log n) = \log \frac{n^2}{n^2-1}$	Poisson identity	§8.3
(115)	$\partial_\beta C^*_\beta < 0$	energy-scale evolution	§8.5

There are three forces on the state space in all – the elementary resonance force, the emergent resonance force, and the metric force, differing in origin and in statistics:

	Elementary resonance force (§7.4)	**Emergent resonance force (§7.5)**	**Metric force (§8)**
Origin	frequency resonance of K	channel amplification of K in the χ_0 direction	metric non-uniformity of H
Acts on	between elementary states	emergent states (and jointly with elementary states)	all excitations
Selectivity	frequency matching	the χ_0 direction	none
Statistics	averaging ($1/N_k$)	summation ($\lvert\text{Aff}\rvert_{\text{eff}}$-fold enhancement)	averaging ($1/p^K$)
Strength	$\alpha_k = \lvert R_k\rvert^2/N_k^{\text{eff}}$	$\alpha_{\text{emg}} = \lvert\text{Aff}\rvert_{\text{eff}}$, $\alpha_{\text{mix}} = \lvert\text{Aff}\rvert_{\text{eff}}\lvert R_{k_1}\rvert^2$	$G = C^*/(N\log^2 N)$
Order of magnitude	$O(1)$	$O(p(p-1))$	$O(p^{-K})$

§9 State Space and p-Channels

§1–§4 constructed the state space from the BC system: for each odd prime p, the p-adic reduction (§2) gives

$$\mathcal{H}_p = L^2(\mathbb{Z}/p^K\mathbb{Z},\, \mathbb{C}), \qquad \dim \mathcal{H}_p = p^K.$$

§5–§8 built spacetime, matter, and forces on top of it: four-dimensional Lorentz spacetime, the two kinds of stable states (elementary and emergent), and the resonance and metric forces. All of these constructions hold for a general odd prime p; different p correspond to different physical content.

We call the complete physical system obtained by fixing one p a **p-channel**: different p are different physical channels, with different numbers of elementary states, different generation numbers, different force spectra, and different gauge groups. This chapter determines the channel corresponding to the observed universe, and establishes, within that channel, the correspondence between characters and particles and forces.

§9.1 Characteristic Quantities of a p-Channel

Determining the channel corresponding to the observed universe requires quantities that can be compared across channels. Among the structures established for general p in the preceding sections, the following quantities are all definite functions of p:

- Spacetime (§5): dimension 4, signature $(1,3)$, the same for all odd primes;
- Generation number (number of doublets, §6.2): $(p-1)/2$, the same for every particle family–free lattice sites and link channels are counted by the same rank, with total always equal to $(p-1)/2$ (§6.4);
- Number of elementary states (number of nontrivial character directions, §6.1): $p-2$;
- Number of emergent states (§6.1): 1, the same for all odd primes;
- Kinds of elementary resonance forces (CRT-factor resonances plus the cross-factor resonance, §7.2): $\omega(p-1)+1$;
- Emergent resonance force (§7.5): 1 kind, the same for all odd primes;
- Metric force (§8): 1 kind, the same for all odd primes;
- Gauge group (§7.2): $\big(U(1) \times SU(q_1^{b_1}) \times \cdots \times SU(q_s^{b_s})\big)/\mathbb{Z}_{p-1}$, where $p-1 = q_1^{b_1} \cdots q_s^{b_s}$.

This set of quantities is called the **characteristic quantities** of the p-channel. For the first few odd-prime channels:

p	Spacetime	Gen.	Elem.	Emg.	Elem. res.	Emg. res.	Metric	Gauge group
3	$4d\,(1,3)$	1	1	1	2	1	1	$U(1) \times SU(2)$
5	$4d\,(1,3)$	2	3	1	2	1	1	$U(1) \times SU(4)$
7	$4d\,(1,3)$	3	5	1	3	1	1	$U(1) \times SU(2) \times SU(3)$
11	$4d\,(1,3)$	5	9	1	3	1	1	$U(1) \times SU(2) \times SU(5)$
13	$4d\,(1,3)$	6	11	1	3	1	1	$U(1) \times SU(4) \times SU(3)$
23	$4d\,(1,3)$	11	21	1	3	1	1	$U(1) \times SU(2) \times SU(11)$
31	$4d\,(1,3)$	15	29	1	4	1	1	$U(1) \times SU(2) \times SU(3) \times SU(5)$

The gauge-group column is throughout modulo the center $\mathbb{Z}_{p-1}$ (§7.2).

Spacetime, the number of emergent states, the emergent resonance force, and the metric force are the same for all odd primes and play no role in distinguishing channels; the differences between channels lie in the generation number, the number of elementary states, the kinds of elementary resonance forces, and the gauge group.

§9.2 The Prime 7 and the p-Channel

The characteristic quantities of the previous subsection vary with p in the theory; in the physical space we inhabit, each entry has a definite value. Listing them item by item, following the entries of the characteristic-quantity table:

- Spacetime: four-dimensional, signature $(1,3)$;
- Generations: three generations of each kind of charged fermion (e-μ-τ, u-c-t, d-s-b);
- Five kinds of elementary particles: neutrinos, leptons, antileptons, up-type quarks, down-type quarks;
- One class of baryons: the proton, the neutron, and the excitation spectrum (composite particles);
- Three gauge forces: electromagnetic, weak, color;
- One nuclear force;
- One gravitational force;
- Gauge group: $U(1)\times SU(2)\times SU(3)$, with charge quantization corresponding to the global form modulo $\mathbb{Z}_6$.

Comparing with the characteristic-quantity table: among the four p-dependent entries, the number of elementary states $p-2=5$, the generation number $(p-1)/2=3$, and the gauge group $p-1=2\times 3$ all require $p=7$, and the number of kinds of elementary resonance forces $\omega(p-1)+1=3$ is consistent with it; the universal entries–baryons, the nuclear force, gravity–agree in count with the number of emergent states, the emergent resonance force, and the metric force (each 1). The only consistent channel is

$$\boxed{p = 7.}$$

The agreement of the counts holds class by class: the five kinds of elementary particles correspond to the five nontrivial character directions.

Beyond the counts there is one further structural condition: the four-case classification of §6.4.7 is complete, with both the closed (order 3) and the open (order 2) self-loop types available, if and only if $p-1=6$; this condition depends on none of the counts above and likewise singles out $p=7$.

That antileptons appear among the five while antiquarks do not is determined by how antiparticles are carried. Antiparticles correspond to the conjugation operation, and the two direct-product parts of the unit group each carry conjugation: the antiparticles of colored particles are carried by the color conjugation of the principal-unit part ($3\leftrightarrow\bar{3}$, conjugation of the color axes, §10.1), which does not occupy a tame direction, so χ_1 and χ_5 each carry an independent kind (up-type and down-type); colorless charged particles have no principal-unit carrier, and their antiparticles are carried by the Galois conjugation of the tame directions, so $\{\chi_2,\chi_4\}$ carries the particle and antiparticle states of the charged leptons; the self-conjugate direction $\chi_3=\bar{\chi}_3$ carries the neutrino, for which particle and antiparticle coincide (Majorana). The count $1+2+1+1=5$ is fixed item by item by color charge, and the agreement with $p-2$ involves no free choice. The specific correspondence is determined by the structures on both sides.

9.2.1 Characters and Particles

The character structure of the $p = 7$ channel was already established in §3–§6: the unit group $(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/6\mathbb{Z}$ has character group $\chi_0, \chi_1, \ldots, \chi_5$, with $\chi_k(3^a) = e^{2\pi i k a/6}$ (3 a primitive root), of orders $1, 6, 3, 2, 3, 6$ respectively; the CRT decomposition $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ gives the coordinates $k \leftrightarrow (k_1, k_2) = (k \bmod 2,\, k \bmod 3)$ (§6.2); under complex conjugation $\chi_k \mapsto \bar{\chi}_k = \chi_{6-k}$, χ_3 is the unique nontrivial real direction, and (χ_2, χ_4) and (χ_1, χ_5) are two complex-conjugate pairs (§3.3).

The real/complex type, the CRT coordinates, and the order of each direction determine, respectively, the charge, the color, and the mass tier of the particle it carries:

- Real/complex $\leftrightarrow$ charge: complex conjugation is a Galois symmetry (§3.3), physically particle–antiparticle conjugation (§6.2). The unique nontrivial real direction χ_3 is invariant under conjugation, so its charge is zero; the two complex-conjugate pairs (χ_2, χ_4) and (χ_1, χ_5) are exchanged under conjugation and carry opposite charges within each pair.
- CRT coordinates $\leftrightarrow$ color: the coordinate pattern is a conjugation invariant (§6.2), separating the two pairs–both components nonzero (χ_1, χ_5) versus a single component (χ_2, χ_4). Its physical reading is given by the carrying mechanism above (§10.1): the antiparticles of the directions with both components nonzero are carried by color conjugation, so these are colored; the single-component directions are paired by Galois conjugation, so these are colorless.
- Order $\leftrightarrow$ mass tier: the order determines the length of the mass path (§6.4), and the orders $2 < 3 < 6$ give three mass tiers; the tree-level mass of a real character degenerates (§6.4), so the mass of χ_3 is suppressed.

Comparing these three properties item by item with the charges, colors, and masses of the five kinds of particles, and then binding the conjugation orientation by the CRT multiplicative structure (argument following the table), the correspondence is uniquely determined:

Direction	**Real/complex**	**CRT coordinates**	**Order**	**Particle**
χ_3	real	$(1,0)$	2	neutrino
χ_2	complex	$(0,2)$	3	lepton
χ_4	complex	$(0,1)$	3	antilepton
χ_1	complex	$(1,1)$	6	up-type quark
χ_5	complex	$(1,2)$	6	down-type quark

The uniqueness argument proceeds in three steps.

Step 1: the first three properties reduce the $5! = 120$ possible assignments to the freedom of conjugation orientation. Real/complex separates χ_3 from the two complex-conjugate pairs; the CRT triple label separates the pair (χ_1, χ_5) from the pair (χ_2, χ_4); the order gives the mass tiers. All three are conjugation invariants and do not distinguish the orientation flip within each pair ($\chi_2 \leftrightarrow \chi_4$ and $\chi_1 \leftrightarrow \chi_5$): there are 4 compatible assignments.

Step 2: the CRT multiplicative structure binds the orientations of the two pairs (the fourth property). The quark directions are the CRT products of the lepton directions with χ_3:

$$\chi_5 = \chi_2 \cdot \chi_3, \qquad \chi_1 = \bar{\chi}_2 \cdot \chi_3,$$

and the down-type/up-type distinction is given by the orientation of the χ_2-content: χ_5, containing a χ_2 component, carries the down type; χ_1, containing a $\bar{\chi}_2$ component, carries the up type. If the lepton orientation is flipped ($\chi_2 \to \chi_4 = \bar{\chi}_2$), then $\chi_4 \cdot \chi_3 = \chi_1$ and $\bar{\chi}_4 \cdot \chi_3 = \chi_5$: the quark orientation is forced to flip along with it; the two pairs cannot flip independently, and the 4 assignments reduce to 2.

Step 3: the remaining two assignments differ by an overall complex conjugation–replacing every direction simultaneously by its conjugate, i.e., the action of the Galois generator σ_{-1}, whose GNS realization is the modular conjugation J (§4.5). Overall conjugation is a symmetry of the state space, not a freedom of the assignment: all physical quantities (masses, couplings, splittings) are conjugation-invariant combinations, so the two assignments give identical physics item by item; their difference is merely the algebraic convention of labeling the same object by χ or by $\bar{\chi}$, of the same nature as the naming convention of taking the Lorentz signature to be $(1, 3)$ or $(3, 1)$ (§5.1). The assignment is therefore unique as a physical object. The trivial direction χ_0 carries no elementary state (§6.1); the pole of $\zeta(\beta)$ drives the collective emergent state only in this direction (§6.5), and emergence is constrained by the color-singlet condition (§10.3): the emergent states in the χ_0 direction correspond to baryons (the proton, the neutron, and the excitation spectrum).

9.2.2 Resonance Channels and Forces

The character labels admit a double reading: as elementary-state directions, the χ_k label particles (the table above); as resonance channels, the χ_k label forces (§7.2). The particle correspondence fixes the first reading, and the second reading then follows:

Resonance channel	**Channel character**	**Gauge factor**	**Force**
$\mathbb{Z}/2\mathbb{Z}$-factor resonance	χ_3	$SU(2)$	weak force
$\mathbb{Z}/3\mathbb{Z}$-factor resonance	χ_2, χ_4	$SU(3)$	color force
cross-factor resonance	χ_1, χ_5	$U(1)$	electromagnetic force
emergent resonance (§7.5)	χ_0	–	nuclear force
metric force (§8)	–	–	gravity

The identification of the forces is given by the structural properties of the channels, compared item by item with the observed characteristics of the forces:

- The involution ι flips the $\mathbb{Z}/2\mathbb{Z}$ component and preserves the $\mathbb{Z}/3\mathbb{Z}$ component (since $-1 = g^3$, ι acts on the exponent as translation by $+3$, i.e., a flip mod 2 and the identity mod 3; §3.3): the symmetry breaking acts solely on the $\mathbb{Z}/2\mathbb{Z}$ channel–the broken force, i.e., the weak force;
- The $\mathbb{Z}/3\mathbb{Z}$ channel is carried by the complex-conjugate pair (χ_2, χ_4); a single direction is not Galois invariant and has no self-adjoint observable (§4): the force whose charge cannot be observed in isolation, i.e., the color force;
- The cross-factor channel is unbroken and unconfined: the long-range Abelian force, i.e., the electromagnetic force;

- The emergent resonance exists only in the χ_0 direction (acting only between emergent states) and is strongly coupled (§7.5): the nuclear force;
- The metric force is universal and, diluted by the p^K modes, the weakest (§8.4): gravity.

Consistency is only a necessary condition. Whether the physical space we inhabit is indeed the $p = 7$ channel depends on whether the energy scales, coupling constants, and mass spectrum given by this channel agree with experiment.

§10 Energy Scales

§9 compared the observed structure with the characteristic quantities of the p-channels; $p = 7$ is the only consistent channel. The test of consistency lies in the numbers. This chapter computes the first set of numbers for the $p = 7$ channel: the energy scales.

§5.3 established the energy-scale formula: for a physical process with participating spectral weight N,

$$E(N) = \sqrt{\frac{N \cdot \log^2 N}{C^*}}\,, \qquad C^* = 1.9543783\,, \tag{117}$$

where C^* is the intrinsic metric constant defined in §3.4, and the output of (117) is in GeV. The group structure of the state space gives three natural mode sets: the unit group (the carrier of the elementary resonance forces, §7), the full group (the carrier of the metric force, §8), and the χ_0 direction (the carrier of the emergent states, §6.5). Each of the three determines an effective mode number N; substituting into (117) yields three energy scales. We compute them in turn.

§10.1 The Electroweak Scale

The elementary resonance forces are defined on the unit group $(\mathbb{Z}/7^k\mathbb{Z})^*$ (§7.1), and their energy scale is determined by the effective mode number of the unit group. The computation proceeds in four steps.

Step 1: the number of degrees of freedom. Each mode is labeled by a K-digit 7-adic address; a structure resolves the first k digits if all of its modes depend only on these k digits of the address– k is then the number of degrees of freedom of that structure. How many digits the elementary resonance forces resolve is uniquely determined by the closure condition of the color structure.

(i) When k digits of the address are resolved, the unit group decomposes by the Teichmüller decomposition (§2.4) as

$$(\mathbb{Z}/7^k\mathbb{Z})^* \;\cong\; \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/7^{k-1}\mathbb{Z}.$$

The characters of the tame part $\mathbb{Z}/6\mathbb{Z}$ are one-dimensional species labels (§9.2): the three-dimensional carrier of color (§7.2) can only reside in the principal-unit part $\mathbb{Z}/7^{k-1}\mathbb{Z}$.

(ii) The canonical filtration of the principal unit group $U^{(1)} = 1 + 7\mathbb{Z}/7^k\mathbb{Z}$ gives $k - 1$ coordinate axes:

$$U_j = 1 + 7^j\mathbb{Z}/7^k\mathbb{Z}, \qquad U_j/U_{j+1} \cong \mathbb{F}_7 \quad (j = 1, \ldots, k-1);$$

a Teichmüller unit $\xi \in \mu_6$ acts on each graded quotient as multiplication by ξ:

$$(1 + 7^j a)^\xi \equiv 1 + 7^j(\xi a) \pmod{7^{j+1}}.$$

Hence the $k - 1$ axes are pairwise isomorphic as μ_6-modules, forming the unique canonical set of directions of the principal-unit part.

(iii) The color directions occupy axes, not values: if color took 3 of the 7 values of a single axis, the choice would have no canonical basis, and the mixing unitary symmetry of 7 values would contradict the gauge group of §7.2. The two members of a weak doublet are identified with each other by the involution ι (§3.3) and occupy no axis; the cross-factor $U(1)$ direction has no multiplet and occupies no axis.

(iv) The mixing unitary group of the $k-1$ equivalent axes must lie inside the gauge group of §7.2, and the unique candidate is $SU(3)$:

$$SU(k-1) = SU(3) \implies k-1 = 3.$$

For $k-1 < 3$ the totally antisymmetric tensor ε^{abc} (§10.3) has no carrier space; for $k-1 > 3$ the surplus equivalent axes contradict the force-species theorem ($\omega(6)+1 = 3$, §7.2). Hence

$$\boxed{K_{\text{EW}} = d_2 + 1 = 4} \tag{118}$$

($d_2 = 3$ is the dimension of the $\mathbb{Z}/3\mathbb{Z}$ factor).

Step 2: the Born value. The mode number of the unit group at K_{EW}-digit resolution:

$$N_{\text{EW}}^{(0)} = \varphi(7^{K_{\text{EW}}}) = \varphi(7^4) = 7^3 \times 6 = 2058. \tag{119}$$

Step 3: the weak-force virtual-process correction. The character χ_3 has order 2 and satisfies $\chi_3^2 = \chi_0$: by the classification of §7.4 it is Type II; its self-loop directly excites the χ_0 direction, so its virtual processes occupy channels in the vacuum direction and must be deducted from the mode number. The self-loops of the Type I directions ($\chi_1^2 = \chi_2 \neq \chi_0$, etc.) do not return to the vacuum and incur no such deduction.

The number of occupied channels is the effective channel number N_W^{eff} of the weak force, given by the Type II correction formula of §7.4:

$$N_W^{\text{eff}} = N_W - d_2|R_1|^2 + |R_2|^2 + c_3|R_3|^2, \tag{120}$$

where $N_W = 18$ is the Born channel number of the weak force (derived in §11.2), $R_k(\beta) = L(\beta, \chi_k)/\zeta(\beta)$ is the resonance ratio (§7.4), the numerical values are taken at the vertex evaluation point $\beta = \frac{1}{2}$ (for the justification see §11.1), $-d_2|R_1|^2$ is the absorption by the color sector ($d_2 = 3$ is the dimension of the $\mathbb{Z}/3\mathbb{Z}$ factor), $+|R_2|^2$ is the return through the electromagnetic sector, and $+c_3|R_3|^2$ is the weak force's own contribution ($c_3 = 2$, χ_3 self-conjugate). Substituting $|R_1|^2 = 0.3448$, $|R_2|^2 = 0.0476$, $|R_3|^2 = 0.6165$:

$$N_W^{\text{eff}} = 18 - 3 \times 0.3448 + 0.0476 + 2 \times 0.6165 = 18.25. \tag{121}$$

Step 4: the effective vacuum mode number and the energy scale. Deducting from the Born value the channels occupied by the weak-force virtual processes:

$$N_{\text{vac}}^{\text{eff}} = \varphi(7^4) - N_W^{\text{eff}} = 2058 - 18.25 = 2039.75. \tag{122}$$

Substituting into (117):

$$\boxed{v_{\text{EW}} = E(2039.75) = \sqrt{\frac{2039.75 \times (\log 2039.75)^2}{1.9543783}} = 246.19 \text{ GeV.}} \tag{123}$$

The experimental value is $v_{\text{EW}}^{\text{exp}} = 246.22\,\text{GeV}$, a deviation of -0.011%.

Beyond the unit group, the next scale comes from the full group.

§10.2 The Planck Scale

The metric force acts on the full group $\mathbb{Z}/7^{K_{\max}}\mathbb{Z}$ (§8.1), and its energy scale is determined by the effective mode number of the full group.

Step 1: the number of degrees of freedom and the bare value. The metric force involves both addition (translation) and multiplication (scaling) and acts universally on all geometric degrees of freedom: the number of degrees of freedom it resolves is the total number $K_{\max}$– the value determined in §2.3 by affine completeness,

$$K_{\max} = \dim_{\mathbb{F}_7}(I) = |\mathrm{Aff}(\mathbb{F}_7)| - 1 = 41, \tag{124}$$

where I is the augmentation ideal of the affine group algebra $\mathbb{F}_7[\mathrm{Aff}(\mathbb{F}_7)]$, and $|\mathrm{Aff}(\mathbb{F}_7)| = 7 \times 6 = 42$. The elementary resonance forces resolve $K_{\mathrm{EW}} = 4$ degrees of freedom (color closure, §10.1), while the metric force resolves all $K_{\max} = 41$: the two numbers are determined separately by two different conditions. The bare value is

$$N_{\mathrm{Pl}}^{\mathrm{bare}} = 7^{41} \approx 4.46 \times 10^{34}. \tag{125}$$

Step 2: the Maschke-defect correction. Maschke's theorem [24] requires the group order to be invertible in the coefficient field, but $|\mathrm{Aff}(\mathbb{F}_7)| = 42 \equiv 0 \pmod 7$: the group algebra $\mathbb{F}_7[\mathrm{Aff}]$ is not semisimple, and the augmentation map cannot completely remove the trivial module S_0. S_0 is glued to S_1 in a non-split way through $\mathrm{Ext}^1(S_0, S_1) = \mathbb{F}_7$, and its projective cover P_0 has the indecomposable Loewy-chain structure

$$P_0 : \quad S_0 \to S_1 \to S_2 \to S_3 \to S_4 \to S_5 \to S_0\,,$$

where S_j corresponds to the character χ_j, with particle identities given by the correspondence of §9.2. The augmentation map removes the head S_0 of the chain, but since P_0 is indecomposable, each nontrivial direction S_j ($j = 1, \ldots, 5$) pulls back part of the vacuum contribution through the Ext coupling, the pull-back amount being the node factor $\mathcal{T}_j$.

In terms of the CRT coordinates (k_1, k_2), the node factor is $\mathcal{T}_j = \mathcal{N}_2(k_1)\,\mathcal{N}_3(k_2)$ (the factor-by-factor node factor of Eq. (52) in §6.4.6): when the $\mathbb{Z}/2\mathbb{Z}$ component vanishes, the direction belongs to a Galois pair and the two states are indistinguishable, so amplitude-level normalization gives $\mathcal{N}_2(0) = 1/\sqrt{2}$; when the $\mathbb{Z}/3\mathbb{Z}$ component vanishes, the three color directions are deconfined and mutually distinguishable, counted at probability level and normalized by the background amplitude at the vertex evaluation point $\beta = \frac{1}{2}$, giving $\mathcal{N}_3(0) = 3/|\zeta(\frac{1}{2})|$ ($|\zeta(\frac{1}{2})| = 1.4604$); when a component is nonzero the factor is 1. Substituting the CRT coordinates of the five directions (the table of §9.2), the product is

$$\prod_{j=1}^{5} \mathcal{T}_j = 1 \times \frac{1}{\sqrt{2}} \times \frac{3}{1.4604} \times \frac{1}{\sqrt{2}} \times 1 = \frac{3}{2 \times 1.4604} = 1.0271. \tag{126}$$

Step 3: the energy scale. The corrected effective mode number is

$$N_{\mathrm{Pl}}^{\mathrm{eff}} = 7^{41} \times 1.0271, \tag{127}$$

and substituting into (117):

$$\boxed{M_{\mathrm{Pl}} = E(N_{\mathrm{Pl}}^{\mathrm{eff}}) = 1.221 \times 10^{19}\ \mathrm{GeV}.} \tag{128}$$

The experimental value is $M_{\text{Pl}}^{\text{exp}} = 1.2209 \times 10^{19}$ GeV; the deviations before and after the correction are -1.3% and $+0.04\%$ respectively.

Beyond the unit group and the full group, the third scale comes from the emergent states in the χ_0 direction.

§10.3 The Nucleon Scale

The participating spectral weight of the emergent states is the spectral weight of the χ_0 direction, $N_{\chi_0} = Z(\beta) \to \zeta(\beta)$ (§6.5), so their energy scale depends on the energy-scale parameter β at which the emergent states sit.

Step 1: the quantization condition for β. In the KMS state the energy variance of the Hamiltonian $H = \log n$ is

$$\mathrm{Var}_\beta(H) = \frac{d^2}{d\beta^2} \log \zeta(\beta),$$

and by the Laurent expansion $\zeta(\beta) = 1/(\beta-1) + \gamma + O(\beta-1)$, the leading order is

$$\mathrm{Var}_\beta(H)\big|_{\text{leading}} = \frac{1}{(\beta-1)^2}.$$

The antisymmetric combination of n colored modes forms a color-singlet emergent state, whose leading-order quantum number of the energy variance equals the mode number n, i.e., the quantization condition $n(\beta-1)^2 = 1$:

$$\beta(n) = 1 + \frac{1}{\sqrt{n}}\,. \tag{129}$$

Step 2: the lowest color singlet. The color-singlet condition requires each of the three color directions to contribute at least one mode (total antisymmetry ε^{abc}), so the lowest level is $n = 3$. Substituting into (129):

$$\beta_B = 1 + \frac{1}{\sqrt{3}} = 1.5774, \qquad \zeta(1.5774) = 2.350.$$

Step 3: the energy scale. Substituting into (117):

$$\boxed{m_p = \sqrt{\frac{2.350 \times (\log 2.350)^2}{1.9543783}} = 937\ \text{MeV.}} \tag{130}$$

The experimental value is $m_p^{\text{exp}} = 938.3\,\text{MeV}$, a deviation of -0.17%. The higher color-singlet levels $n = 4, 5, \ldots$ give the baryon excitation spectrum; its spectral table and fine structure are given in §13.

The three energy scales are given by the single formula (117) and three mode sets:

Scale	N	**Predicted**	**Experiment (deviation)**
Electroweak	$\varphi(7^4) - N_W^{\text{eff}} = 2039.75$	246.19 GeV	246.22 GeV (-0.011%)
Planck	$7^{41} \times 1.0271$	1.221×10^{19} GeV	1.2209×10^{19} GeV ($+0.04\%$)
Nucleon	$\zeta(1 + 1/\sqrt{3}) = 2.350$	937 MeV	938.3 MeV (-0.17%)

§11 Force Coupling Constants

§10 gave the energy scales of the $p = 7$ channel; this section computes the second set of numbers: the coupling constants of the five forces. §7.4 established, for general p, the coupling formula of the elementary resonance forces

$$\alpha_k = \frac{|R_k|^2}{N_k^{\mathrm{eff}}}, \qquad R_k(\beta) = \frac{L(\beta, \chi_k)}{\zeta(\beta)}, \tag{131}$$

§7.5 gave the joint coupling of the emergent resonance force, and §8.4 gave the metric-force constant. We now substitute $p = 7$ step by step: first the evaluation point and the numerical values of the resonance ratios (§11.1), then the Born channel numbers (§11.2), then the virtual-process corrections (§11.3) yielding the numerical values of the three elementary resonance forces (§11.4), and finally the emergent resonance force (§11.5) and the metric-force constant (§11.6).

§11.1 Vertex Amplitudes

The resonance ratio $R_k(\beta)$ in the coupling-constant formula (131) is a function of the energy-scale parameter β, so the numerical computation must first fix the evaluation point. The evaluation point is determined by the modular structure (§4.5): the KMS state undergoes a phase transition at $\beta = 1$ (§5.2), the physical point at which resonance occurs; the vertex is an object at the $\Delta^{1/2}$ level of the standard form (the modular matrix element connecting a mode with its conjugate mode, i.e., the vertex response of §4.5), and the Tomita–Takesaki polar decomposition $S = J\Delta^{1/2}$ gives the amplitude-level weight $n^{-\beta/2} = n^{-1/2}$; the evaluation point $\beta/2 = 1/2$ is forced by the uniqueness argument of §4.5. Physical reading: probability is the squared modulus of amplitude (the Born rule), and the amplitude level corresponding to the probability-level weight n^{-1} is $n^{-1/2}$. The vertex amplitudes are therefore given by the values of the L-functions at $\beta = \frac{1}{2}$, normalized by the vacuum amplitude $\zeta(\frac{1}{2})$:

$$R_k = \frac{L(\frac{1}{2}, \chi_k)}{\zeta(\frac{1}{2})}, \qquad \zeta(\tfrac{1}{2}) = -1.4604. \tag{132}$$

The L-values and resonance ratios of the six characters of $p = 7$ at $\beta = \frac{1}{2}$:

k	$\mathrm{ord}(\chi_k)$	$L(\frac{1}{2}, \chi_k)$	$\lvert R_k\rvert^2$
0	1	-0.9084	0.386928
1	6	$0.71394 + 0.47490i$	0.344760
2	3	$0.31009 - 0.07264i$	0.047562
3	2	1.14659	0.616448
4	3	$0.31009 + 0.07264i$	0.047562
5	6	$0.71394 - 0.47490i$	0.344760

The χ_0-direction contribution appearing in the virtual-process corrections is the linear resonance ratio $R_0 = 1 - 7^{-1/2} = 0.622$ (§7.4), not its squared modulus. These are the numerical values used in §10.1.

§11.2 Born Channel Numbers

The starting point for the denominator N_k^{eff} is the Born channel number– the total number of indistinguishable equivalent propagation paths in direction k. It is the product of three degeneracy factors:

$$N_i = |\ker(\chi_{k_i})| \cdot m \cdot c_i\,, \tag{133}$$

where $|\ker(\chi_k)| = (p-1)/\mathrm{ord}(\chi_k)$ is the path degeneracy of the character kernel (§7.4), $m = (p-1)/2 = 3$ is the coset degeneracy of the involution ι (three doublets, §6.2, §7.4), and $c_i = 1+[\chi_i = \bar{\chi}_i]$ is the self-conjugation doubling (for a real character the forward and backward directions are indistinguishable).

The choice of vertex character is determined by the distinction between the two character data of a force. A force involves two different characters, each answering a different question.

First, the **channel character**–the kind of force. An excitation of a force changes the CRT coordinates of a mode, and the channel is labeled by the **coordinate difference** (§7.2): a color-force excitation steps the $\mathbb{Z}/3\mathbb{Z}$ coordinate by ± 1, so its channel is $\chi_2 \oplus \chi_4$; a weak-force excitation flips the $\mathbb{Z}/2\mathbb{Z}$ coordinate, so its channel is χ_3; the cross-factor global resonance is $U(1)$. An excitation changing both coordinates simultaneously (a coordinate difference of type $\chi_1 \oplus \chi_5$) does not preserve the CRT factor structure and constitutes no independent channel (§7.2). This level is the content of the force table of §9.2.2 and answers which forces exist.

Second, the **vertex character**–how strongly the force couples. Resonance takes place on matter modes: the vertex amplitude is the response $|R_k|^2$ of the participating modes to the vacuum, with k the residence of the matter family participating in the force (the particle table of §9.2): the participants of the color resonance are the colored modes, of residence χ_1/χ_5, vertex $|R_1|^2$ ($|R_5| = |R_1|$, equal moduli within a Galois pair); the participants of the electromagnetic resonance are the purely charged modes (charged but colorless), of residence χ_2/χ_4, vertex $|R_2|^2$; the participants of the weak resonance are the purely weak modes (participating only in the weak force), of residence χ_3, vertex $|R_3|^2$.

The channel sits on the coordinate difference, the vertex on the coordinate: the former is determined by how the force is transmitted, the latter by the matter that participates in it; they are read off from the difference and the point of the same CRT structure, each uniquely. That the color-force channel is $\chi_2 \oplus \chi_4$ while its vertex is $|R_1|^2$ is the direct expression of gluons changing the color coordinate while quarks sit on the color coordinate.

Force	χ_k	ord	$\lvert\ker\rvert$	m	c	N_i
color	χ_1	6	1	3	1	3
electromagnetic	χ_2	3	2	3	1	6
weak	χ_3	2	3	3	2	18

The weak-force value $N_W = 18$ is the Born channel number used in §10.1. The Born-level couplings $\alpha_i^{(0)} = |R_{k_i}|^2/N_i$ do not yet include the virtual-process contributions and differ appreciably from the low-energy experimental values; we now correct them term by term.

§11.3 Virtual-Process Corrections

§7.4 established the general formula for the virtual-process corrections; the correction type is determined by the destination of the self-loop χ_i^2:

Force	Self-loop	Destination	Type
color	$\chi_1^2 = \chi_2$	electromagnetic sector	Type I
electromagnetic	$\chi_2^2 = \chi_4 = \bar{\chi}_2$	conjugate sector	Type I (phase reversal)
weak	$\chi_3^2 = \chi_0$	vacuum	Type II

Step 1: the Type I correction (color and electromagnetic forces). By the Type I formula of §7.4, $N_i^{\text{eff}} = N_i + V_0 - 2|R_i|^2$, where V_0 is the connected value of the background fluctuation (§7.4): the bare value $R_0 = 0.622$ contains a disconnected part $\Pi_{\text{nc}} = R_0 - |R_3|^2 \approx 0.006$, and after its removal $V_0 = |R_3|^2 = 0.616448$ (χ_3 being the unique Type II direction). Numerically:

$$\begin{aligned} \text{color:} \quad & N_s^{\text{eff}} = 3 + 0.616448 - 2 \times 0.344760 = 2.926928, \\ \text{electromagnetic:} \quad & N_{\text{EM}}^{\text{eff}} = 6 + 0.616448 - 2 \times 0.047562 = 6.521324. \end{aligned}$$

For the color force the self-energy slightly exceeds the fluctuation, so N decreases and the coupling strengthens; for the electromagnetic force the fluctuation far exceeds the self-energy, so N increases and the coupling weakens.

Step 2: the Type II correction (weak force). After the self-loop of χ_3 falls into the vacuum, the re-excitation of each sector is determined term by term by character multiplication:

- The color sector absorbs $-d_2|R_1|^2$: each of the $d_2 = 3$ color channels absorbs one share $|R_1|^2$ of propagation probability;
- The electromagnetic sector returns $+|R_2|^2$: the cross process $\chi_2\chi_3 = \chi_5$ returns one share of propagation probability through the electromagnetic channel;
- The weak force itself contributes $+c_3|R_3|^2$: for a real character ($\chi_3 = \bar{\chi}_3$) the forward and backward directions of the self-loop are indistinguishable, so $c_3 = 2$.

All three coefficients are determined by character identities, with no free choice:

$$N_3^{\text{eff}} = 18 - 3 \times 0.344760 + 0.047562 + 2 \times 0.616448 = 18.246178. \tag{134}$$

Step 3: the Dyson iteration [36] (electromagnetic force). $\chi_2^2 = \bar{\chi}_2$ is a phase reversal: in the corrected propagation environment the self-loop couples again, producing an alternating geometric series (§7.4). The first-order shift is $x = 3\alpha_W^{(0)} - 2\alpha_{\text{EM}}^{(0)}$: $+3\alpha_W^{(0)}$ comes from weak vacuum excitation ($\chi_3^2 = \chi_0$ opens virtual channels on the $m = 3$ cosets), $-2\alpha_{\text{EM}}^{(0)}$ from electromagnetic antiscreening ($\chi_2^2 = \bar{\chi}_2$ closes channels). The ratio of second order to first order is

$$r = \frac{\alpha_{\text{EM}}^{(0)}\left(2\alpha_W^{(0)} + \alpha_{\text{EM}}^{(0)}\right)}{3\alpha_W^{(0)} - 2\alpha_{\text{EM}}^{(0)}} = 0.006972, \tag{135}$$

with inputs taken as the Born-level bare couplings $\alpha_{\rm EM}^{(0)} = |R_2|^2/6$ and $\alpha_W^{(0)} = |R_3|^2/18$ (inputs of the correction layer are all taken at Born level, the same evaluation level as the mass phase correction of §12; for its derivation see §6.4). The iteration acts only on the virtual-process shift $\Delta N = N^{\rm eff} - N$:

$$N_{\rm EM}^{\rm full} = 6 + \frac{6.521324 - 6}{1.006972} = 6.517715. \tag{136}$$

The color and weak forces have no phase reversal, so $N^{\rm full} = N^{\rm eff}$. The same coupling thus splits into two levels: the Born-level bare value $\alpha^{(0)} = |R_k|^2/N_k$ (before correction) and the corrected level $\alpha = |R_k|^2/N^{\rm full}$ (matching the $q = 0$ observation); the mass phase correction of §12 takes the Born-level value.

§11.4 Numerical Values of the Elementary Resonance Forces

The corrected coupling constants $\alpha_i = |R_{k_i}|^2/N_i^{\rm full}$:

Force	Type	$N_i^{\rm full}$	$\lvert R_k\rvert^2$	$1/\alpha_i$	Experiment (deviation)
color	I	2.9269	0.344760	8.49	8.48 (+0.08%)
electromagnetic	I+Dyson	6.5177	0.047562	137.036	137.036 (+0.0003%)
weak	II	18.2462	0.616448	29.60	29.60 (−0.004%)

§11.5 The Emergent Resonance Force

The nuclear force is the force between emergent states. §7.5 gave the joint coupling of the emergent-state and elementary-state directions: the elementary-state direction supplies the resonance amplitude $|R_k|^2$, while the emergent-state direction supplies an inclusive summation over independent channels (enhancement rather than dilution).

Step 1: the amplitude. The purest participants exchanged between nucleons sit in the χ_1 direction, so the amplitude is shared with the color force: $|R_1|^2 = 0.3448$.

Step 2: the channel number. The emergent states sit in the χ_0 direction and do not distinguish unit-group elements; the totality of internal operations visible to them forms the affine group over $\mathbb{F}_7$:

$$\mathrm{Aff}(\mathbb{F}_7) = \{x \mapsto ax + b : \ a \in \mathbb{F}_7^*, \ b \in \mathbb{F}_7\}, \qquad |\mathrm{Aff}(\mathbb{F}_7)| = 42.$$

$|\mathrm{Aff}| = 42 \equiv 0 \pmod 7$ makes Maschke's theorem fail, and the Loewy-chain mechanism of §10.2 acts here as well: in the metric force $|\mathrm{Aff}|$ appears in the denominator, and the Ext gluing increases the effective number; in the nuclear force $|\mathrm{Aff}|$ appears in the numerator, and the same gluing decreases the effective number:

$$|\mathrm{Aff}|_{\rm eff} = \frac{|\mathrm{Aff}|}{\prod_{j=1}^{5} \mathcal{T}_j} = 42 \times \frac{2 \times 1.4604}{3} = 40.89. \tag{137}$$

Step 3: the coupling constant.

$$\boxed{\alpha_{\rm nuclear} = |\mathrm{Aff}|_{\rm eff} \times |R_1|^2 = 40.89 \times 0.3448 = 14.10.} \tag{138}$$

The experimental value is $g_{\pi NN}^2/(4\pi) = 14.0 \pm 0.3$, a deviation of +0.7%; the values before and after the Loewy correction are 14.48 (+3.5%) and 14.10 (+0.7%) respectively.

§11.6 The Metric-Force Constant

The coupling constant of the metric force is given by the formula of §8.4, with N taken as the value $N_{\rm Pl}^{\rm eff} = 7^{41} \times 1.0271$ determined in §10.2:

$$\boxed{G = \frac{C^*}{N_{\rm Pl}^{\rm eff} \cdot \log^2 N_{\rm Pl}^{\rm eff}} = 6.70 \times 10^{-39}\ \mathrm{GeV}^{-2}.} \tag{139}$$

The experimental value is $G_N^{\rm exp} = 6.71 \times 10^{-39}\,\mathrm{GeV}^{-2}$[46], a deviation of -0.09%.

The coupling constants of all five forces are given by the resonance ratios and channel counting:

Force	Formula	Predicted	Experiment (deviation)
color	$\lvert R_1\rvert^2/N_s^{\rm full}$	$\alpha_s^{-1} = 8.49$	8.48 (+0.08%)
electromagnetic	$\lvert R_2\rvert^2/N_{\rm EM}^{\rm full}$	$\alpha_{\rm EM}^{-1} = 137.036$	137.036 (+0.0003%)
weak	$\lvert R_3\rvert^2/N_3^{\rm full}$	$\alpha_W^{-1} = 29.60$	29.60 (−0.004%)
nuclear	$\lvert \mathrm{Aff}\rvert_{\rm eff}\,\lvert R_1\rvert^2$	$\alpha_{\rm nuclear} = 14.10$	14.0 ± 0.3 (+0.7%)
gravity	$C^*/(N_{\rm Pl}^{\rm eff} \log^2 N_{\rm Pl}^{\rm eff})$	$G = 6.70 \times 10^{-39}$	6.71×10^{-39} (−0.09%)

§12 Mass Spectrum

After the energy scales (§10) and the coupling constants (§11), this section computes the last set of numbers: the mass spectrum. §6.4 established the unified mass formula for the elementary states in the unit-group directions,

$$m_f = v \times \prod_k |L(\tfrac{1}{2}, \chi_{s_k})|^{G_k \cdot C_k} \times \prod_k \mathcal{T}_{s_k} \times \mathrm{Gen}_f^{(g)}, \tag{140}$$

the factorization of the KMS evaluation $m = v \cdot \varphi_\beta(P_f)$ along the transition path: the product of the L-values at the stations (with exponent Galois degree $\times$ covering number), the node normalization factors $\mathcal{T}$, and the generational factor Gen. For the emergent states in the full-group direction, §6.5 gives

$$m_{\mathrm{em}} = \sqrt{\frac{\zeta(\beta) \cdot \log^2 \zeta(\beta)}{C^*}}. \tag{141}$$

The numerical inputs all come from the preceding sections: the scale parameter $v = 246.19\,\mathrm{GeV}$ (§10.1), the L-function values from the table of §11.1, and the four kinds of node-factor values–Galois projection $\div\sqrt{d_1}$, color counting $\times d_2$, background normalization $\div|\zeta(\frac{1}{2})|$, vacuum dilution $\div N$–from §6.4. The generational factors are given by the lattice projection

$$\rho_j = \left(\frac{f_j}{f_0}\right)^2, \qquad f_j = 1 + \sqrt{G}\,\cos\!\left(\delta + \frac{2\pi j}{m}\right) \tag{142}$$

(§6.4; $G = 2$ is the Galois degree, $m = 3$ the generation number). The phase δ is determined in two steps.

The bare phase. For a character χ define the Gauss sum and the Jacobi sum

$$\tau(\chi) = \sum_{a=1}^{6} \chi(a)\, e^{2\pi i a/7}, \qquad J(\chi, \chi) = \frac{\tau(\chi)^2}{\tau(\chi^2)}.$$

$J(\chi_2, \chi_2)$ has the exact value $-1 - d_2\,\omega_3$ ($\omega_3 = e^{2\pi i/3}$), so $|J|^2 = 7$ and $\arg J = -\arctan(d_2^{3/2})$. The functional equation at the central point gives $\arg L(\frac{1}{2}, \chi) = \frac{1}{2}\arg\tau(\chi)$; from $\chi_2^2 = \bar\chi_2$ and $\tau(\chi_2)\tau(\bar\chi_2) = 7$ one gets $\arg J = 3\arg\tau(\chi_2)$, hence

$$\delta_2 = \arg L(\tfrac{1}{2}, \chi_2) = \frac{\arg J(\chi_2, \chi_2)}{\varphi(7)} = \frac{-\arctan(d_2^{3/2})}{2d_2} = -13.18°. \tag{143}$$

The electromagnetic self-energy correction. The mass phase of a charged particle acquires one share of the Born-level bare coupling:

$$\delta = \delta_2 + \alpha_{\mathrm{EM}}^{(0)}, \qquad \alpha_{\mathrm{EM}}^{(0)} = \frac{|R_2|^2}{N_{\mathrm{EM}}} = \frac{0.0476}{6} = 0.0079\ (= 0.454°), \tag{144}$$

i.e., $\delta = -12.73°$. The form of the correction term is fixed by the following four steps. First, the channel is unique: the transition generator connecting the two members of a chirality pair is uniquely determined by the selection rule ($\chi_2 \cdot \chi_2 = \chi_4$, $\chi_4 \cdot \chi_4 = \chi_2$), i.e., the electromagnetic self-loop $\chi_2^2 = \bar\chi_2$ of §7.4– the self-energy insertion and the chirality cross coupling are one and the same channel. Second, the insertion operator is antidiagonal in the $\{\chi_2, \bar{\chi}_2\}$ basis, with vanishing diagonal matrix elements: the first-order correction vanishes, and the lowest nonvanishing correction is second order with two vertices, each insertion carrying one share of $\alpha_{\rm EM}^{(0)}$ (inputs of the correction layer are taken at Born level, as in §11.3). Third, the normalized insertion operator is an involution ($\hat{V}^2 = 1$), whose eigenbasis is precisely the symmetrized combinations $\chi_2^{(\pm)}$ (§6.4.2)– an insertion operator with vanishing diagonal in the flavor basis diagonalizes in this basis, with eigenvalues ±1: the mass eigenstates are eigenstates of the insertion operator, and the insertion flow (the sequence of insertions acting continuously along the dwell segment) only accumulates phase on them without changing the modulus– the modulus correction is already accounted for by $N^{\rm eff}$ (§11.3); the two are the two parts of the polar decomposition of one and the same correction. The n insertions within one dwell segment are indistinguishable, so the counting divides by $n!$, and the series sums to $e^{i\alpha}$: the phase is strictly linear in $\alpha_{\rm EM}^{(0)}$, with coefficient 1. Fourth, the sign: the entry station must be an even character (§6.4.3), and the background coupling selects the symmetric combination, i.e., the $+1$ eigenvector, so the phase is $+\alpha_{\rm EM}^{(0)}$.

Fifth, the second-order correction. The generator multiplication of the insertion pair yields a unique second-order channel $\chi_2 \cdot \bar{\chi}_2 = \chi_0$: the virtual quantum pair returns through the trivial direction. χ_0 carries no doublet label and is the uniform vector in the m-dimensional generational evaluation space, so the amplitude component returning to the generation fixed by the endpoints is $1/\sqrt{m}$; the χ_0 direction carries no insertion flow (the same clause as the electric-neutrality exemption), and no phase accumulates during that segment of the propagation. Combining first and second order, the effective phase rate per dwell cycle is

$$\alpha_{\rm eff} = \alpha_{\rm EM}^{(0)} \Big(1 - \frac{\alpha_{\rm EM}^{(0)}}{\sqrt{m}}\Big) = 0.0078907, \tag{145}$$

and the phase correction and the frequency shift (next paragraph) are evaluated at the same rate: $\delta = \delta_2 + \alpha_{\rm eff} = -12.732°$. The electrically neutral neutrino has no electromagnetic insertion and carries none of these terms. Numerically:

$$\rho_0 = 1, \qquad \rho_1 = 0.0595, \qquad \rho_2 = 2.88 \times 10^{-4}.$$

Frequency shift. The phase accumulation of the third through fifth steps proceeds in units of the dwell cycle: the eigenmode dwells continuously under the time evolution, acquiring the phase $\alpha_{\rm eff}$ per cycle, so the phase grows linearly in the evolution parameter–a frequency shift. In the eigenmode phase $e^{i\omega t}$, ω points in the negative energy-level direction; the added phase has the opposite sign (fourth step), so the frequency reading decreases:

$$m_{\rm obs} = m_{\rm arith} \times \big(1 - \alpha_{\rm eff}\big), \tag{146}$$

where $m_{\rm arith}$ is the arithmetic value of Eq. (140). This factor and the phase correction $\delta = \delta_2 + \alpha_{\rm eff}$ live at two different layers: the latter enters the lattice projection ρ_j and changes only the intergenerational ratios; the former multiplies the overall scale and changes no ratio. Its range of applicability is the eigenmodes of the convolution operator that carry an insertion flow: the electrically neutral neutrino has no insertion flow and does not carry this factor; the emergent states and channel excitations are not eigenmodes of the convolution operator– their masses are read out from the energy-scale formula (§10.3) and from weight equipartition (§14.3) respectively, and are not in this class; the mass formulas of the charged elementary states below all contain this factor; the factor columns of the individual tables list only the spectral weight, the node factor, and the generational factor, without this factor.

The two-component projection chain of the locked-link phase (§6.4) gives the Cabibbo angle [37]. The phase is a readout of the decoherent segment, accumulates no insertion phase, and takes the bare value δ_2; the modulus of the readout is a real transmission in the post-transition vacuum and takes the corrected-level value (§6.4, Eq. (61)):

$$\lambda = \tfrac{1}{2}|\sin 2\delta_2| \times \left(1 + \tfrac{d_2}{2}\,\alpha_{\mathrm{EM}}^{(0)}\right) = 0.2221 \times 1.0119 = 0.2247\,. \tag{147}$$

We now compute the stable states class by class.

§12.1 Charged Leptons

The charged leptons sit in the χ_2 direction (order 3, CRT coordinates $(0,1)$), with transition path $\chi_0 \to \chi_2$. χ_2 is a complex character ($G = 2$), and the residence path forms a single probability closure ($C = 2$), so the spectral weight is

$$|L_2|^4 = (0.3185)^4 = 1.029 \times 10^{-2};$$

the CRT coordinate $k_1 = 0$ triggers the Galois projection, giving the node factor $\mathcal{T} = 1/\sqrt{d_1} = 0.707$; the three generations are distinguished by the lattice projections ρ_0, ρ_1, ρ_2, with τ as the reference generation. Masses:

$$m_\tau = v \times |L_2|^4 \times \tfrac{1}{\sqrt{2}} \times \rho_0 \times \left(1 - \alpha_{\mathrm{eff}}\right) = 1776.93\,\mathrm{MeV}\,, \tag{148}$$
$$m_\mu = m_\tau \times \rho_1 = 105.66\,\mathrm{MeV}\,, \tag{149}$$
$$m_e = m_\tau \times \rho_2 = 0.5110\,\mathrm{MeV}\,. \tag{150}$$

Particle	Spectral weight	Node factor	Gen. factor	Predicted	Experiment (dev.)
τ	$\lvert L_2\rvert^4$	$1/\sqrt{d_1}$	ρ_0	1776.93 MeV	1776.93 MeV (-0.0002%)
μ	$\lvert L_2\rvert^4$	$1/\sqrt{d_1}$	ρ_1	105.655 MeV	105.658 MeV (-0.003%)
e	$\lvert L_2\rvert^4$	$1/\sqrt{d_1}$	ρ_2	0.51101 MeV	0.51100 MeV ($+0.002\%$)

Two structural checks (§6.4.5, §6.4.7) are completed numerically here. First, the conjugate connection of the vertex: replacing it by the diagonal evaluation $\langle\chi^{(+)}|K|\chi^{(+)}\rangle = \mathrm{Re}\,L_2$ turns the spectral weight into $(\mathrm{Re}\,L_2)^4$, and the deviation of m_τ reaches -10%; the conjugate connection gives $|L_2|^4$, with deviation -0.0002%: the diagonal choice is excluded. Second, the unit weight of the χ_0 component of the response vector: scanning w in $f_j = w + \sqrt{G}\cos\theta_j$, the masses of μ and e are reproduced exactly at $w = 1.00003$ and $w = 1.0000$ respectively, so two independent observations fix $w = 1$ simultaneously.

§12.2 Down-Type Quarks

The down-type quarks sit in the χ_5 direction (order 6, CRT coordinates $(1,2)$; §9.2). The generator decomposition $(1,2)=(1,0)+(0,2)$ is unique (§6.4), and the path is the open path $\chi_0\to\chi_2\to\chi_3$ ($\chi_2\cdot\chi_3=\chi_5$: the residence is reached by the combination of the generators; the entry takes the even generator χ_2). Spectral weight: the χ_2 station contributes $|L_2|^4$ (the same as for the leptons); for b (the reference generation) the path is coherent, the readout of the final generator χ_3 of the open path is a vertex response, with covering number $C=1$ (exponent 1); for s,d the intergenerational mediator χ_3 lies on the path, the propagation breaks into decoherent segments (§6.4), each segment covered once, so $C=2$ and the exponent is 2.

Node factors: b is coherent, so the endpoint zero-component factors take effect– the Galois projection of the entry χ_2 ($k_1=0$) and the color counting and background normalization of the exit χ_3 ($k_2=0$),

$$\mathcal{T}_b=\frac{d_2}{\sqrt{d_1}\,|\zeta(\frac{1}{2})|}=\frac{3}{\sqrt{2}\times 1.4604}=1.452;$$

s,d are decoherent segments; the segment contains no $\Delta^{1/2}$ vertex, so the entry projection and the color counting are exempted simultaneously, each segment retaining only one share of background normalization (§6.4):

$$\mathcal{T}_{s,d}=\frac{1}{|\zeta(\frac{1}{2})|^2}=0.469.$$

Generational factors: the two-station path occupies one dimension, leaving $m-1=2$ free lattice sites (§6.4)– b takes ρ_0, s takes ρ_1; d exceeds the quota, borne by the Born readout of the locked link, taking $\rho_1\lambda^2$ ($\lambda=0.2247$ is the corrected-level value of the link readout, Eq. (147)). Masses:

$$m_b=v\times|L_2|^4L_3\times\mathcal{T}_b\times\rho_0\times\big(1-\alpha_{\text{eff}}\big)=4185\,\text{MeV}\,, \tag{151}$$
$$m_s=v\times|L_2|^4L_3^2\times\mathcal{T}_{s,d}\times\rho_1\times\big(1-\alpha_{\text{eff}}\big)=92.1\,\text{MeV}\,, \tag{152}$$
$$m_d=v\times|L_2|^4L_3^2\times\mathcal{T}_{s,d}\times\rho_1\lambda^2\times\big(1-\alpha_{\text{eff}}\big)=4.65\,\text{MeV}\,. \tag{153}$$

Particle	Spectral weight	Node factor	Gen. factor	Predicted	Experiment (dev.)
b	$\lvert L_2\rvert^4L_3$	$d_2/(\sqrt{d_1}\lvert\zeta\rvert)$	ρ_0	4185 MeV	4183 MeV (+0.05%)
s	$\lvert L_2\rvert^4L_3^2$	$1/\lvert\zeta\rvert^2$	ρ_1	92.1 MeV	93.5 MeV (−1.5%)
d	$\lvert L_2\rvert^4L_3^2$	$1/\lvert\zeta\rvert^2$	$\rho_1\lambda^2$	4.65 MeV	4.70 MeV (−1.1%)

The mass ratio of b to τ is the Georgi–Jarlskog factor: $m_b/m_\tau=d_2L_3/|\zeta(\frac{1}{2})|=3\times 1.1466/1.4604=2.355$, experimental value 2.354, deviation +0.1%.

§12.3 Up-Type Quarks

The up-type quarks sit in the χ_1 direction (order 6, CRT coordinates $(1,1)$). The generator decomposition $(1,1)=(1,0)+(0,1)$ is unique, with generators χ_3 and $\chi_4=\bar{\chi}_2$ ($\chi_4\cdot\chi_3=\chi_1$), and the path is the closed path $\chi_0\to\chi_2\to\chi_3\to\chi_1$, ending at the residence (§6.4); the closed path forms a single probability closure, all stations have $C = 2$, and the spectral weight passes through all three L-values:

$$|L_2|^4 \cdot L_3^2 \cdot |L_1|^4, \qquad |L_1|^4 = (0.8575)^4 = 0.5406$$

($|L_4| = |L_2|$, equal moduli within a Galois pair; the real-character station χ_3 has $G = 1$, $C = 2$, giving L_3^2). Node factors: the Galois projection of the entry χ_2 ($k_1 = 0$); χ_3 is an intermediate station, transparent with no factor (§6.4); the exit χ_1 has both coordinates nonzero, factor 1: $\mathcal{T} = 1/\sqrt{d_1}$.

Generational factors: the three-station path occupies two dimensions, leaving $m - 2 = 1$ free lattice site, and the two surplus generations are borne by the self-loop channels of the two links (§6.4(b), the four cases)– c is the reference, ρ_0; t goes through the closed self-loop of the χ_2 link ($\chi_2^2 = \bar{\chi}_2$), with dwell enhancement $\div\alpha_{\mathrm{EM}}$; u goes through the open self-loop of the χ_3 link ($\chi_3^2 = \chi_0$), with emission cost $\times\alpha_W$ and flip readout $\times\lambda^2$. Masses:

$$m_c = v \times |L_2|^4 L_3^2 |L_1|^4 \times \tfrac{1}{\sqrt{2}} \times \rho_0 \times \big(1 - \alpha_{\mathrm{eff}}\big) = 1263\,\mathrm{MeV}\,, \tag{154}$$

$$m_t = m_c/\alpha_{\mathrm{EM}} = 173.0\,\mathrm{GeV}\,, \tag{155}$$

$$m_u = m_c \times \lambda^2 \alpha_W = 2.15\,\mathrm{MeV}\,. \tag{156}$$

Particle	Spectral weight	Node factor	Gen. factor	Predicted	Experiment (dev.)
c	$\lvert L_2\rvert^4 L_3^2 \lvert L_1\rvert^4$	$1/\sqrt{d_1}$	ρ_0	1263 MeV	1273 MeV (−0.8%)
t	$\lvert L_2\rvert^4 L_3^2 \lvert L_1\rvert^4$	$1/\sqrt{d_1}$	$\rho_0/\alpha_{\mathrm{EM}}$	173.0 GeV	172.57 GeV (+0.3%)
u	$\lvert L_2\rvert^4 L_3^2 \lvert L_1\rvert^4$	$1/\sqrt{d_1}$	$\rho_0\lambda^2\alpha_W$	2.15 MeV	2.16 MeV (−0.3%)

In the table, the c and b rows are compared with the $\overline{\mathrm{MS}}$-scheme masses $m_Q(m_Q)$, while the t row is compared with the direct measurement [45]; the two schemes differ. The factors of the two surplus generations provide a zero-free-parameter cross-check: in the ratio $m_t\, m_u/m_c^2 = \lambda^2\alpha_W/\alpha_{\mathrm{EM}}$, α_{EM} and α_W are taken from §11 and λ from Eq. (147), none of them fit parameters of the mass spectrum; the prediction is $m_c^2\lambda^2\alpha_W/(m_t\, m_u\, \alpha_{\mathrm{EM}}) = 1$, experimental value 1.02.

§12.4 Neutrinos

The neutrinos sit in the indirect channel of the χ_3 direction. χ_3 is a real character; after doublet symmetrization the seed cancels exactly (§6.4), the direct channel is closed, and the mass must be acquired through the indirect projection relayed by χ_0.

Spectral weight: the χ_0 relay makes the signal pass through the χ_2 intermediate station twice, each passage counting one probability closure, so the covering number is $C = 2 + 2 = 4$ and the exponent $G \cdot C = 8$:

$$|L_2|^8 = 1.06 \times 10^{-4}.$$

Node factors: the χ_2 station is passed twice, Galois projection $(\div\sqrt{d_1})^2 = \div d_1$; the χ_0 station is passed twice, vacuum dilution $(\div N)^2$, with $N = \varphi(7^4) = 2058$ (§10.1):

$$\mathcal{T}_\nu = \frac{1}{d_1 N^2} = \frac{1}{2 \times (2058)^2} = 1.18 \times 10^{-7}.$$

The N^2 dilution is the origin of the extreme smallness of the neutrino masses: the signal passes through the vacuum twice, each time shared among 2058 modes.

Generational factors: the indirect channel has no coherent path, so the three generations are not given by the lattice projections ρ_j but determined by the doublet chain weights (§6.4(c)). The generators f_j of the circulant matrix are taken from the χ_2 intermediate station ($G = 2$, of the same origin as the spectral weight): all coherent structure of the indirect channel is carried by the relay path. The eigenvalues of the circulant matrix are invariant under translation of the phase δ–the electromagnetic phase correction does not enter the neutrino intergenerational ratios, consistent with the electric-neutrality exemption. The chain weights take exponent $m = 3$:

$$w_k = \zeta_{a_k}(3)\,\zeta_{7-a_k}(3) = \frac{1}{7^6}\,\zeta_{\mathrm{H}}\Big(3, \frac{a_k}{7}\Big)\,\zeta_{\mathrm{H}}\Big(3, \frac{7-a_k}{7}\Big);$$

the condition $V_k^3 \equiv -1 \pmod 7$ on the doublet products $V_k = a_k(7-a_k) \in \{6, 10, 12\}$ holds identically by Euler's criterion ($7 \equiv 3 \bmod 4$, §6.4(c)). The circulant matrix $\mathcal{C} = \mathrm{circ}(f_0, f_2, f_1)$ and the chain weights jointly give the mass matrix $\mathcal{M} = \mathcal{C}\,\mathrm{diag}(w_1, w_2, w_3)\,\mathcal{C}^T$, with eigenvalues

$$\varepsilon_1 = 3.296 \times 10^{-3}, \qquad \varepsilon_2 = 6.476 \times 10^{-3}, \qquad \varepsilon_3 = 3.292 \times 10^{-2}.$$

$\chi_3 = \bar{\chi}_3$ makes the forward and return paths indistinguishable, so the generational factor acquires the Majorana symmetry factor $\frac{1}{2}$. Masses:

$$m_{\nu_k} = v \times |L_2|^8 \times \frac{1}{d_1 N^2} \times \frac{\varepsilon_k}{2}\,. \tag{157}$$

Particle	Spectral weight	Node factor	Gen. factor	Predicted	Experiment (dev.)
ν_3	$\lvert L_2\rvert^8$	$1/(d_1 N^2)$	$\varepsilon_3/2$	50.6 meV	∼50 meV (+1.3%)
ν_2	$\lvert L_2\rvert^8$	$1/(d_1 N^2)$	$\varepsilon_2/2$	9.96 meV	—
ν_1	$\lvert L_2\rvert^8$	$1/(d_1 N^2)$	$\varepsilon_1/2$	5.07 meV	—

The absolute mass sum is $\Sigma m_\nu = 65.7$ meV, within the sensitivity of CMB-S4; the ratio of mass-squared differences is $\Delta m^2_{31}/\Delta m^2_{21} = 34.5$, experimental value 33.8 ± 0.9 [49], a deviation of 0.8σ.

§12.5 Emergent States

The masses of the emergent states are given by Eq. (141), computed in three steps (derivation in §10.3).

Step 1: the scale parameter β. In the KMS state the leading order of the energy variance of the Hamiltonian $H = \log n$ is $1/(\beta-1)^2$; the antisymmetric combination of n colored modes forms a color-singlet emergent state, whose leading-order quantum number equals the mode number n, and the quantization condition $n(\beta-1)^2 = 1$ gives

$$\beta(n) = 1 + \frac{1}{\sqrt{n}}\,. \tag{158}$$

Step 2: the lowest color singlet. The color-singlet condition $\sum_i c(a_i) \equiv 0 \pmod 3$ together with total antisymmetry ε^{abc} requires one mode of each of the three colors (§13.2), so the lowest level is $n = d_2 = 3$:

$$\beta_B = 1 + \frac{1}{\sqrt{3}} = 1.5774, \qquad \zeta(\beta_B) = 2.350.$$

Step 3: the mass. Substituting the spectral weight $N_{\chi_0} = \zeta(\beta_B)$ into (141):

$$m_p = \sqrt{\frac{\zeta(\beta_B)\,\log^2 \zeta(\beta_B)}{C^*}} = \sqrt{\frac{2.350 \times (\log 2.350)^2}{1.9543783}} = 937 \text{ MeV}, \qquad (159)$$

experimental value 938.3 MeV, a deviation of -0.17%. The neutron–proton mass difference and the baryon excitation spectrum at the higher color-singlet levels $n = 4, 5, \ldots$, together with the fine structure, are given in §13.

§12.6 Summary Table of the Mass Spectrum

Comparison of all masses of the two kinds of stable states:

Particle	Spectral weight	Node factor	Gen. factor	Predicted	Experiment (dev.)
τ	$\|L_2\|^4$	$1/\sqrt{d_1}$	ρ_0	1776.93 MeV	1776.93 (-0.0002%)
μ	$\|L_2\|^4$	$1/\sqrt{d_1}$	ρ_1	105.655 MeV	105.658 (-0.003%)
e	$\|L_2\|^4$	$1/\sqrt{d_1}$	ρ_2	0.51101 MeV	0.51100 ($+0.002\%$)
b	$\|L_2\|^4 L_3$	$d_2/(\sqrt{d_1}\|\zeta\|)$	ρ_0	4185 MeV	4183 ($+0.05\%$)
s	$\|L_2\|^4 L_3^2$	$1/\|\zeta\|^2$	ρ_1	92.1 MeV	93.5 (-1.5%)
d	$\|L_2\|^4 L_3^2$	$1/\|\zeta\|^2$	$\rho_1\lambda^2$	4.65 MeV	4.70 (-1.1%)
c	$\|L_2\|^4 L_3^2 \|L_1\|^4$	$1/\sqrt{d_1}$	ρ_0	1263 MeV	1273 (-0.8%)
t	$\|L_2\|^4 L_3^2 \|L_1\|^4$	$1/\sqrt{d_1}$	$\rho_0/\alpha_{\text{EM}}$	173.0 GeV	172.57 ($+0.3\%$)
u	$\|L_2\|^4 L_3^2 \|L_1\|^4$	$1/\sqrt{d_1}$	$\rho_0\lambda^2\alpha_W$	2.15 MeV	2.16 (-0.3%)
ν_3	$\|L_2\|^8$	$1/(d_1 N^2)$	$\varepsilon_3/2$	50.6 meV	~50 ($+1.3\%$)
ν_2	$\|L_2\|^8$	$1/(d_1 N^2)$	$\varepsilon_2/2$	9.96 meV	—
ν_1	$\|L_2\|^8$	$1/(d_1 N^2)$	$\varepsilon_1/2$	5.07 meV	—
p	$\zeta(\beta_B)$	$n = 3,\ \beta_B = 1 + 1/\sqrt{3}$		937 MeV	938.3 (-0.17%)

The deviations of the nine charged fermions do not exceed 1.5%, with the three charged leptons at experimental precision; the absolute neutrino mass $\Sigma m_\nu = 65.7$ meV is a prediction awaiting verification. With this, the masses of both kinds of stable states of §6–elementary and emergent–have all been computed.

§13 Baryons, Quarks, and Color Confinement

§9.2 identified baryons with the emergent states in the χ_0 direction, and §10–§12 obtained, under this correspondence, the nucleon energy scale, the baryon excitation spectrum, and the nuclear-force coupling. The standard-physics description of baryons is built on a different picture: a baryon is composed of three quarks and is a composite particle; about 99% of its mass does not come from its constituents; a single quark has never been observed (color confinement); the nuclear force is positioned as a residue of the strong force, not among the fundamental forces. This chapter argues: the baryon is a fundamental particle–a single collective condensate in the χ_0 direction, whose mass and stability are given by the condensate as a whole and are not decomposable; the "three quarks" are its internal projection under the color-singlet condition, a readout of decomposition rather than pre-existing constituents; the identities of color confinement and the nuclear force are thereby fixed. The criteria for a fundamental particle are three: it is a single state, not a bound system of other particles; its mass is given by the whole itself and is not decomposable into a sum of parts; its stability is protected by its own structure and does not depend on an externally imposed conservation law. In what follows, the collective structure in the χ_0 direction (§6.2) is called the condensate layer, and the elementary-state structure in the nontrivial character directions (§6.2–§6.3) the constituent layer.

§13.1 The Direction Selection of Emergence

This section determines the location and composition of the emergent state in the state space; the argument proceeds in three steps.

Step 1: the emergent direction is unique. §6.5 defined the spectral weight of the character direction χ_k,

$$N_{\chi_k}(\beta) \;=\; \sum_n \chi_k(n)\, n^{-\beta}, \tag{160}$$

and the criterion for emergence is a divergent spectral weight (§6.5). Computing (160) direction by direction: for $k \neq 0$ the sum is restricted to the unit group,

$$N_{\chi_k}(\beta) = L(\beta, \chi_k), \qquad \lim_{\beta \to 1^+} L(\beta, \chi_k) = L(1, \chi_k) \in (0, \infty) \tag{161}$$

(Dirichlet's theorem: $L(1, \chi)$ is finite and nonzero for nontrivial characters); for $k = 0$, χ_0 performs no sieving on the full group, and the spectral weight is the partition function itself,

$$N_{\chi_0}(\beta) = Z(\beta) \xrightarrow{K\to\infty} \zeta(\beta) = \frac{1}{\beta - 1} + \gamma + O(\beta - 1) \xrightarrow{\beta\to 1^+} \infty. \tag{162}$$

Comparing (161) with (162): the divergent spectral weight occurs in one and only one direction, χ_0. The equivalent global statement is the Artin factorization $\zeta_{\mathbb{Q}(\mu_7)}(\beta) = \zeta(\beta) \prod_{j=1}^{5} L(\beta, \chi_j)$– all divergence is carried by the ζ factor. The emergent state resides in the single direction χ_0.

Step 2: there is no single-mode excitation in the χ_0 direction. A stable single-mode excitation is an eigenvector of K (§6.1): the spectrum of K is simple, and the eigendirections are exactly $\tilde{\chi}_0, \tilde{\chi}_1, \ldots, \tilde{\chi}_5$. Among them $\tilde{\chi}_0$ is the identity function on the unit group– the direction of the KMS background itself; an excitation must be orthogonal to the background, so $\tilde{\chi}_0$ does not represent an excitation. The remaining $p-2$ nontrivial directions constitute the elementary-state spectrum (§6.1), whose spectral weights are all finite by (161). Therefore the nontrivial content of the χ_0 direction has no single-mode realization and can only be a collective degree of freedom of the full group (§6.2).

Step 3: combining. Emergence can only occur at χ_0, and χ_0 carries no single mode, so the emergent state is formed by the collective condensation of the lattice modes of the full group, and the projection of the condensing combination onto the χ_0 direction must be nonzero. What participates in the condensation are the lattice modes of the state space, not any particles: the emergent state is a collective condensate in the χ_0 direction, not a bound system–the first criterion holds. The internal structure of this state is fixed by the projection condition; see the next section.

§13.2 The Color-Singlet Condition

The internal structure of the emergent state is constrained by the projection condition in the χ_0 direction. This section gives the projection condition and determines the lowest combination that satisfies it.

Each lattice point $a \in (\mathbb{Z}/7\mathbb{Z})^*$ decomposes in CRT coordinates: the $\mathbb{Z}/3\mathbb{Z}$ coordinate is written as the color $c(a)$, and the $\mathbb{Z}/2\mathbb{Z}$ coordinate (the involution $\iota(a) = 7-a$) gives the spin projection (§6.2):

Color $c \in \mathbb{Z}/3\mathbb{Z}$	Elements $\{a, 7-a\}$	Spin projection
0	$\{1, 6\}$	$(+\frac{1}{2}, -\frac{1}{2})$
1	$\{3, 4\}$	$(-\frac{1}{2}, +\frac{1}{2})$
2	$\{2, 5\}$	$(+\frac{1}{2}, -\frac{1}{2})$

Each color sector contains exactly two elements, interchanged by the involution.

Step 1: the projection condition. Suppose the condensing combination consists of n modes $a_1, \ldots, a_n$, with total color charge $C = \sum_i c(a_i)$. The generator t of $\mathbb{Z}/3\mathbb{Z}$ multiplies the combined wavefunction by the phase ω^C ($\omega = e^{2\pi i/3}$), and the projection onto the χ_0 direction is the average over $\mathbb{Z}/3\mathbb{Z}$:

$$P_{\chi_0}\Psi \;\propto\; \frac{1}{3}\sum_{t=0}^{2}\omega^{tC}\,\Psi \;=\; \delta_{C\equiv 0 \pmod 3}\,\Psi. \tag{163}$$

The projection of the combination onto the χ_0 direction is nonzero if and only if

$$\boxed{\sum_{i=1}^{n} c(a_i) \equiv 0 \pmod 3.} \tag{164}$$

A combination satisfying (164) is called a **color singlet**; a combination with $C \not\equiv 0$ falls entirely in colored directions and cannot emerge by §13.1. The emergent state must be a color singlet.

Step 2: excluding $n = 1, 2$. Checking case by case via (164):

- $n = 1$: $c \neq 0$ violates (164); $c = 0$ is a single mode, already excluded by Step 2 of §13.1;
- $n = 2$: $c_1 + c_2 \equiv 0$ requires $c_2 = -c_1$, and in $\mathbb{Z}/3\mathbb{Z}$ one has $-1 \equiv 2$ and $-2 \equiv 1$, so the solutions are $(0, 0)$, $(1, 2)$, $(2, 1)$; $(0, 0)$ is confined to a single sector (two modes already exhaust that sector), the $(1, 2)$ type is a color–anticolor pair, satisfying color neutrality, but it cannot simultaneously satisfy the antisymmetrization requirement in color and the symmetrization requirement in spin (next step); it corresponds to meson-type combinations, which are not in the lowest stable emergent sequence;
- $n = 3$: $(c_1, c_2, c_3) = (0, 1, 2)$ satisfies (164); a same-color triple (c, c, c), though satisfying $3c \equiv 0$, is excluded because each color sector has only two elements, so three same-color modes cannot be drawn.

Covering all three colors, one per color, is the unique realization at $n = 3$, and also the lowest combination containing all three colors.

Step 3: antisymmetry. The spin of a mode is half-integer (the $\mathbb{Z}/2\mathbb{Z}$ coordinate, §6.2), obeying Fermi statistics, so the $n = 3$ combined wavefunction is antisymmetric under mode exchange; the color part with one of each of the three colors has only one antisymmetric tensor–ε^{abc}–so the color wavefunction is ε^{abc} and the spin–flavor part is symmetric. Combining the three steps:

$$n \geq 3, \qquad n = 3: \text{ one per color, color-antisymmetric } \varepsilon^{abc}. \tag{165}$$

Decomposing this color singlet along the constituent directions, the mode content at each color position is read out via the correspondence of §9.2 (the χ_1 direction is written u, the χ_5 direction d): the readouts of the two lowest combinations are uud and udd, i.e., the proton and the neutron. This decomposition readout is exactly what standard physics calls the "three quarks": the color wavefunction of $\varepsilon^{abc} q^a q^b q^c$ is the same structure as (165). What the color-singlet condition gives is the shape that the internal projection must present–three colors, three shares– not a list of constituents: having internal structure is not the same as being a composite particle. The attribution of mass, quantum numbers, and lifetime is determined by the two-layer structure of the next section.

§13.3 Mass, Quantum Numbers, and Stability

The attribution of mass, quantum numbers, and stability is determined by one structural fact about the condensate layer and the constituent layer: the two layers do not interpenetrate. The argument runs in two directions.

First, the structure of the constituent directions cannot enter χ_0. The matrix element of the resonance coupling is given by the selection rule of §7.1: the coupling matrix element of a translation a acting on the direction χ_k is

$$g_k(a) \ \propto \ \chi_k(a) - 1, \tag{166}$$

and there is no force between frequency-mismatched modes. Along the χ_0 direction, $\chi_0(a) \equiv 1$ holds for all a,

$$g_0(a) \ \propto \ \chi_0(a) - 1 \ \equiv \ 0 : \tag{167}$$

the entire gauge structure of the constituent directions–charge, flavor, transitions– has identically vanishing matrix elements in the χ_0 direction.

Second, quantities on χ_0 can only come from the condensate. Mass is the KMS evaluation on the full projection (§6.4); the projection of the emergent state is the full-group χ_0-direction projection P_{em} (§6.5),

$$\varphi_\beta(P_{\mathrm{em}}) \;=\; \frac{1}{Z(\beta)} \sum_{n=1}^{p^K} n^{-\beta}\, \langle n|P_{\mathrm{em}}|n\rangle, \tag{168}$$

and the expression contains only the Boltzmann weight $n^{-\beta}$ of $H = \log n$, with no $L(\beta, \chi_k)$ ($k \neq 0$): the KMS time evolution $\sigma_t = e^{itH}$ depends only on H, and the character structure of the constituent directions does not enter the KMS condition.

Taken together: constituent quantities cannot enter χ_0 (Eq. (167)), and χ_0 quantities do not originate from the constituents (Eq. (168)). From this, the origins of mass, quantum numbers, and stability are determined one by one.

Mass belongs to the whole. Substituting $N_{\chi_0} = \zeta(\beta)$ into the energy-scale formula of §5.3, with β fixed by the color-singlet level n ($n(\beta-1)^2 = 1$, §10.3), $n = 3$:

$$m_p = \sqrt{\frac{\zeta(\beta_B)\log^2\zeta(\beta_B)}{C^*}} = 937\text{ MeV}, \qquad \beta_B = 1 + \frac{1}{\sqrt{3}}. \tag{169}$$

By the Euler product

$$\zeta(\beta) = \prod_{q\ \text{prime}} \left(1 - q^{-\beta}\right)^{-1}, \tag{170}$$

ζ is a product of contributions from all primes; its logarithm is the infinite series $\sum_q \sum_m q^{-m\beta}/m$, which cannot be written as a finite sum of single-prime contributions: 937 MeV is a quantity of the whole. The constituent readout $m_u + m_u + m_d \approx 9.0$ MeV (§12.2–§12.3) is the projection of 937 along the nontrivial character directions, not an additional contribution superposed on it.

Quantum numbers belong to the projection. $\chi_0(a) \equiv 1$ carries no character label, so the condensate as a whole has no charge and no flavor (another reading of Eq. (167): the χ_0 direction has zero charge under every gauge channel); electric charge and flavor are carried by the projection content along the χ_1 and χ_5 directions, and the quantum numbers of the emergent state equal the sum of the projection readouts:

$$Q_p = Q_u + Q_u + Q_d = \tfrac{2}{3} + \tfrac{2}{3} - \tfrac{1}{3} = +1, \qquad Q_n = \tfrac{2}{3} - \tfrac{1}{3} - \tfrac{1}{3} = 0. \tag{171}$$

Stability belongs to the whole. Decay must lower the mass, and the mass resides in the condensate layer; the energy levels $E(n)$ of the condensate layer are strictly increasing in n:

$$\frac{dE^2}{dn} = \frac{d}{d\beta}\left[\frac{\zeta\log^2\zeta}{C^*}\right]\cdot\frac{d\beta}{dn} = \underbrace{\frac{\zeta'}{C^*}\left(\log^2\zeta + 2\log\zeta\right)}_{<0}\cdot\underbrace{\left(-\frac{1}{2n^{3/2}}\right)}_{<0} \;>\; 0 \tag{172}$$

($\zeta' < 0$; for $n \geq 3$ one has $\beta \leq 1 + 1/\sqrt{3}$, $\zeta \geq 2.35$, $\log\zeta > 0$). $n = 3$ is the lowest color singlet (Eq. (165)), and the condensate layer has no lower final state; the constituent directions can change quantum numbers but, by Eq. (167), cannot touch the mass, and their transitions are further suppressed by the generational mass hierarchy (§12). The stability of the proton is given by the two structural facts “$n = 3$ is lowest” and “the constituent directions have no χ_0 matrix elements”, without assuming baryon-number conservation.

At this point all three criteria are met: it is a single collective condensate in the χ_0 direction (§13.1); its mass is given by the whole and is not decomposable (Eq. (170)); its stability is protected by structure, without an externally imposed conservation law (Eqs. (172) and (167)).

$$\boxed{\begin{array}{c}\text{Baryons are fundamental particles;}\\ \text{“quarks” are the readouts of their internal projection.}\end{array}} \tag{173}$$

The observational counterpart: lattice QCD[50] gives that about 99% of the proton mass comes from field energy, with constituent rest masses accounting for about 1%; baryon quantum numbers are additive in quarks; the lower bound on the proton lifetime is $\tau_p > 2.4 \times 10^{34}$ yr. The imprint of the projection content on observables is tested in the next section.

§13.4 The Neutron–Proton Mass Difference

$m_n - m_p = 1.293$ MeV, only 0.14% of the proton mass; this splitting determines the neutron's β decay and the hydrogen-dominated composition of the universe. By the division of labor of the preceding section, the condensate layer gives p and n the same mass, and the splitting can only come from the projection content– the reality of the projection is tested on this number. The computation proceeds in three steps.

Step 1: the condensate layer cannot produce a splitting. p and n are both $n = 3$ color singlets; the condensate spectral weight $\zeta(\beta)$ is real-valued, giving the same 937 MeV. A stronger conclusion: χ_1 and χ_5 are complex conjugates of each other, $L(\beta, \chi_5) = \overline{L(\beta, \chi_1)}$; for holomorphic f, $f(\bar{z}) = \overline{f(z)}$, so for real β,

$$\operatorname{Re} f\big(L(\beta, \chi_5)\big) = \operatorname{Re} \overline{f\big(L(\beta, \chi_1)\big)} = \operatorname{Re} f\big(L(\beta, \chi_1)\big). \tag{174}$$

The difference between uud and udd is merely the interchange of the roles of χ_1 and χ_5: any holomorphic combination of L-functions takes the same real part on both and can produce no splitting. The splitting can only come from a non-holomorphic quantity.

Step 2: phases accumulate share by share according to the projection content. The phase $\theta = \arg L$ is non-holomorphic: $\arg \bar{z} = -\arg z$. Each share of the projection contributes a phase along its own direction: u (χ_1) contributes $+\theta$ and d (χ_5) contributes $-\theta$ $(\theta = \arg L(\beta_B, \chi_1) = 0.330)$:

$$p = uud: \;\; 2\theta - \theta = +\theta, \qquad n = udd: \;\; \theta - 2\theta = -\theta, \tag{175}$$

differing by 2θ: the existence and the sign of the splitting are given by the projection content.

Step 3: combining the two layers. The observables of the condensate layer are supported on the χ_0 component, those of the constituent layer on the nontrivial components, and the expectation of the cross terms contains the character orthogonality sum

$$\sum_{a \in (\mathbb{Z}/7\mathbb{Z})^*} \chi_k(a) = 0 \qquad (k \neq 0), \tag{176}$$

which vanishes identically: joint observables are the algebraic sum of the two layers' contributions, with no cross terms. The phase difference of the projection content is read out through the unique vertex connecting p and n–the weak vertex of β decay–with normalization energy scale v_{EW} (§10.1):

$$\Delta m^{(L)} = \frac{2m_p^2}{v_{\mathrm{EW}}}\tan\theta = \frac{2\times 937^2\times 0.343}{246\,200\text{ MeV}} = +2.44\text{ MeV}; \tag{177}$$

the condensate-layer contribution is the Coulomb self-energy difference of the charge distributions, shared among the $N_{\mathrm{EM}} = 6$ electromagnetic channels (§11.2):

$$\Delta m^{(\mathrm{EM})} = -\frac{\alpha_{\mathrm{EM}}\, m_p}{N_{\mathrm{EM}}} = -\frac{937}{137.0\times 6} = -1.14\text{ MeV}. \tag{178}$$

The two terms have opposite signs and comparable magnitudes:

$$\boxed{m_n - m_p = 2.44 - 1.14 = 1.30\text{ MeV},} \tag{179}$$

the experimental value is 1.293 MeV, a deviation of +0.5%. The existence, size, and sign of the splitting are given by the projection content, while the common baseline 937 is given by the condensate; whether the projection content can appear as independent particles is addressed in §13.6.

§13.5 The Nuclear Force

Besides leaving the imprint of the mass splitting, the projection content also fixes the vertex of the force between nucleons. Standard physics positions the internucleon force as follows: the strong force is the $SU(3)$ gauge force between quarks and gluons, one of the four fundamental forces; the nuclear force is the short-range force between color-neutral nucleons, not among the fundamental forces, and is interpreted as a residual effect of the strong force between color-neutral objects–in analogy with molecules being electrically neutral while intermolecular van der Waals forces persist.

In the present framework, the nuclear force is an independent force. In the force classification of §7–§8 it constitutes its own class–the emergent resonance force (§7.5): emergent states all reside in the χ_0 direction, so frequencies match automatically, and the joint coupling with the constituent direction k is given by the formula of §7.5:

$$\alpha_{\mathrm{mix}}(k) = |\mathrm{Aff}|_{\mathrm{eff}}\cdot|R_k|^2, \tag{180}$$

where the channel count is enhanced by inclusive summation over independent channels rather than diluted by averaging; taking $k = 1$ and including the Loewy correction, $\alpha_{\mathrm{nuclear}} = |\mathrm{Aff}|_{\mathrm{eff}}\,|R_1|^2 = 40.9\times 0.345 = 14.1$ (§11.5). It is independent of the color force at three levels: channel (χ_0 versus $\mathbb{Z}/3\mathbb{Z}$), statistics (summation versus averaging), and strength formula ((180) versus $|R_k|^2/N_k^{\mathrm{eff}}$); the value 14.1 is not derived from α_s.

The source of the "residual" impression can be located in the structure: the vertex amplitude of the nuclear force is taken from the χ_1 direction where the projection content resides, sharing the same $|R_1|^2 = 0.345$ with the color force– the common amplitude origin makes the nuclear force appear observationally as a companion phenomenon of the color force. Only the vertex amplitude is shared; the channel, the statistics, and the strength are the nuclear force's own.

§13.6 Color Confinement

The preceding two sections used in several places the fact that the projection content does not appear as independent particles; this section gives the complete argument. The observational content of color confinement consists of two facts: an isolated color charge has never appeared as a particle; color charge has never been measured in isolation. The two facts concern existence and observation respectively. The preceding parts have already fixed the criteria for each: readouts are given by self-adjoint operators, and observable values are invariant under the Galois group (§4.2); the existence of a macroscopically stable particle requires a divergent spectral weight–the condensation criterion (§6.5). Color charge is checked against these below; the argument has three parts.

Part 1: operators carrying color charge have no self-adjoint elements. In the Galois multiplicative action (§3.3, §4.2), take the generator g of the $\mathbb{Z}/3\mathbb{Z}$ subgroup; its unitary realization on $\mathcal{H}_p$ is

$$(Uf)(x) = f(g^{-1}x), \qquad U^*U = I. \tag{181}$$

Definition: an operator T carries color charge $c \in \mathbb{Z}/3\mathbb{Z}$ if

$$U\,T\,U^{-1} = \omega^c\,T, \qquad \omega = e^{2\pi i/3}. \tag{182}$$

This grading has the same origin as the color coordinate of the lattice in §13.2: the transition operator $|a\rangle\langle b|$ is multiplied by $\omega^{\,c(a)-c(b)}$ under conjugation by U, so the color charge of an operator is the lattice color difference it transfers. Taking $*$ on both sides of (182):

$$U\,T^*\,U^{-1} = \big(U\,T\,U^{-1}\big)^* = \overline{\omega^c}\,T^* = \omega^{-c}\,T^*, \tag{183}$$

i.e., T^* carries color charge $-c$. The self-adjointness condition $T = T^*$ requires

$$c \equiv -c \pmod 3 \iff 2c \equiv 0 \pmod 3 \iff c \equiv 0 \tag{184}$$

($\gcd(2,3) = 1$). The only self-adjoint operator carrying nonzero color charge is zero: every measurable operator is color-neutral, and any measurement reads out from a colored state only its color-neutral part. Contrast with the $\mathbb{Z}/2\mathbb{Z}$ factor: an operator carrying weak charge $s = 1$ satisfies $2s = 2 \equiv 0 \pmod 2$, compatible with self-adjointness–weak charge is observable. The dividing line for confinement is the parity of the CRT factor: odd-order factors confine, even-order factors do not.

Part 2: color directions have no Galois-rational readout. This is the observable-value formulation of the same fact. The color channel is carried by the complex-conjugate pair (χ_2, χ_4) (§9.2), $\bar\chi_2 = \chi_4$. By the rationality argument of §4.2: observable values must be invariant under the Galois group, while

$$\sigma_{-1}\big(L(\beta,\chi_2)\big) = L(\beta,\chi_4) = \overline{L(\beta,\chi_2)} \neq L(\beta,\chi_2), \tag{185}$$

so the eigenvalue of a single color direction is not Galois-invariant and cannot serve as a readout; the minimal Galois-invariant combination is the norm

$$N\big(L(\beta,\chi_2)\big) = L(\beta,\chi_2)\,\overline{L(\beta,\chi_2)} = |L(\beta,\chi_2)|^2 : \tag{186}$$

any readout necessarily binds χ_2 with χ_4; color can enter observables only in pairs. This is the other face of the same fact as Part 1: real characters ($\chi = \bar\chi$) correspond to self-adjoint admissibility and Galois rationality; the characters of the color channel are all complex and satisfy neither; the weak channel's χ_3 is real, $L(\beta, \chi_3) \in \mathbb{R}$, satisfying both.

Part 3: colored states have no spectral weight relative to the background. The background of observation is the KMS state: the existence of a particle is measured by its spectral weight relative to the thermal-equilibrium background (§6.5). Combinations with total color charge $C \not\equiv 0$ (including the single quark, $n = 1$) have vanishing projection onto the χ_0 direction (Eq. (163)); their spectral weight falls in the nontrivial directions and is the finite $|L(\beta, \chi)|$ by Eq. (161); the weight of the background itself is $\zeta(\beta)$. A macroscopically stable state requires a divergent spectral weight (the condensation criterion of §6.5), and the finite $|L|$ fails this for all $\beta > 1$: colored excitations cannot condense. At the phase transition where condensation occurs, the suppression strengthens to infinity:

$$\frac{|L(\beta, \chi)|}{\zeta(\beta)} \xrightarrow{\beta \to 1^+} 0. \tag{187}$$

The only combinations that can exist as particles are those satisfying the color-singlet condition (164).

Taking the three parts together, color charge fails item by item on both the observation side and the existence side: no self-adjoint element (Eq. (184), parity), no Galois-rational readout (Eq. (185), complex conjugation), no divergent spectral weight (Eq. (161), uniqueness of the pole). The first two are algebraic facts, holding for all β; the third holds for all $\beta > 1$ (finite weight cannot condense), strengthening to infinite suppression at the phase transition (Eq. (187)). This gives the precise meaning of "quarks are confined": the constituent content genuinely exists inside the color singlet, but cannot be observed as isolated particles– no operator can read out its color charge, and no divergent spectral weight can carry it, so it cannot be separated out of the baryon. That emergence is restricted to χ_0 and that colored states cannot exist independently are the two readings of the same set of facts (ζ diverges, L is finite).

§14 Bosons

The two classes of stable states established in §6–elementary states and emergent states–yield, via the correspondence of §9–§13, the fermions and the baryons. The observed particle spectrum has one more family of members: the photon, W and Z, the gluons, the π meson, and the Higgs particle. This chapter determines their identities within the framework. The $p-2$ elementary-state directions are already filled by the fermions (§9.2), and the bosons occupy no new character directions; they are another class of excitations of the existing structure– energy quanta propagating on resonance channels, and amplitude oscillations of the χ_0 condensate. They are determined channel by channel below, with all mass values coming entirely from quantities already fixed in §10–§12.

§14.1 Channel Excitations

A resonance force is the coupling of frequency-matched modes through the convolution operator K (§7.1), with propagation given by the Green kernel $L(\beta, \chi)$; the transmission of force does not presuppose particle exchange. What observation calls "gauge bosons" are, in the framework, channel excitations: energy quanta propagating on resonance channels. Channels are labeled by characters (the force table of §9.2.2), and the properties of a channel excitation are read from the channel character:

- Spin: the propagation mode of a channel along the $\mathbb{Z}$ direction is vector-type, $J = 1$;
- Charge: the excitation carries the charge of the channel character– excitations of real channels are electrically neutral and observable, excitations of complex channels carry color;
- Mass: whether the channel is broken by the χ_0 condensate– excitations of unbroken channels are massless, excitations of the breaking-direction channel acquire mass.

The three channels (§9.2.2) are checked in turn as follows.

§14.2 The Photon

The excitation of the cross-factor channel ($U(1)$, labeled by χ_1, χ_5). The involution ι flips the $\mathbb{Z}/2\mathbb{Z}$ component and preserves the $\mathbb{Z}/3\mathbb{Z}$ component (§3.3, §9.2), and the breaking acts on the $\mathbb{Z}/2\mathbb{Z}$ factor; the overall resonance of the cross-factor channel is not broken, its excitation is massless, and the force is long-range–this is the photon. The channel is Abelian ($U(1)$), and the excitation carries no charge of its own: the photon is electrically neutral, and photons have no direct coupling to one another. Masslessness and electric neutrality make it satisfy the observation criteria; the photon is a directly detected channel excitation.

§14.3 W and Z

The excitations of the weak channel ($SU(2)$, labeled by χ_3). The involution flips the $\mathbb{Z}/2\mathbb{Z}$ component, and the χ_0 condensation proceeds along the breaking direction χ_3 (the unique nontrivial real character, §6.2; for the phase transition see §5.2), so the weak-channel excitations acquire mass and the force is short-range. The mass computation proceeds in four steps.

Step 1: the metric energy of the condensate. By the energy-scale definition $E(N)^2C^* = N\log^2 N$ (§5.3), the total H^2-weight carried by the condensate is the identity

$$v_{\mathrm{EW}}^2\, C^* = N_{\mathrm{vac}}^{\mathrm{eff}} \log^2 N_{\mathrm{vac}}^{\mathrm{eff}}. \tag{188}$$

Step 2: equal sharing among channels. A weak-channel excitation is an oscillation of the condensate amplitude modulated by the χ_3 phase mode; the modulation multiplies only a phase and $|\chi_3(n)| = 1$, so the mode-by-mode squared amplitude is unchanged, and the distribution of the H^2-weight among the N_W^{eff} equivalent paths of the weak sector obeys KMS equal weighting (§7.4, the same lemma as the α_k dilution); a single excitation occupies one orthogonal path mode and carries one share:

$$M_W^2 = \frac{v_{\mathrm{EW}}^2\, C^*}{N_W^{\mathrm{eff}}}, \qquad \boxed{M_W = v_{\mathrm{EW}}\sqrt{\frac{C^*}{N_W^{\mathrm{eff}}}} = 80.6\ \mathrm{GeV},} \tag{189}$$

the experimental value is 80.37 GeV, a deviation of +0.25% ($N_W^{\mathrm{eff}} = 18.246$: the excitation is a physical state in the post-transition vacuum, so the corrected-level count is used).

Step 3: the neutral component. The probability exchange between the weak sector and the cross-factor sector is measured by the Born-level structure ratio (namely the two sectors transferred by the Type II redistribution of §11.3):

$$\sin^2\theta_W = \frac{d_2\,|R_2|^2}{|R_3|^2} = 0.2315, \tag{190}$$

the share leaking into the unbroken cross-factor channel carries no condensate weight, and the massive neutral combination retains a $\cos^2\theta_W$ share of the channel:

$$M_Z^2 = \frac{v_{\mathrm{EW}}^2\, C^*}{N_W^{\mathrm{eff}}\cos^2\theta_W}, \qquad \boxed{M_Z = \frac{M_W}{\cos\theta_W} = 91.9\ \mathrm{GeV},} \tag{191}$$

the experimental value is 91.19 GeV, a deviation of +0.78%; the orthogonal combination has all its weight in the unbroken channel, i.e., the zero mass of the photon (§14.2).

Step 4: cross-check. Combined with the independent derivation chain of the condensate oscillation (§14.6, $m_H = v_{\mathrm{EW}}/2$), the unadjusted ratio $M_W/m_H = 2\sqrt{C^*/N_W^{\mathrm{eff}}} = 0.654$ is predicted, versus the experimental 0.642 (1.9%). The Standard Model tree-level relation $M_W = gv/2$ ($g^2 = 4\pi\alpha_W$) is numerically compatible with Eq. (189) (80.2 versus 80.6), but g and 4π are not fundamental quantities in this framework. χ_3 is a real character, so the channel excitations are observable with charge ($W^\pm$) or neutral (Z); W and Z are directly detected channel excitations.

§14.4 Gluons

The excitations of the color channel ($SU(3)$, labeled by χ_2, χ_4).

Zero mass. The breaking direction is uniquely χ_3 and does not touch the $\mathbb{Z}/3\mathbb{Z}$ factor (§9.2.2); the condensate has no phase gradient on the $\mathbb{Z}/3\mathbb{Z}$ factor: by the sharing mechanism of Eq. (189), the color channel's share of the H^2-weight is zero, and the gluon is massless–the same mechanism as the zero mass of the photon (§14.2).

The octet. The independent excitation modes of the color channel are the traceless operators of the color sector, counted $d_2^2 - 1 = 8$– the dimension of the $SU(d_2)$ factor of the gauge group (§7.2); the trace part is the color-blind direction, belonging to the χ_0 side.

The excitations of the color channel genuinely exist: the color-force vertex amplitude $|R_1|^2 = 0.345$ enters the formulas for α_s and α_{nuclear} (§11), mode transitions inside color singlets are transmitted by it, and three-jet events are its indirect imprint. But the color channel is carried by a complex-conjugate pair, its excitations carry color charge, and the three arguments of §13.6 apply item by item: operators carrying nonzero color charge have no self-adjoint elements (Eq. (184)), complex directions have no Galois-rational readout (Eq. (185)), and colored excitations have no divergent spectral weight (Eq. (161)). The gluon is therefore in the same situation as the quark: genuinely existing inside color singlets, but confined– not observable as an isolated particle. The observational record agrees: the photon, W, and Z have all been directly detected, while a free gluon has never been observed, its evidence coming entirely from indirect imprints inside hadrons. The contrast with the photon is the arithmetic formulation of Abelian versus non-Abelian: excitations of the cross-factor channel carry no charge of their own and can appear alone; excitations of the color channel carry color and can exist only inside color-neutral combinations.

§14.5 The π Meson

The energy quantum exchanged in transitions between emergent states (the nuclear force of §13.5). The π resides in the color-singlet combination $\chi_1 \cdot \bar{\chi}_1 = \chi_0$: itself color-neutral, hence observable; $\chi_0(g^3) = 1$ does not change sign, so the spin is $J = 0$–a scalar boson.

The mass is composed from the energy scales of the two layers, with each of the three factors having its own origin. First, the transition spans two orthogonal layers (§13.4): its projection decomposes as the tensor product of the condensate layer (energy scale m_p) and flavor (the constituent layer), and KMS evaluation multiplies over tensor products– $m_\pi^2 \propto m_p \times$ (flavor weight); the geometric mean is a direct consequence of the product of the two layers' projections. Second, the flavor weight: the π transmits $u \leftrightarrow d$ flavor exchange; the flavor projection is supported on the two slots of the same doublet, with $P_u \perp P_d$ and $P_{\text{flavor}} = P_u + P_d$, and KMS evaluation adds over orthogonal projections– the flavor weight is $m_u + m_d = 6.8\,\text{MeV}$ (§12). Third, the rearrangement can route through d_2 color channels; superselection forbids interference, and probability-level inclusive summation gives the factor d_2 (the same lemma as the distinguishable counting of node factors in §6.3):

$$\boxed{m_\pi = \sqrt{d_2\, m_p\, (m_u + m_d)} = \sqrt{3 \times 937 \times 6.80} = 138.3\ \text{MeV},} \tag{192}$$

the experimental value is $m_{\pi^\pm} = 139.6\,\text{MeV}$, a deviation of -0.9%. In the chiral limit $m_u, m_d \to 0$, $m_\pi \to 0$: the π becomes a massless Goldstone-type excitation.

The mass of the π simultaneously fixes the range of the nuclear force: the propagator decays as $e^{-m_\pi r}$ (r the propagation distance), with characteristic length $1/m_\pi \approx 1.4\,\text{fm}$– the reason the nuclear force is short-range is that its carrier is massive.

§14.6 The Higgs

The amplitude oscillation mode of the χ_0 condensate itself– unlike the previous four sections, it is attached not to any resonance channel but to the condensate itself. Five steps follow: formation, properties, frequency, correspondence with observation, identity.

Step 1: formation of the oscillation. The linear resonance ratio in the χ_0 direction is $R_0(\beta) = 1 - p^{-\beta}$ (§7.4), which at the phase-transition point takes the amplitude-level value

$$R_0 = 1 - 7^{-1/2} = 0.622 > \tfrac{1}{2}: \tag{193}$$

the occupation probability of the KMS weight in the χ_0 direction exceeds one half, the vacuum expectation is nonzero, and condensation occurs; the condensate amplitude is given by the energy scale of the effective mode number, $v_{\mathrm{EW}} = E(N^{\mathrm{eff}}_{\mathrm{vac}}) = 246.19\,\mathrm{GeV}$ (§10.1). What observation calls the Higgs mechanism[38, 39] is this phase transition; no additional scalar field is needed. The condensate has amplitude oscillations about the new ground state; the oscillation mode is labeled by the condensation direction and the breaking direction, each of which is unique: condensation requires a divergent spectral weight (§6.5), and divergence occurs only at χ_0 (§13.1); the post-breaking vacuum expectation is real, so the breaking direction must be a real character (§6.2), and the reality condition $\chi_k = \bar{\chi}_k$ is equivalent to

$$2k \equiv 0 \pmod 6 \iff k \in \{0, 3\}, \tag{194}$$

leaving only χ_3 after excluding the trivial $k = 0$. The two labeling data are each unique, so there is exactly one oscillation mode: there is no second Higgs.

Step 2: properties of the oscillation. Three properties are read directly from the structure:

- Scalar: $\chi_0(a) \equiv 1$ carries no character label; the oscillation has no spin direction, $J = 0$;
- Electrically neutral: the oscillation carries no charge of any channel (the coupling matrix elements of χ_0 to all gauge channels vanish, §13.3);
- Coupling proportional to mass: the condensate amplitude v_{EW} enters all mass formulas (fermions $m_f \propto v_{\mathrm{EW}}$, §12; W, Z with $M \propto v_{\mathrm{EW}}$, §14.3), so the oscillation of the amplitude couples to every particle that acquires mass, with strength proportional to its mass.

Step 3: the frequency of the oscillation. The frequency is fixed by the condensate amplitude and the quartic self-coupling:

$$m = v_{\mathrm{EW}}\sqrt{2\lambda_H}. \tag{195}$$

The self-coupling λ_H is the Born probability of the four-oscillation vertex–the probability that the oscillation simultaneously occupies the scalar components of all three doublets. The computation has three steps: in the two-dimensional space of each doublet the scalar component (the symmetric combination) occupies one dimension, with occupation probability $1/d_1 = 1/2$ (the doublet structure of §6.2); the three doublets D_1, D_2, D_3 are pairwise disjoint (§6.2), so the occupation events are independent; the probability of occupying all three simultaneously is the product of the three factors:

$$\lambda_H = \left(\frac{1}{d_1}\right)^{d_2} = \left(\frac{1}{2}\right)^3 = \frac{1}{8}, \tag{196}$$

a Born-level probability of the same kind as the gauge couplings $\alpha_k = |R_k|^2/N_k$ (§7.4). Substituting into (195):

$$\boxed{m = v_{\mathrm{EW}}\sqrt{2/8} = \frac{v_{\mathrm{EW}}}{2} = 123.1\ \mathrm{GeV}.} \tag{197}$$

Step 4: correspondence with the Higgs. Observationally, the LHC detected a scalar particle of mass 125.2 GeV (the Higgs particle): $J = 0$, electrically neutral, coupling strength proportional to particle mass. The three properties agree one by one with the structural readouts of Step 2, and the mass deviates from Eq. (197) by -1.7%: what observation calls the Higgs particle is this condensate oscillation mode. The Standard Model introduces a fundamental scalar field to realize the symmetry breaking; its three external parameters are all corollaries in this framework: the quadratic term $\mu^2 < 0$ of the potential corresponds to the KMS phase transition at $\beta = 1$ (§5.2), the vacuum expectation $v \neq 0$ corresponds to Eq. (193), and the quartic self-coupling λ corresponds to Eq. (196).

Step 5: not a fundamental particle. Checking the criteria of §13 item by item:

- It is not an independent state–the oscillation is attached to the condensate; if the condensate disappears ($\beta \leq 1$, symmetry restored), the oscillation does not exist;
- Its mass is not its own spectral weight– Eq. (195) is a combination of the condensate amplitude and an occupation probability, containing no L-value or pole belonging to the oscillation itself;
- It corresponds to no fundamental field–all parameters of the scalar field are already corollaries (Step 4).

The Higgs is therefore not a fundamental particle but the oscillation sequence of the condensate– sharing with the baryon the same phase transition in the χ_0 direction: the spectral weight of the condensate gives the baryons (§13), and the amplitude oscillation of the condensate gives the Higgs.

The identities of the bosons and the numerical comparison:

Particle	**Identity**	**Channel/direction**	J	**Mass**	**Observation**
γ	Channel excitation	cross-factor $U(1)$	1	0	Direct detection
$W^{\pm}$	Channel excitation	weak χ_3	1	80.6 GeV (+0.25%)	Direct detection
Z	Channel excitation	weak χ_3	1	91.9 GeV (+0.78%)	Direct detection
Gluon	Channel excitation	color χ_2, χ_4	1	–	Confined
π	Transition quantum	$\chi_1\bar{\chi}_1 \to \chi_0$	0	138.3 MeV (−0.9%)	Direct detection
Higgs	Condensate oscillation	χ_0	0	123.1 GeV (−1.7%)	Direct detection

The bosons occupy no character directions: channel excitations are attached to resonance channels, the π to transitions between emergent states, and the Higgs to the χ_0 condensate.

§15 Spectral Completeness

Matter and force each have a definition in this framework: matter consists of the stable states whose form is invariant under time evolution–the eigenmodes of K (elementary states, §6.1) and the emergent states driven by divergent spectral weight (§6.5); force is the resonance of frequency-matched modes through K (§7.1) and the metric force of H (§8). §9–§14 matched them one by one, within the $p = 7$ channel, to the observed particles and forces. This chapter lists, under the same set of definitions, all the solutions at $p = 7$: the complete set of solutions of the definitions at $p = 7$ corresponds one-to-one with the observed particles and forces, with no remainder on either side.

§15.1 Species of Fundamental Particles

The first class, elementary states: the eigenmodes of K (§6.1). K is diagonal in the character basis (§3.2), and the eigendirections are exhausted by the character group; $(\mathbb{Z}/7\mathbb{Z})^*$ is a cyclic group of order 6, and the dual group is isomorphic to it,

$$\left|\widehat{(\mathbb{Z}/7\mathbb{Z})^*}\right| = \varphi(7) = 6 : \tag{198}$$

the eigendirections are exactly $\chi_0, \ldots, \chi_5$, with no seventh. $\tilde{\chi}_0$ is the direction of the KMS background itself and does not represent an excitation (§13.1); the elementary states are the five nontrivial directions, and complex conjugation $\chi_k \mapsto \chi_{6-k}$ partitions them into three orbits

$$\{\chi_3\}, \qquad \{\chi_2, \chi_4\}, \qquad \{\chi_1, \chi_5\} \tag{199}$$

(fixed points $k \equiv -k \pmod 6 \iff k \in \{0, 3\}$). Via the three criteria of §9.2, the three orbits give four particle classes: χ_3 gives the neutrinos, χ_2 and χ_4 give the charged leptons and antileptons, χ_1 gives the up-type quarks and χ_5 the down-type quarks; each class splits into three generations along the doublet structure of the involution (§6.2),

$$\underbrace{4}_{\text{classes}} \times \underbrace{3}_{\text{generations}} = 12 \tag{200}$$

fermions.

The second class, emergent states: collective condensates driven by divergent spectral weight. The divergent spectral weight occurs in one and only one direction, χ_0 (the unique pole of ζ in $\operatorname{Re}\beta > 0$, §13.1): the emergent direction is unique, and the emergent sequence is unique– the baryon spectrum at color-singlet levels $n \geq 3$ (§13). There is no second pole, hence no second emergent sequence.

The complete set of solutions for stable states:

Direction	Order	Stability	Particle	Generations	Source
$\{\chi_3\}$	2	Spectral simplicity	Neutrinos	3	§9.2, §12.4
$\{\chi_2, \chi_4\}$	3	Spectral simplicity	Charged leptonsś	3	§9.2, §12.1
$\{\chi_1, \chi_5\}$	6	Spectral simplicity	Up-/down-type quarks	3	§9.2, §12.2–12.3
$\{\chi_0\}$	1	Pole	Baryon sequence $n \geq 3$	–	§13

All four orbits are used up: in total 12 elementary-state excitations and one emergent sequence. The possibility of enlargement is excluded by (198): a fourth generation would require each orbit to carry four doublet shares, while the generation number $(p-1)/2 = 3$ is fixed by the number of involution orbits (§6.2); a new particle class would require a new character orbit, i.e., $\varphi(p) > 6$, contradicting (198). The same batch of nontrivial orbits admits another, component reading: the thermodynamic components that do not participate in the $\beta = 1$ condensation constitute dark matter, with ratio $(p-2) + \varphi(p-1)/(p-1) = 16/3$ (§16)– the particle reading and the component reading are two evaluations of the same structure, not double counting. With the particle species fixed, the forces among them are fixed with the channels; see the next section.

§15.2 Species of Forces and Bosons

The definition of force gives two mechanisms (§7.1, §8), solved class by class at $p = 7$.

The first class, resonance forces: couplings of frequency-matched modes through K, with channels labeled by characters. The resonance channels between elementary states are fixed by the CRT decomposition (§7.2): $6 = 2 \times 3$, the prime factorization is unique, and the factor-preserving unitary group is

$$\mathcal{G} \cong \big(U(1) \times SU(2) \times SU(3)\big)/\mathbb{Z}_6 \tag{201}$$

(the central quotient $\mathbb{Z}_6 \cong \mathbb{Z}_2 \times \mathbb{Z}_3$ is the imprint of the CRT decomposition, §7.2; this is the global form of the Standard Model gauge group), with exactly

$$\omega(6) + 1 = 3 \tag{202}$$

independent channels: two factor resonances ($SU(2)$ weak, $SU(3)$ color) and one cross-factor resonance ($U(1)$ electromagnetic). A fourth resonance channel would require 6 to have a third prime factor, contradicting $6 = 2 \times 3$; a transition changing both CRT factors simultaneously does not preserve the factor structure and does not constitute an independent channel (§7.2). The resonance channel between emergent states is unique: the emergent direction is unique (§15.1) $\Rightarrow$ emergent states all reside at χ_0 and their frequencies match automatically; their resonance is one force– the nuclear force (§13.5).

The second class, the metric force: the metric non-uniformity of $H = \log n$ (§8). The metric is uniquely given by H (§3.4), so there is exactly one metric force–gravity.

The species of forces are hereby closed:

$$\boxed{\omega(6) + 1 + 1 + 1 = 5 : \quad \text{electromagnetic, weak, color, nuclear, gravity.}} \tag{203}$$

The carriers of the forces are fixed with the channels (§14); whether a channel excitation can be observed as a particle is checked item by item by the three criteria on the channel character (§13.6):

- Self-adjointness: self-adjoint operators carrying nonzero charge exist only in even-order factors (the solutions of $2c \equiv 0$, §13.6);
- Galois rationality: readouts of real character directions belong to $\mathbb{R}$; complex directions must enter in pairs through the norm (§13.6);

- Spectral weight: massless or mass-acquiring excitations can propagate; colored excitations have no divergent spectral weight (§13.6).

The results of the channel-by-channel check: the cross-factor channel is unbroken, its excitation massless and chargeless–γ, observable; the weak channel χ_3 is real and broken by the condensate, its excitations massive–W, Z, observable; the color channel χ_2, χ_4 is complex, its excitations colored–gluons, existing but confined; the transition quantum between emergent states is color-neutral–π, observable. The complete set of solutions for forces and carriers:

Force	**Channel**	**Carrier**	**Observation**
Electromagnetic	cross-factor $U(1)$	γ	Direct detection
Weak	χ_3, $SU(2)$	W, Z	Direct detection
Color	$\chi_2\chi_4$, $SU(3)$	Gluons	Confined
Nuclear	χ_0 emergent resonance (vertex χ_1)	π	Direct detection
Gravity	Metric (H)	–	Macroscopic phenomena

The numerical values of the coupling constants and carrier masses are given in §11 and §14. Beyond forces and carriers, one mode remains to be counted–the oscillation of the condensate itself; see the next section.

§15.3 The Condensate Oscillation and the Higgs

The condensate oscillation is not a force and occupies no character direction, but it is an independent class of excitation in the spectrum, and its counting must be closed separately. Its formation, properties, and identity have been fully determined in §14.6: the condensation direction and the breaking direction are each unique, so there is exactly one oscillation mode; by the criteria of §13 it is not a fundamental particle. It therefore does not enter the fermion list and is counted separately as one mode in the completeness count.

§15.4 Completeness

The solutions of the three sections combine into the completeness statement.

On the state-space side: 6 character directions (Eq. (198): one background-cum-condensation direction and five elementary-state directions), 3 elementary resonance channels (Eq. (202)), one pole, and one metric– each item has been given its physical correspondent in §15.1–§15.3– the solutions admitted by the definitions have all been used in the correspondence, with no structure left over.

On the observational side, the two master tables together:

- 12 fermions (Eq. (200));
- 1 emergent sequence;
- 4 observable bosons (γ, W, Z, π);
- 1 condensate oscillation mode;
- the color-sector degrees of freedom (quarks and gluons: existing, confined);
- 5 interactions (Eq. (203)),

and every item of the observed spectrum has been given its origin from the definitions of stable states or forces, with no gap.

Closure yields a falsifiable statement: within the $p = 7$ channel there is no structural slot to accommodate a fourth generation of fermions (Eq. (198)), a sixth interaction (Eq. (203)), or a second Higgs (the double uniqueness); the experimental discovery of any such object would refute the channel identification of §9.

The structure and the inventory are hereby closed.

§16 β Evolution

All the preceding parts worked at fixed β: §1–§9 built spacetime, matter, and force on the state space, §10–§12 computed the energy scales, coupling constants, and mass spectrum once the channel is fixed, and §13–§15 established the baryons, the bosons, and spectral completeness; throughout, β appeared as a parameter. This chapter lets β vary: the progression of the KMS state φ_β with β is the evolution of the system.

§16.1 The Pole

The partition function $Z(\beta) = \sum_{n=1}^{p^K} n^{-\beta}$ tends to $\zeta(\beta)$ in the limit $K \to \infty$ (§6.1). As $\beta \to 1^+$ three thermodynamic quantities diverge simultaneously, and the triple divergence is driven entirely by the simple pole of ζ at $\beta = 1$.

First, the partition function. The Laurent expansion of ζ is

$$\zeta(\beta) = \frac{1}{\beta - 1} + \gamma + O(\beta - 1), \tag{204}$$

with $\gamma = 0.5772\ldots$ the Euler–Mascheroni constant, so $Z \to \infty$.

Second, the energy density. In the KMS state the occupation probability of mode n is $p_n = n^{-\beta}/\zeta(\beta)$,

$$\rho(\beta) = \langle H \rangle_\beta = \sum_n p_n \log n = -\frac{\zeta'(\beta)}{\zeta(\beta)}. \tag{205}$$

By $\zeta'(\beta) = -(\beta-1)^{-2} - \gamma_1 + O(\beta-1)$ ($\gamma_1 = -0.0728\ldots$ the first Stieltjes constant),

$$\rho(\beta) = \frac{1}{\beta - 1} - \gamma + O(\beta - 1) \;\to\; +\infty. \tag{206}$$

Third, the entropy. From $S = -\sum_n p_n \log p_n = \beta\rho(\beta) + \log\zeta(\beta)$, substituting (204) and (206):

$$S(\beta) = \frac{\beta}{\beta - 1} + \log\frac{1}{\beta - 1} + O(1) \;\to\; +\infty. \tag{207}$$

Apart from the simple pole at $s = 1$, $\zeta(s)$ is analytic in the entire complex plane, and its zeros produce no divergence of the partition function: the system has one and only one divergence point. A general Dirichlet series can have several singularities on the boundary of its half-plane of convergence; the unique pole of ζ is a special property of the most uniform coefficient distribution ($a_n \equiv 1$). The relation between internal time and β is

$$dt = \frac{d\beta}{\langle H \rangle_\beta} = \frac{\zeta(\beta)}{-\zeta'(\beta)}\, d\beta, \tag{208}$$

and as $\beta \to 1^+$, $dt/d\beta \to 0$: internal time freezes at the origin. At $\beta = 1$ the partition function, the energy density, and the entropy diverge simultaneously; this is the phase-transition point of the system, i.e., the starting point of the evolution; observational cosmology calls it the Big Bang.

§16.2 Initial Conditions

Observation gives three features of the early universe: highly uniform temperature ($\Delta T/T \sim 10^{-5}$), spatial curvature near zero ($\Omega_k = 0.001 \pm 0.002$), and a nearly scale-invariant fluctuation power spectrum ($n_s = 0.965 \pm 0.004$). All three are given by the structure of the phase transition at $\beta = 1$.

First, uniformity. On the interval $\beta \leq 1$ the KMS state is unique (§5.2): before the transition the system is in a single global thermal-equilibrium state, and no causally disconnected regions exist. After β crosses 1 the symmetry breaks spontaneously and the KMS state splits into a family of extremal states parametrized by $(\mathbb{Z}/p\mathbb{Z})^*$, but the pre-splitting uniformity is inherited as the initial condition–uniform temperature everywhere is the legacy of the KMS uniqueness at $\beta \leq 1$.

Second, flatness. The logarithmic metric $d(m,n) = |\log(m/n)|$ (§3.4) gives the spacing of neighboring modes

$$d(n, n{+}1) = \log\Big(1 + \frac{1}{n}\Big) = \frac{1}{n} + O\Big(\frac{1}{n^2}\Big) \;\to\; 0: \tag{209}$$

the logarithmic spacing of neighboring integers decreases monotonically to zero as n grows, and the large-scale geometry is asymptotically flat (the quantitative curvature statement is given by the Ollivier–Ricci curvature of §16.6[29]).

Third, the fluctuation spectrum and non-Gaussianity. For $\beta > 1$ the KMS state splits into the extremal states φ_{β,σ_c} ($c \in (\mathbb{Z}/p\mathbb{Z})^*$), and the differences between distinct extremal states produce density fluctuations. Character projection gives the Fourier components of the fluctuations

$$\delta\varphi_k(c) = \frac{\tau(\bar{\chi}_k)\, \chi_k(c)\, L(\beta, \chi_k)}{\varphi(p)\, \zeta(\beta)}, \qquad k = 1, \ldots, p-2, \tag{210}$$

where $\tau(\chi_k)$ is the Gauss sum. The KMS state is equally weighted on $(\mathbb{Z}/p\mathbb{Z})^*$, and the power spectrum

$$P_k = \frac{p\,|L(\beta, \chi_k)|^2}{\varphi(p)^2\, \zeta(\beta)^2} \tag{211}$$

is approximately equal for all k–this is near scale invariance. The three-point correlation is fixed by character orthogonality: $\langle \delta\varphi_{k_1} \delta\varphi_{k_2} \delta\varphi_{k_3} \rangle \propto \delta_{k_1+k_2+k_3 \equiv 0 \pmod{\varphi(p)}}$, and the non-Gaussianity parameter is

$$f_{\mathrm{NL}} = \lim_{\beta\to\infty} \frac{\varphi(p)\, \zeta(\beta)}{\sqrt{p}\, |L(\beta, \chi_2)|} = \frac{\varphi(p)}{\sqrt{p}} \xrightarrow{p=7} \frac{6}{\sqrt{7}} = 2.27, \tag{212}$$

where $\sqrt{p}$ comes from the Gauss sum. The fluctuations are produced at the transition, and the non-Gaussianity is measured today: in the current universe $\beta_{\text{now}} \gg 1$, both ζ and $|L|$ have converged to 1, and f_{NL} is a pure arithmetic constant, independent of the value of β. Planck 2020 measures $f_{\mathrm{NL}}^{\mathrm{local}} = -0.9 \pm 5.1$[48], and the prediction lies within 1σ; the target precision $\sigma(f_{\mathrm{NL}}) \sim 1$ of CMB-S4 suffices for a direct test.

Uniformity, flatness, and the fluctuation spectrum are all given by the KMS structure and the character structure, without using an inflationary mechanism.

§16.3 Cooling

The temperature is $T = 1/\beta$, and the evolution proceeds by discrete steps.

Step 1: the discrete temperature step. The smallest resolvable step of β is fixed by the resolution limit of KMS states: the KMS states at β and $\beta + \Delta\beta$ are distinguishable if and only if the occupation probability of at least one mode changes by more than $1/N$ (N the effective mode number); the maximum of $dp_n/d\beta = -p_n(\log n - \rho(\beta))$ is $O(1)$, so $\Delta\beta = 1/N$. Taking $N = N_{\rm Pl}^{\rm eff} \approx 4.6 \times 10^{34}$ at the gravitational scale (§10.2), $\Delta\beta \approx 2.2 \times 10^{-35}$–the continuum approximation is extremely accurate.

Step 2: the discrete time step. Each step advances internal time by

$$\Delta t = \frac{\Delta\beta}{\rho(\beta)} : \tag{213}$$

$\Delta\beta$ is constant while Δt varies with β– at early times ρ is enormous and Δt minute; at late times $\rho \to 0$ and Δt enormous.

Step 3: the continuum limit. As $\Delta\beta \to 0$, (213) gives $dt = d\beta/\rho(\beta)$; near $\beta = 1$, $\rho \approx 1/(\beta - 1)$ (Eq. (206)), and integration yields $t = (\beta - 1)^2/2$, i.e.,

$$\boxed{\beta(t) = 1 + \sqrt{2t}\,, \qquad T(t) = \frac{1}{1 + \sqrt{2t}}\,.} \tag{214}$$

Three regimes: at $t = 0$, $T = 1$ (the transition temperature); at early times $t \ll 1$, $T \approx 1 - \sqrt{2t}$ (rapid cooling); at late times $t \gg 1$, $T \propto t^{-1/2}$ (power-law cooling). $T \propto t^{-1/2}$ agrees with the Friedmann temperature law of the radiation-dominated era, but (214) uses neither the scale factor $a(t)$ nor the Friedmann equations– the cooling law is determined entirely by the pole behavior of $\zeta(\beta)$.

§16.4 The Emergence of Spacetime and Matter

After the cooling crosses $\beta = 1$, the structures built in the preceding parts fall into place in order. The transition breaks time-translation symmetry, giving the arrow of time (§5.2); the four-dimensional Lorentz structure (§5.1) provides the carrying geometry for all subsequent excitations. For $\beta > 1$ the KMS evaluation is finite and stable states can be defined: the eigenmodes in the unit-group directions of the species layer (elementary states, §6.1) and the collective emergent states in the χ_0 direction (§6.5) appear during the cooling–matter forms; the resonance forces and the metric force (§7–§8) are activated together with the stable states. All structures of the preceding parts lie on the $\beta > 1$ side; the remainder of this chapter computes the numbers along this timeline.

The sections above use only the ζ pole and the KMS structure and hold for all p channels; the numbers in the sections below make essential use of the $p = 7$ fixed in §9 (the mechanism of the endpoint section holds for all channels, with the numbers taken at $p = 7$).

§16.5 The Baryon-to-Photon Ratio

Emergent states condense in the χ_0 direction (§6.5). The fraction of matter retained by the condensation, $\eta = n_b/n_\gamma$, is fixed by four links: the production of net charge (the source), the registration into the baryon part, the equilibrium allocation, and the vacuum dilution– Sakharov's three conditions (B violation, CP violation, departure from equilibrium) each have their counterpart here. The computation proceeds in five steps.

Step 1: uniqueness of the source. A vertex producing net charge must map the excitation of some direction to its conjugate excitation, and that direction must return to itself under complex conjugation: $\chi = \bar{\chi}$. For complex characters, particle and antiparticle reside in two separate directions paired by Galois, and character multiplication conserves their charge difference coordinate by coordinate, producing no net charge; the nontrivial self-conjugate character is unique–χ_3 (§6.2), whose only charge is lepton number (§9.2). Hence the source of net charge is the $\Delta L = 2$ vertex in the χ_3 direction, unique, and the net charge produced is $B - L$. The colored directions χ_1, χ_5 are both complex characters, and $\Delta B = 2$-type vertices have no carrying direction: $B - L$ violation occurs only through the lepton direction, and neutron–antineutron oscillation does not exist in this framework. Event counting: each source event deposits $|\Delta L| = 2$ units; $\chi_3 = \bar{\chi}_3$ makes the two orientations of the vertex the same physical process, so the counting carries the Majorana factor $\frac{1}{2}$ (§12.4)–each countable event deposits a net $2 \times \frac{1}{2} = 1$ unit.

Step 2: the registration amplitude. The source acts on the character directions, while baryon number resides in the χ_0 condensate; the unique operation connecting the two is the μ_p^* descent chain– the unique irreversible operation in the system ($\mu_p^* \mu_p = 1 \neq \mu_p \mu_p^*$). Starting from the $K_{\mathrm{EW}} = 4$ address digits distinguished by the elementary resonance forces (§10.1),

$$\mathbb{Z}/p^4 \xrightarrow{\mu_p^*} \mathbb{Z}/p^3 \xrightarrow{\mu_p^*} \mathbb{Z}/p^2 \xrightarrow{\mu_p^*} \mathbb{Z}/p \xrightarrow{\mu_p^*} \mathbb{Z}/1,$$

the Maschke defect (§10.2) prevents the augmentation map from removing the trivial mode completely, adding a final step into the vacuum, for a total of $\varphi(p) - 1 = 5$ layers. The action formula on the graded quotients, $(1 + 7^j a)^n \equiv 1 + 7^j(na) \pmod{7^{j+1}}$, shows that the lattice label n acts as $\times n$ simultaneously on all five layers; the descent chain changes only the 7-adic address digits, and n coprime to 7 is conserved in the descent. A single registration event is therefore the action of the single isometry $\mu_{(7n)^5}$, whose KMS scaling $\varphi_\beta(\mu_m \cdot \mu_m^*) = m^{-\beta} \varphi_\beta(\cdot)$ is evaluated at the transition point $\beta = 1$; summing over n gives

$$7^{-5} \sum_{7 \nmid n} \chi_5(n)\, n^{-5} = \Gamma_1\, L(5, \chi_5), \qquad \Gamma_1 = p^{-(\varphi(p)-1)} = 7^{-5} = 5.95 \times 10^{-5}. \tag{215}$$

The exponent of Γ_1 and the argument of the L-function have a common origin: they are respectively the 7-adic part and the coprime part of the same five-layer transfer weight; the argument 5 of L is the layer multiplicity, and the evaluation point is $\beta = 1$.

Step 3: the CP asymmetry. χ_3 is a real character and carries no phase of its own; the asymmetry comes from the dressing of the source vertex by complex characters and the interference with the tree level. Of the $\varphi(p) = 6$ characters, $p - 3 = 4$ are complex, with participation rate

$$\Gamma_2 = \frac{p-3}{p-1} = \frac{2}{3}; \tag{216}$$

the interference phase is given by the phase part of the registration weight (215) ($\chi_5 = \bar{\chi}_1$, the two having the same modulus):

$$\Gamma_3 = \sin\bigl(\arg L(5,\chi_1)\bigr) = \frac{|\mathrm{Im}\, L(5,\chi_1)|}{|L(5,\chi_1)|} = \frac{0.0295}{0.9864} = 0.0299, \tag{217}$$

where the normalization uses the sector's own modulus $|L|$ rather than the global partition function– the CP asymmetry is an intrinsic property of the sector.

Step 4: the equilibrium allocation. A chemical potential is the twisting parameter of the KMS state along a conserved-charge direction; μ_p belongs to the algebra, so the charge direction altered by the descent chain admits no chemical potential, and the constraints below are existence conditions of the KMS state, maintained up to the transition point. The retained fraction $\kappa = B/(B-L)$ is solved from three constraints together. Constraint 1 (descent-chain balance): the five layers are three wild color axes plus the tame closure (§10.1), and one descent is metered as three shares of color address and one share of species address, $N_c\,\mu_q + \mu_\ell = 0$ ($N_c = 3$ the order of the color factor). Constraint 2 (vertex conservation): the conservation law $\chi_{\mathrm{in}} = \chi_{\mathrm{out}}$ (§7.1) imposed on the vertices of the mass path (§6.4), with the condensate quantum written μ_H: $\mu_{u_R} = \mu_q + \mu_H$, $\mu_{d_R} = \mu_q - \mu_H$, $\mu_{e_R} = \mu_\ell - \mu_H$; the equal moduli of the Gauss sums guarantee that all three generations equilibrate simultaneously. Constraint 3 (charge neutrality): the KMS uniqueness at $\beta \le 1$ makes the state Galois-invariant, so all gauge charge densities vanish, $\sum_i Y_i \tilde{\mu}_i = 0$. The hypercharges are fixed by framework data: χ_3 is self-conjugate with zero charge, and its mass vertex gives $Y_H = -Y_\ell$; the linearization of vertex conservation transmits Y direction by direction; background neutrality $\mathrm{Tr}\,Y = 0$ gives $Y_q = Y_H/3$; taking $Y_H = 1$ as the charge unit yields $Y = (\frac{1}{3}, \frac{4}{3}, -\frac{2}{3}, -1, -2, 1)$ (in the order $q, u_R, d_R, \ell, e_R, H$). Substituting $N_f = (p-1)/2 = 3$ (the generation number, §6.2) and the 2 condensate-oscillation components:

$$3(8\mu_q + 4\mu_H) + 2\mu_H = 0 \implies \mu_H = -\frac{12}{7}\mu_q,$$

whence $B = 12\mu_q$, $L = -\frac{153}{7}\mu_q$, $B - L = \frac{237}{7}\mu_q$:

$$\kappa = \frac{B}{B-L} = \frac{84}{237} = \frac{28}{79} = 0.354. \tag{218}$$

Step 5: vacuum dilution and assembly. The baryon signal is shared among $N_{\mathrm{vac}}^{\mathrm{eff}} = 2039.75$ (§10.1) independent vacuum modes–this is the root cause of $\eta \sim 10^{-10}$. η involves only the $\mathbb{Z}/2\mathbb{Z}$ direction (matter versus antimatter); the color degrees of freedom are symmetric between matter and antimatter, so the dilution factor is divided by the color number $N_c = 3$:

$$N_{\mathrm{vac}}^{\mathrm{eff}}/N_c = 2039.75/3 = 679.9.$$

Assembling:

$$\boxed{\eta = \frac{\Gamma_1\Gamma_2\Gamma_3\,\kappa}{N_{\mathrm{vac}}^{\mathrm{eff}}/N_c} = \frac{5.95\times10^{-5}\times0.667\times0.0299\times0.354}{679.9} = 6.18\times10^{-10}.} \tag{219}$$

Planck 2020 measures $(6.12 \pm 0.04) \times 10^{-10}$[47], a deviation of $+1.0\%$. The extreme smallness of η comes from a triple suppression: the exponential decay of the five-layer registration (Γ_1), the CP phase tending to zero as the layer multiplicity grows (Γ_3), and the dilution by the effective mode number at the electroweak scale ($N_{\text{vac}}^{\text{eff}}/N_c$). Converted through BBN nucleosynthesis[51],

$$\Omega_b h^2 = 3.66 \times 10^7 \cdot \eta = 0.0226, \tag{220}$$

Planck 2020 measures 0.02237 ± 0.00015, a deviation of $+1.0\%$.

The above chain yields an existence criterion for the channel parameter p. The source of Step 1 requires a lift in which particle and antiparticle coincide on the self-conjugate direction; this lift is realizable if and only if $V_a^m \equiv -1 \pmod p$ (§6.4(c)), i.e., $m = (p-1)/2$ is odd, equivalent to $p \equiv 3 \pmod 4$ (-1 a quadratic nonresidue). For channels with $p \equiv 1 \pmod 4$, the particles of the self-conjugate direction are Dirac-type and the $\Delta L = 2$ vertex does not exist: with no source, $B-L$ is identically zero, the equilibrium allocation gives $B = \kappa(B-L) = 0$, matter and antimatter annihilate completely, and $\eta = 0$ holds exactly. Channels with a matter surplus must therefore belong to the class $p \equiv 3 \pmod 4$; $p = 7$ satisfies this criterion. The same piece of arithmetic simultaneously fixes the type of the neutrino: the existence of matter requires the existence of the source vertex, i.e., the neutrinos in the χ_3 direction must be Majorana-type[35] (this is what fixes the $\frac{1}{2}$ factor of §12.4); if experiment determines the neutrino to be Dirac-type, the baryon-to-photon-ratio mechanism of this section is refuted.

§16.6 Components

The components of the universe are given by the character sector structure.

Step 1: dark matter. $L(s,\chi_0) = \zeta(s)(1-p^{-s})$ has a pole at $s = 1$; the χ_0 direction participates in the $\beta = 1$ transition and constitutes the visible-matter sector; the $L(s,\chi_j)$ of the remaining $p-2 = 5$ nontrivial characters are analytic at $s = 1$, do not participate in the transition, and freeze after it into invisible sectors. These sectors participate in the metric force (universal over the full group, §8) but not in the $p = 7$ gauge forces– precisely the defining feature of dark matter. The KMS state acts transitively on the Galois group, so the nontrivial sectors each contribute the same energy density: the leading term is $p-2 = 5$; the conjugation structure of the five dark sectors (two conjugate pairs $\chi_1 \leftrightarrow \chi_5$, $\chi_2 \leftrightarrow \chi_4$ plus one self-conjugate χ_3) gives the correction term $\varphi(p-1)/(p-1) = 1/3$; energy density, as a positive linear functional, superposes linearly, and the two terms add:

$$\boxed{\frac{\Omega_{\text{DM}}}{\Omega_b} = (p-2) + \frac{\varphi(p-1)}{p-1} = 5 + \frac{1}{3} = \frac{16}{3} = 5.333.} \tag{221}$$

Planck 2020 measures 5.364 ± 0.066, a deviation of -0.6%. From $\Omega_b h^2 = 0.0226$ and $h = 0.674$, $\Omega_b = 0.0498$,

$$\Omega_m = \Omega_b \times \left(1 + \frac{16}{3}\right) = 0.0498 \times \frac{19}{3} = 0.315,$$

in agreement with Planck 2020's 0.315.

Step 2: the sign of the dark energy. There are two independent arguments for $\Lambda > 0$. The arithmetic argument: the residue of $\zeta(s)$ at $s = 1$ is 1, which via the von Mangoldt explicit formula $\psi(x) = x - \sum_\rho x^\rho/\rho - \cdots$ becomes the coefficient of the linear term of the prime-counting function– the existence and positive density of the primes correspond to $\Lambda > 0$: the positivity of the cosmological constant and the infinitude of the primes have a common origin. The geometric argument: on the BC metric space $(\mathbb{Z}_{>0}, d(m,n) = |\log(m/n)|)$, take the symmetric simple random walk $\mu_n = \frac{1}{2}\delta_{n-1} + \frac{1}{2}\delta_{n+1}$; the Ollivier–Ricci curvature

$$\kappa_{\mathrm{OR}}(n, n{+}1) = 1 - \frac{W_1(\mu_n, \mu_{n+1})}{d(n, n{+}1)}$$

is strictly negative for all $n \geq 2$: the optimal coupling gives $W_1 = \frac{1}{2}[\log \frac{n}{n-1} + \log \frac{n+2}{n+1}]$, and $\kappa_{\mathrm{OR}} < 0$ is equivalent to $\frac{n(n+2)}{(n-1)(n+1)} > \frac{(n+1)^2}{n^2}$, which cross-multiplies and simplifies to $2n + 1 > 0$, always true; asymptotically $\kappa_{\mathrm{OR}} \sim -1/n^2$. Space has hyperbolic geometry, consistent with de Sitter[44] ($\Lambda > 0$); the two arguments are mutually independent and point jointly to $\Lambda > 0$. At the same time $\kappa_{\mathrm{OR}} \to 0$ gives large-scale asymptotic flatness, i.e., $k = 0$ (Planck $\Omega_k = 0.001 \pm 0.002$).

Step 3: the dark-energy density. Spatial flatness closes the Friedmann constraint[43]:

$$\Omega_\Lambda = 1 - \Omega_m = 0.685, \tag{222}$$

in agreement with Planck 2020's 0.685; substituting $H_0 = 67.4$ km/s/Mpc,

$$\boxed{\Lambda = \frac{3\Omega_\Lambda H_0^2}{c^2} = 2.85 \times 10^{-122}\ \ell_{\mathrm{Pl}}^{-2},} \tag{223}$$

consistent with the observed value.

The conversion to absolute densities uses H_0 (equivalently T_{CMB}) as the observational anchor– it serves only to locate the present moment and enters no structural derivation; the ratios η and $\Omega_{\mathrm{DM}}/\Omega_b$ do not depend on this anchor; Ω_Λ uses the anchor through the conversion $\Omega_b = \Omega_b h^2/h^2$.

§16.7 The Present and the Endpoint

The evolution is parametrized by β, and the current position is read from the temperature. The temperature of §16.3 is in units of the transition temperature (dimensionless); the calibration of the physical temperature is given by the energy-scale ceiling M_{Pl} (§10.2), $\beta = M_{\mathrm{Pl}}/T$. Substituting $T_{\mathrm{CMB}} = 2.725\,\mathrm{K}$,

$$\beta_{\mathrm{now}} \approx 5 \times 10^{31}.$$

The endpoint lies at $\beta \to \infty$. $n^{-\beta} \to 0$ for all $n \geq 2$, $\zeta(\beta) \to 1$, the ground-state occupation probability $p_1 = 1/\zeta \to 1$, and the entropy $S \to 0$– all excited states decay, leaving only $|1\rangle$. The heat death is not the disordered maximum-entropy state but the ordered lowest-energy state: all prime channels close, and all characters cease to be distinguishable.

Whether the endpoint is reachable depends on whether the degrees of freedom are finite. The $\mathbb{N}^*$ of the standard BC system is infinite, the heat capacity is positive at every $\beta > 1$, and the cooling never completes–$\beta = \infty$ is unreachable. The mode number of the physical system is truncated at $N_{\mathrm{Pl}}^{\mathrm{eff}} \approx 4.6 \times 10^{34}$ (§10.2): in the truncated system the occupation probability of the highest excited state $|N\rangle$ is $p_N = N^{-\beta}/Z_N(\beta) \leq N^{-\beta}$, so there exists a finite β_{HD} at which the occupation probabilities of all excited states fall below any given ε. At large β the energy density is dominated by the $n = 2$ mode, $\langle H \rangle_\beta \approx (\log 2)\, 2^{-\beta}$, and the internal time required for each further step of cooling grows exponentially; taking $\beta_{\mathrm{HD}} \approx N_{\mathrm{Pl}}/\log 2$, the total internal time

$$t_{\mathrm{HD}} = \int_1^{\beta_{\mathrm{HD}}} \frac{d\beta}{\langle H \rangle_\beta} \approx \frac{2^{\beta_{\mathrm{HD}}}}{(\log 2)^2} = \frac{e^{N_{\mathrm{Pl}}}}{(\log 2)^2} :$$

of doubly exponential order, but strictly finite; with infinitely many modes the integral diverges and the heat death is unreachable. Currently $\beta_{\mathrm{now}}/\beta_{\mathrm{HD}} \sim 10^{-3}$– the evolution is about one thousandth complete.

§16.8 Channels and Evolution

The evolution of the state space under β gives the complete history of the universe from birth to endpoint: the divergence at the ζ pole at $\beta = 1$ is the Big Bang, cooling advances along β, spacetime, matter, and force emerge after the transition, the baryons and the components take shape in turn, and within finite time the system reaches ground-state locking.

p and β are the two labels of the framework: different p are different physical worlds, and different β are different moments of the same world. The state space thereby gives a multiverse framework– horizontally, a family of worlds labeled by primes; vertically, each world's history from the pole to ground-state locking; the physical space we inhabit is the $p = 7$ channel at the moment it has advanced to β_{now}.

§17 Conclusion

This paper has constructed the state space from the Bost–Connes system and argued that its structural properties naturally describe the physical world: spacetime, matter, forces, and cosmic evolution are not objects referred to outside the state space, but readouts of the structure of the state space. This chapter collects the core formulas and the summary table of numerical results, and lists the testable predictions.

§17.1 Core Formulas

The derivation chain starts from the BC system (§1):

$$\mathcal{A}_{\mathrm{BC}} = C^*(\mathbb{Q}/\mathbb{Z}) \rtimes \mathbb{N}^*, \qquad H = \log n, \qquad \sigma_t = e^{itH}.$$

Its p-adic reduction yields the state space, and the five axioms of quantum mechanics follow as consequences of its structure (§2–§4):

$$\mathcal{H}_p = L^2(\mathbb{Z}/p^K\mathbb{Z}).$$

Translation-invariant propagation on the state space is carried by the convolution operator, which is diagonal in the character basis; the elementary states, the emergent states, and the force channels are all read from its eigenvalues (§3.2, §6–§7):

$$K\chi_k = L(\beta, \chi_k)\,\chi_k.$$

The Born rule gives the measurement probability–the overlap between the KMS state and the character directions; the evaluation point $\beta/2 = \frac{1}{2}$ is fixed by the modular structure, and the resonance ratio R_k thereby enters all subsequent couplings and corrections (§4.5):

$$P(\chi_k) = \frac{|R_k|^2}{\sum_j |R_j|^2}, \qquad R_k = \frac{L(\frac{1}{2}, \chi_k)}{\zeta(\frac{1}{2})}.$$

The intrinsic constant C^* is given by the metric structure of the Hamiltonian (§3.4):

$$C^* = \sum_{n\geq 1} |\nabla H(n)|^2.$$

For a process participating with spectral weight W, its energy scale is defined by normalization against C^*; the three choices of W–the unit group, the full group, and the spectral weight of the χ_0 direction– give the electroweak, Planck, and nucleon energy scales respectively (§5.3, §10):

$$E(W)^2\, C^* = W \log^2 W.$$

Mass is the evaluation of the KMS state on the complete projection; elementary states factorize along the transition path into L-values, node factors, and generational factors, while emergent states are read out from the spectral weight via the energy-scale formula (§6.4–§6.5, §12–§13):

$$m_f = v \times \prod_k \left|L(\tfrac{1}{2}, \chi_{s_k})\right|^{G_k C_k} \times \prod_k \mathcal{T}_{s_k} \times \mathrm{Gen}_f, \qquad m_{\mathrm{em}} = \sqrt{\frac{\zeta(\beta)\log^2\zeta(\beta)}{C^*}}.$$

The equations of the two forces are combined into one table: the resonance force acts on the direction degrees of freedom (§7.6), and the metric force acts on the position degrees of freedom (§8):

Equation	**Meaning**
Resonance force (§7.6)	
$\Delta(k) = \Phi(\beta)\left[J(k+1) - J(k)\right]$	Resonance constraint: distribution in frequency space
$\dfrac{\partial}{\partial\beta}\log R_k = -\sum_{n\geq 2}\dfrac{\Lambda(n)\,[\chi_k(n)-1]}{n^\beta}$	Green-kernel evolution: couplings run with the energy scale
$\dfrac{\partial A_k}{\partial\beta} + \nu_k\,A_k + \sqrt{\alpha_k}\sum_{i+j\equiv k} f_{ijk}\,A_iA_j = \mathcal{J}_k$	Nonlinear resonance: amplitude self-coupling
$\dfrac{\partial}{\partial\beta}\log\zeta(\beta) = -\sum_{n\geq 2}\dfrac{\Lambda(n)}{n^\beta}$	ζ evolution: participation scale of the emergent state
$\alpha_k = \lvert R_k\rvert^2/N_k^{\mathrm{eff}}$	Coupling constant: resonance ratio diluted by the number of equivalent paths (§11)
$\alpha_{\mathrm{mix}} = \lvert\mathrm{Aff}\rvert_{\mathrm{eff}}\,\lvert R_k\rvert^2$	Emergent resonance coupling: dilution replaced by channel summation
Metric force (§8)	
$F_{\mathrm{grav}}(n) = -\nabla H(n) = -\log(1+1/n)$	Force field: negative gradient of the potential
$\sigma_t(F_{\mathrm{grav}}) = F_{\mathrm{grav}}$	Static background
$\partial F_{\mathrm{grav}}/\partial k = 0$	Universality: independent of the frequency directions
$w_\beta(n) = e^{-\beta H(n)}/Z(\beta)$	Equilibrium distribution: the response of matter
$-\Delta(\log n) = \log[n^2/(n^2-1)]$	Poisson identity: source density
$\partial_\beta C^*_\beta < 0$	Energy-scale evolution
$G = C^*/(N\log^2 N)$	Metric-force constant (N is the mode count of the full group, §11.6)

Here $\Delta(k)$ is the resonance deviation, $J(k)$ the order-jump source, $\mathcal{J}_k(\beta)$ the matter source term, $\nu_k = -\partial_\beta \log R_k$, and $f_{ijk} = \delta_{i+j,k}$ is nonzero only within the same CRT factor; when the linear part stands alone it reduces to the Green-kernel evolution, and the $k = 0$ limit is the ζ evolution. The Poisson source density is strictly positive and decays with n as $1/n^2 + O(1/n^4)$ (the Newton limit).

§17.2 Summary Table of Numerical Results

The characteristic quantities of the formulas of the previous subsection vary with p: the generation number $(p-1)/2$, the number of elementary states $p-2$, the number of force types $\omega(p-1)+1$, and the gauge group $\big(U(1)\times\prod_i SU(q_i^{b_i})\big)/\mathbb{Z}_{p-1}$. Comparing with observation: three generations of fermions, five elementary particles, three gauge forces, and $(U(1)\times SU(2)\times SU(3))/\mathbb{Z}_6$– the four conditions jointly single out $p=7$ as the unique compatible channel (§9). With $p=7$ fixed, all numerical outputs are compared with experiment [45] in a single table:

Quantity	**Predicted**	**Experiment**	**Deviation**	**Section**
v_{EW}	246.19 GeV	246.22 GeV	−0.011%	§10.1
M_{Pl}	1.221×10^{19} GeV	1.2209×10^{19} GeV	+0.04%	§10.2
m_p	937 MeV	938.3 MeV	−0.17%	§10.3
α_s^{-1}	8.49	8.48	+0.08%	§11.4
$\alpha_{\mathrm{EM}}^{-1}$	137.036	137.036	+0.0003%	§11.4
α_W^{-1}	29.60	29.60	−0.004%	§11.4
$\alpha_{\mathrm{nuclear}}$	14.10	14.0 ± 0.3	+0.7%	§11.5
G	6.70×10^{-39} GeV^{-2}	6.71×10^{-39} GeV^{-2}	−0.09%	§11.6
m_τ	1776.93 MeV	1776.93 MeV	−0.0002%	§12.1
m_μ	105.655 MeV	105.658 MeV	−0.003%	§12.1
m_e	0.51101 MeV	0.51100 MeV	+0.002%	§12.1
m_b	4185 MeV	4183 MeV	+0.05%	§12.2
m_s	92.1 MeV	93.5 MeV	−1.5%	§12.2
m_d	4.65 MeV	4.70 MeV	−1.1%	§12.2
m_c	1263 MeV	1273 MeV	−0.8%	§12.3
m_t	173.0 GeV	172.57 GeV	+0.3%	§12.3
m_u	2.15 MeV	2.16 MeV	−0.3%	§12.3
m_{ν_3}	50.6 meV	∼50 meV	+1.3%	§12.4
m_{ν_2}	9.96 meV	—	—	§12.4
m_{ν_1}	5.07 meV	—	—	§12.4
M_W	80.6 GeV	80.37 GeV	+0.25%	§14.3
M_Z	91.9 GeV	91.19 GeV	+0.78%	§14.3
m_π	138.3 MeV	139.6 MeV	−0.9%	§14.5
m_H	123.1 GeV	125.20 GeV	−1.7%	§14.6
m_γ	0	0	—	§14.2
Gluon	0 (confined)	—	—	§14.4
m_b/m_τ	2.355	2.354	+0.1%	§12.2
$\Delta m_{31}^2/\Delta m_{21}^2$	34.5	33.8 ± 0.9	0.8σ	§12.4
m_n-m_p	1.30 MeV	1.293 MeV	+0.5%	§13.4
η	6.18×10^{-10}	$(6.12\pm0.04)\times10^{-10}$	+1.0%	§16.5
$\Omega_b h^2$	0.0226	0.02237 ± 0.00015	+1.0%	§16.5
$\Omega_{\mathrm{DM}}/\Omega_b$	$16/3=5.333$	5.364 ± 0.066	−0.6%	§16.6
Ω_Λ	0.685	0.685	—	§16.6
f_{NL}	2.27	-0.9 ± 5.1	within 1σ	§16.2

The three energy scales deviate by less than 0.2%; the five coupling constants are within 0.7%, of which the three gauge couplings are within 0.1%; the nine charged-fermion masses are within 1.5%; the four boson masses are within 1.7%; the nuclear quantities are within 1%; the cosmological parameters are within 1%; and f_{NL} and the neutrino mass-squared-difference ratio are within 1σ. None of these outputs contains a free parameter; the only external input is the observational anchor H_0 (equivalently T_{CMB}) that locates the present moment, which enters only the conversion of $\Omega_b h^2$ and Ω_Λ and takes no part in any structural derivation (§16).

§17.3 Testable Predictions

Beyond the numerical outputs of the summary table, the derivation chain yields a set of qualitative and quantitative predictions that can be decided by future observation, listed here in the order of the main text:

1. Cabibbo angle $\lambda = 0.2247$: the corrected-level readout of the locked link (§12.1). The three classes of $|V_{us}|$ determinations are currently in tension with one another: $K_{\ell 3}$ decays give 0.2233, $K_{\ell 2}$ and the global CKM fit give 0.2250, and unitarity inferred from superallowed β decays gives 0.2277 [45]; our value lies on the $K_{\ell 2}$/fit side, differing from the fit value by -0.13%.
2. Neutrino absolute mass spectrum $(5.07, 9.96, 50.6)\,\mathrm{meV}$, $\Sigma m_\nu = 65.7\,\mathrm{meV}$, normal ordering, given by the indirect-channel mechanism (§12.4); the combined sensitivity of CMB-S4 [56] and DESI [57] covers this value.
3. The baryon is an emergent state: a single collective condensate in the χ_0 direction; it is an elementary particle, not a bound system of three quarks (§13).
4. The proton is stable: its stability is protected by the condensate structure itself, does not rely on an imposed conservation law, and no decay channel exists (§13) [53].
5. Color confinement is a structural property rather than a dynamical effect; free quarks cannot be observed in isolation (§13).
6. The Higgs is not an elementary particle but an oscillation sequence of the condensate (§14.6).
7. Within the $p = 7$ channel there is no structural position to accommodate a fourth generation of fermions, a sixth interaction, or a second Higgs (§15.4).
8. Primordial non-Gaussianity $f_{\mathrm{NL}} = 6/\sqrt{7} = 2.27$, an arithmetic constant independent of the current value of β (§16.2); the current Planck measurement is -0.9 ± 5.1, and the CMB-S4 target precision is $\sigma(f_{\mathrm{NL}}) \sim 1$.
9. Baryon asymmetry $\eta = 6.18 \times 10^{-10}$, current deviation $+1.0\%$ (§16.5); the combined Planck and BBN constraints are approaching 0.1% precision.
10. Neutrinos are of Majorana type: the source of the baryon-to-photon ratio requires the lift identifying particle and antiparticle in the χ_3 direction, and its realizability is equivalent to $p \equiv 3 \pmod 4$; the existence of matter, the Majorana nature of the neutrino, and the quadratic non-residuosity of -1 are given by one and the same piece of arithmetic (§16.5); the type will be decided by neutrinoless double-β-decay experiments [52].
11. All colored directions are complex characters; vertices of $\Delta B = 2$ type have no direction to carry them, and neutron–antineutron oscillation does not occur [54]; the violation of $B - L$ proceeds only through the lepton directions (§16.5).
12. Dark-energy equation of state $w_{\mathrm{DE}} = -1$: Λ arises from structural properties of the discrete geometry (negative Ollivier–Ricci curvature and the residue of

ζ), and contains no dynamical scalar field (§16.6); DESI [55] and Euclid [58] reach precision $\Delta w \sim 0.01$.

All of the above lie within the sensitivity of current or next-generation experiments.

§17.4 Closing Remarks

This paper started from the BC system and obtained the state space by p-adic reduction; its structure yields, in turn, quantum mechanics, spacetime, matter, forces, and cosmic evolution, with the numerical outputs collected in the summary table (§17.2). That the state space can carry this description is rooted in three basic facts about the physical world:

- Excitations are countable. Energy exchange has a smallest unit: blackbody radiation [30] and the photoelectric effect [31] show that fields exchange energy in quanta, and the spectra of bound systems are discrete; the excitation number of a single mode takes non-negative integer values, and measurement reduces to counting.
- Energy is additive. When non-interacting systems are combined, the total energy is the sum of the parts; energy is an extensive quantity, and the first law of thermodynamics and calorimetry rest on this.
- Independent composition multiplies. When mutually independent systems form a whole, the state of the whole is formed by pairing one choice from each side: two systems with microstate counts W_1 and W_2 form a whole with W_1W_2; the Boltzmann entropy [32] $S = k\log W$ is thereby additive; probabilities of independent events multiply, partition functions of independent systems multiply, and in quantum mechanics this is written as the tensor product of state spaces (§1).

These three facts coincide, item by item, with the foundations of arithmetic: countability supplies the objects, and merging and independent pairing are the only two ways countable objects combine. Under counting, merging gives addition, $|A \sqcup B| = |A| + |B|$; independent pairing gives multiplication, $|A \times B| = |A| \cdot |B|$; the distributive law $a(b+c) = ab + ac$ is the compatibility condition between the two. $(\mathbb{N}, +, \times)$ is thereby completely determined, and the unique factorization theorem holds within it:

$$n = \prod_p p^{a_p}, \qquad H = \log n = \sum_p a_p \log p,$$

with the primes as the irreducible elements under multiplication–the excitations that cannot be further divided. Every system in which excitations are countable, energy is additive, and independent composition multiplies realizes the unique factorization of $\mathbb{N}^*$ and thus carries the BC system. Its state space structure thereby naturally describes the physical world:

$$\mathcal{A}_{\mathrm{BC}} = C^*(\mathbb{Q}/\mathbb{Z}) \rtimes \mathbb{N}^*, \qquad \mathcal{H}_p = L^2(\mathbb{Z}/p^K\mathbb{Z}).$$

Data availability

No datasets were generated or analyzed during the current study. All results are obtained by analytic derivation.

Conflict of interest

On behalf of all authors, the corresponding author states that there is no conflict of interest.

§A Derivation and uniqueness of the evaluation rules

Stable states form in the KMS heat bath: energy propagates station by station along the torus, attenuating at each step, and grows into a standing wave; when a quantum number that the path is required to display is incompatible with the mode held at the endpoint, the wave train decoheres there and the phase is carried away by the vacuum. Mass is the accumulated attenuation of this formation process. Where the formation process projects, when it is forced to decohere, and how many times it decoheres are determined by the conservation laws and the complementarity of the system: the system carries three symmetries—the modular flow (the intrinsic evolution of the KMS equilibrium state), parity (the involution ι), and complex conjugation (the Galois action)—together with one grading (the p-adic valuation), and propagation and projection must be compatible with them; meanwhile position and mode are complementary kinds of information and cannot be sharpened simultaneously. This appendix shows that the eighteen evaluation rules of the main text are precisely the complete inventory of these physical constraints: the conservation laws give the selection rules, complementarity gives the decoherence criterion, and the destination of the phase gives the decoherence count; the modular structure of the spectral-weight KMS state supplies the precise formulation and the verification. The argument is carried out for all odd primes p; the concrete numerical examples take $p = 11$ and $p = 13$, while the physical channel $p = 7$ is identified in the main text.

§A.1 State space and the KMS state

Take the section of the state space at the unit-group level, $\mathcal{H} = L^2\big((\mathbb{Z}/p\mathbb{Z})^*\big)$, of dimension $n := p-1$. The mode basis (the character basis of the main text) consists of the Dirichlet characters $\{\chi_k\}_{k=0}^{n-1}$ (the eigenbasis of the convolution operator K); the lattice basis consists of the group elements $\{\delta_a\}_{a\in(\mathbb{Z}/p\mathbb{Z})^*}$; the transition coefficients between the two bases are $\langle\delta_a|\chi_k\rangle = \chi_k(a)/\sqrt{n}$. The involution $\iota : a \mapsto p - a$ and complex conjugation $\chi \mapsto \bar{\chi}$ are introduced as in the main text. The spectral-weight KMS state is

$$\varphi(A) = \mathrm{Tr}(\rho A), \qquad \rho = \frac{1}{Z}\sum_k w_k P_{\chi_k}, \qquad w_k = \big|L(\tfrac{1}{2}, \chi_k)\big|^2, \quad Z = \sum_k w_k,$$

with modular flow $\sigma_t = \mathrm{Ad}(\rho^{it})$. The full algebra is $\mathcal{M} = B(\mathcal{H}) \cong M_n(\mathbb{C})$.

There are only two external inputs, both results established in the main text:

1. The two modes of superposition (the Born rule of §4 of the main text): partial amplitudes of a coherent segment superpose at amplitude level (evaluation at $\beta = \frac{1}{2}$), partial intensities of a decoherent segment superpose at intensity level (evaluation at $\beta = 1$).

2. The evaluation level (main text §6.4, §11): real processes in the post-transition vacuum read at the corrected level, inputs internal to a virtual process read at the Born level; this level assignment is derived in the main text from the temporal structure of the process (the unique summation of the insertion series and the time average of the insertion flow, main text §6.4).

Sections §A.2–§A.4 answer three questions in turn: where projection takes place, when decoherence is forced to occur, and how many times it occurs. §A.5 derives all eighteen rules from the three answers, §A.6 establishes completeness and uniqueness, and §A.7 states the boundary conditions.

§A.2 Conservation sectors of projection

A projection of the standing wave extracts some observable of the system and must therefore be compatible with its conservation structure: only processes that do not disturb thermal equilibrium can leave a definite weight within coherent evolution. In operator language, a projection, a correction, or a decoherence event is expressed as a conditional expectation compatible with φ, and the compatibility criterion (Takesaki, 1972) is: a φ-preserving conditional expectation $E : \mathcal{M} \to \mathcal{N}$ exists if and only if the target subalgebra is globally invariant under the modular flow, $\sigma_t(\mathcal{N}) = \mathcal{N}$. This section uses the criterion to inventory all projection locations permitted by the conservation laws.

A.2.1 Action of the modular flow on matrix units

Compute directly on the matrix units $e_{kl} = |\chi_k\rangle\langle\chi_l|$ of the mode basis: $\rho^{it} = Z^{-it} \sum_k w_k^{it} P_{\chi_k}$, hence

$$\sigma_t(e_{kl}) = \rho^{it} e_{kl} \rho^{-it} = \left(\frac{w_k}{w_l}\right)^{it} e_{kl}.$$

The modular flow does not move the cell (k, l); it only endows it with the phase $(w_k/w_l)^{it}$. Two immediate consequences:

1. For any cell set $S \subset \{0, \ldots, n-1\}^2$, the subspace $\mathcal{A}_S = \operatorname{span}\{e_{kl} : (k,l) \in S\}$ always satisfies $\sigma_t(\mathcal{A}_S) = \mathcal{A}_S$; it is a subalgebra if and only if S contains the diagonal, is closed under transposition, and is closed under cell composition $(k,l)(l,m) = (k,m)$.

2. If E is the cell truncation onto $\mathcal{A}_S$ (keep inside S, set zero outside), then since ρ is diagonal and truncation preserves the diagonal, $\operatorname{Tr}(\rho E(A)) = \operatorname{Tr}(\rho A)$ holds for all A: truncation automatically preserves φ. Complete positivity of E follows from $E = \sum_j V_j \cdot V_j^*$ (with V_j the row-block projections of S), and the bimodule property of a conditional expectation is verified entry by entry.

A.2.2 Degeneracy structure of the spectral weights

Subalgebras that are not of cell type may also be invariant; the criterion reduces to the weight spectrum. The eigenvalue family of σ_t is $\{(w_k/w_l)^{it}\}$, and invariant subalgebras are controlled by the equality classes of weight ratios. Equalities of weights have two sources:

First, identities. The functional equation gives $\overline{L(\frac{1}{2}, \chi)} = \epsilon\, L(\frac{1}{2}, \bar{\chi})$ $(|\epsilon| = 1)$, hence

$$w_k = \left|L(\tfrac{1}{2}, \chi_k)\right|^2 = \left|L(\tfrac{1}{2}, \bar{\chi}_k)\right|^2 = w_{k'} \qquad (\chi_{k'} = \bar{\chi}_k) :$$

the weights of a conjugate pair coincide identically, for all p, independently of numerics.

Second, accidental equalities. Weight equalities between non-conjugate characters have no known source; for a given p this is a verifiable numerical condition. Write

- ND1: the weights of non-conjugate characters are pairwise distinct;
- ND2: among all ratios w_k/w_l ($k \neq l$), those belonging to different conjugate-class pairs are pairwise distinct and distinct from 1.

Verification is given in §A.7. The argument below assumes ND1 and ND2.

A.2.3 The chain of invariant subalgebras

Under ND1, the fixed points of σ_t (the centralizer $\mathcal{M}^\varphi$) are spanned by the cells with $w_k = w_l$: the full diagonal, plus the two antidiagonal cells of each complex conjugate pair $\{\chi, \bar{\chi}\}$. Real characters ($\chi = \bar{\chi}$) contribute no block. With m_c the number of complex conjugate pairs,

$$\dim \mathcal{M}^\varphi = n + 2m_c.$$

For an odd prime p there are exactly two real characters (χ_0 and the quadratic character), so $m_c = (n-2)/2$ and $\dim \mathcal{M}^\varphi = 2n - 2 = 2p - 4$.

The cell-type subalgebras carrying projection structure form a chain under inclusion. Take the parity grading $\epsilon(k) \in \{0, 1\}$, $\chi_k(-1) = (-1)^{\epsilon(k)}$; the even-difference cell set $S_{\text{even}} = \{(k,l) : \epsilon(k) = \epsilon(l)\}$ contains the diagonal and is closed under transposition and composition (additivity of the grading), so its span $\mathcal{A}_{\text{even}}$ is a subalgebra, of dimension $n^2/2$. Thus

$$\boxed{\mathcal{A}_{\text{diag}}\ (n)\ \subset\ \mathcal{M}^\varphi\ (2n-2)\ \subset\ \mathcal{A}_{\text{even}}\ \tfrac{n^2}{2}\ \subset\ \mathcal{M}\ (n^2)}$$

$p = 11$: $10 \subset 18 \subset 50 \subset 100$; $p = 13$: $12 \subset 22 \subset 72 \subset 144$ (for both examples the invariance, the dimensions, and the exclusion property of the next subsection are verified numerically, §A.7).

Each of the three levels carries an explicit conditional expectation, verified one by one:

1. The mode-diagonal algebra $\mathcal{A}_{\text{diag}}$: E_{diag} = diagonal truncation. Amplitude-level projection (the extraction of L-values) takes place at this level.
2. The centralizer $\mathcal{M}^\varphi$: $E_\varphi(A) = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} \sigma_t(A)\,dt$. Acting cell by cell, this average retains exactly the cells with $w_k = w_l$ and eliminates the rest (the phases average to zero), so the image is $\mathcal{M}^\varphi$, and φ-preservation follows from the truncation property. Corrections (the two-vertex virtual processes between conjugate pairs) occur only inside the 2×2 blocks of this level; real characters have no block, so real-character directions receive no correction—the corresponding clause of the main text is obtained here. This agrees with the common fact of perturbation theory: nondegenerate vibrational modes do not mix, only degenerate (resonant) modes mix.

3. The even-difference algebra $\mathcal{A}_{\text{even}}$: the parity operator in the mode basis is $\Pi = \operatorname{diag}\bigl(\chi_k(-1)\bigr)$. Transforming to the lattice basis:

$$\langle\delta_a|\Pi|\delta_b\rangle = \frac{1}{n}\sum_k \chi_k(-1)\chi_k(a)\bar{\chi}_k(b) = \frac{1}{n}\sum_k \chi_k(-ab^{-1}) = \delta_{b,-a} :$$

in the lattice basis Π is the reflection $a \mapsto p-a$, i.e. the involution ι. Hence

$$E_{\text{even}}(A) = \tfrac{1}{2}\bigl(A + \iota A \iota\bigr)$$

is the conditional expectation onto the Π-even part (that is, $\mathcal{A}_{\text{even}}$); it preserves φ because Π commutes with ρ. The mode-basis matrix elements of the generation projection $\delta_b + \delta_{p-b}$ are

$$\langle\chi_k|\,(\delta_b{+}\delta_{p-b})\,|\chi_l\rangle = \frac{1}{n}\Bigl(\chi_l(b)\bar{\chi}_k(b){+}\chi_l(-b)\bar{\chi}_k(-b)\Bigr) = \frac{\chi_l\bar{\chi}_k(b)}{n}\Bigl(1{+}\chi_l\bar{\chi}_k(-1)\Bigr),$$

which vanish identically on odd-difference cells ($\chi_l\bar{\chi}_k$ odd): the display of the generation quantum number falls entirely within the even-difference level, and the even-character entry condition follows. This is isomorphic to the parity selection rule of atomic spectra: transitions forbidden by parity cannot be excited. Decoherence takes place at this level.

The chain does not exhaust the invariant subalgebras: on the 2×2 block of a conjugate pair σ_t acts as the identity ($w_k = w_{\bar{k}}$) and the restriction of ρ is a scalar, so every subalgebra inside the block is invariant and every projection in any direction preserves φ. This does not change the counting: whichever direction the intra-block basis takes, the center of a two-dimensional commutative subalgebra supplies exactly one binary quantum number, and §A.4.3 uses only the count. The intra-block basis direction is fixed by the eigenbasis of the only nontrivial framework operator inside the block, the insertion operator (main text §12); that the mass states are the $\pm$ symmetrized combinations is thereby determined.

A.2.4 Exclusion of the lattice projections

The mode-basis matrix elements of a lattice projection δ_a, $\langle\chi_k|\delta_a|\chi_l\rangle = \chi_l(a)\bar{\chi}_k(a)/n$, vanish nowhere. Consider the smallest σ-invariant subalgebra $\mathcal{N}_a$ containing δ_a:

Step one, Fourier separation. The (k,l) component of $\sigma_t(\delta_a)$ carries the phase $(w_k/w_l)^{it}$; weighted averages over t (finitely many t's, with linear combinations approximating indicator functions) split δ_a into classes according to the value of the ratio w_k/w_l. Under ND2, each class with ratio $\neq 1$ is exactly the cell block of one conjugate-class pair (c,c'), with component $P_c\,\delta_a\,P_{c'}$ (P_c the projection onto the subspace of conjugacy class c); the class with ratio $=1$ consists of all centralizer cells—the diagonal falls in together with the block antidiagonals, because $w_k = w_{\bar{k}}$ is precisely the identity of §A.2.2—yielding the single component $\sum_c P_c\delta_a P_c$.

Step two, the exact range of single-point generation. δ_a has rank one, $\delta_a = |v\rangle\langle v|$, so every component is a compression of δ_a onto conjugacy-class subspaces: $P_c\delta_a P_{c'} = |v_c\rangle\langle v_{c'}|$ with $v_c := P_c v$. Products act only within the span of the vector family $\{v_c\}$, giving

$$\mathcal{N}_a = \mathcal{M}(W_a) \oplus \mathbb{C}\,\mathbf{1}_{W_a^\perp}, \qquad W_a = \operatorname{span}\{v_c\}, \quad \dim W_a = m_c + 2 :$$

single-point generation does not reach the full algebra. From this structure, the dimension of the commutant of $\mathcal{N}_a$ is $1+(n-m_c-2)^2$, i.e. $5, 17, 26$ for $p = 7, 11, 13$; the numerical verification agrees in all three cases (§A.7, Note 1).

Step three, generation of the complete position frame. The smallest σ-invariant subalgebra containing all $\{\delta_a\}_{a\in(\mathbb{Z}/p\mathbb{Z})^*}$ reaches the full algebra: within one complex class $c = \{k, \bar{k}\}$ the components of $v_c^{(a)}$ are $(\bar{\chi}_k(a), \chi_k(a))$, and as a runs over the unit group two distinct characters are linearly independent, so $\{v_c^{(a)}\}_a$ spans each class subspace; products of the cross-class components $|v_c^{(a)}\rangle\langle v_{c'}^{(a)}|$ then produce all matrix units, hence $\mathcal{N}_{\text{all}} = \mathcal{M}$. In all three examples the all-a commutant is scalar (§A.7, Note 1). The criterion for a pair of lattice points $\{a, a'\}$ to suffice: for every complex class k, $2kt \not\equiv 0 \pmod{n}$, where t is the discrete logarithm of $a'a^{-1}$; for $p = 13$ the pair $(2, 3)$ fails on the two complex classes with k even, the predicted commutant dimension is $1 + 2^2 = 5$, and the numerics agree—the lemma is therefore stated with all lattice points.

Hence the complete position frame is contained in no proper invariant subalgebra: within coherent evolution there exists no admissible projection that sharpens all lattice points simultaneously. On the other hand $\mathcal{N}_a$ itself is σ-invariant and the Takesaki condition is satisfied, so projection onto a single position is admissible, consistent with the generation projection $\delta_b + \delta_{p-b}$ of §A.2.3, level 3: single-point readout is admissible, simultaneous sharpening of the complete frame is not.

At this point the three symmetries have each carved out a conservation sector: amplitude-level projection lives in the mode-diagonal sector, corrections live in the zero-frequency (degenerate) sector of the modular flow, and decoherence lives in the parity-conservation sector; outside the conservation sectors there is no admissible projection. The three symmetries are not chosen: complex conjugation and the modular flow are extracted uniquely from the algebraic structure by the polar decomposition $S = J\Delta^{1/2}$ (main text §4), and ι is the same involution that builds the spacetime doublet in §5 of the main text. With the locations settled, the next section turns to the paths on which decoherence must occur.

§A.3 The decoherence criterion

Decoherence is a consequence of complementarity: the spatial detail of a standing wave and the vibrational mode it locks cannot both be had. When the generation quantum number that the path is required to display is position-type information while the endpoint locks a single mode, the two cannot be sharpened simultaneously, and the generation quantum number can only be displayed in one decoherence event. This section turns that consequence into a criterion; two facts are needed: the uniform distribution after a sharp mode projection, and the endpoint structure of paths.

A.3.1 Sharp mode projection and uniform distribution

Suppose the final station performs an amplitude-level sharp projection onto the mode χ_k; the state afterwards is $|\chi_k\rangle$ (up to phase). Its lattice distribution is

$$\left|\langle\delta_a|\chi_k\rangle\right|^2 = \frac{|\chi_k(a)|^2}{n} = \frac{1}{n} \qquad \forall a :$$

uniform everywhere. A standing wave locking a single mode necessarily spreads out in space—any non-uniform intergenerational weight ρ_j (j the generation index) has nothing to act on downstream of a sharp mode projection: non-uniform spatial weights must be displayed upstream of the sharp projection.

A.3.2 Endpoint structure of paths

The three final-state types of §6.4 of the main text: the final station of an open path is an amplitude-level mode projection (covering number $C = 1$); the final state of a closed path is $|\Omega\rangle$, whose construction carries no lattice information; the evaluation of a single-station path is itself one intensity-level closure.

A.3.3 Case check over path topologies

1. Single-station path. The evaluation contains a single squared modulus and the result is already a classical weight; the generational factor ρ_j appears directly as a ratio of classical weights, and no extra decoherence is needed for any j.

2. Closed path. The final state $|\Omega\rangle$ has no lattice information and no position projection occurs at the endpoint; the generation descent is carried by self-loops inside the path, the closing squared modulus performs an inclusive summation over the intermediate quantum numbers of the self-loop, the quantum number is displayed inside the summation, and no independent decoherence is produced.

3. Open path, $j = 0$. The final station is a sharp mode projection; the $j = 0$ weight is the σ-symmetric combination (the invariant component of the uniform distribution), compatible with the post-projection uniform distribution; no decoherence is needed.

4. Open path, $j \geq 1$. By §3.1 the distribution downstream of the final station is uniform and ρ_j has nowhere to appear; the display must move upstream of the final station, and by Input 1 it can only proceed at intensity level. Upstream there is no pre-existing intensity-level cross-section (the intermediate stations of the path are identity transfers, §A.2.1), so the wave train must decohere once upstream; the location of the decoherence is confined by §A.2 to the parity-conservation level.

Combining the four cases:

$$\boxed{\text{decoherence is forced} \iff \text{the path topology is open} \wedge \text{the generation index } j \geq 1}$$

The criterion holds for all odd primes. As an application, check the nine charged fermions of the $p = 7$ channel identified in the main text one by one: τ, μ, e are single-station (case 1), c, t, u are closed paths (case 2), b is an open path with $j = 0$ (case 3), and s, d are open paths with $j \geq 1$ (case 4). The decoherence assignment

$$(\tau, \mu, e, b, s, d, c, t, u) \longmapsto (0, 0, 0, 0, 1, 1, 0, 0, 0)$$

agrees bit for bit with the assignment used by all mass formulas of the main text; the assignment is determined by the residence character. With existence settled, the remaining question is the count.

§A.4 Decoherence counting

Decoherence is a physical event: the phase lost by the wave train must be carried away by some excitation in the vacuum; if no excitation carries the phase away, the wave train is still coherent and nothing has happened. The number of decoherence events is therefore counted by the destination of the phase; this can be written as a precise identity. The environment state carrying the phase will be called the record state—from here on, "record" refers exclusively to this physical object.

A.4.1 The phase-transfer criterion

Insert at the cut a decomposition $\{P_x\}$ ($\sum_x P_x = \mathbf{1}$), with the phase of the intermediate quantum number x carried by the environment state $|\varepsilon_x\rangle$. The process splits into segment A (before the cut) and segment B (after the cut), with joint amplitude

$$|W\rangle = \sum_x BP_xA|\text{in}\rangle \otimes |\varepsilon_x\rangle.$$

Taking the squared modulus for a specified final state $|\text{out}\rangle$ and tracing over the environment, with $M_x = \langle\text{out}|BP_xA|\text{in}\rangle$:

$$P = \sum_{x,x'} M_x\bar{M}_{x'}\,\langle\varepsilon_{x'}|\varepsilon_x\rangle.$$

Two extreme cases:

1. Orthogonal environment states, $\langle\varepsilon_{x'}|\varepsilon_x\rangle = \delta_{xx'}$: the cross terms vanish, $P = \sum_x |M_x|^2$, partial intensities superpose at intensity level—the wave train decoheres here once;
2. Identical environment states, $|\varepsilon_x\rangle \equiv |\varepsilon\rangle$: $P = \left|\sum_x M_x\right|^2 = \left|\langle\text{out}|BA|\text{in}\rangle\right|^2$, the insertion equals the identity operation and the evaluation is unchanged—the phase has nowhere to go and the wave train stays coherent.

Decoherence is inseparable from the destination of the phase: a decoherence whose phase has not been carried away changes no observable result and amounts to not having happened. Intermediate overlaps do not occur: the basis of the record states lies in some center (§A.4.2), the quantum-number operator is self-adjoint and conserved, so record states in different eigensectors remain strictly orthogonal to the end of the process; distinctions within one sector are carried by no conserved quantity and are washed out by the modular-flow average in the KMS state (the same mechanism as level 2 of §A.2.3), leaving the record states effectively identical. The effective overlap therefore takes values in $\{0, 1\}$ only, and the two extreme cases exhaust all possibilities.

A.4.2 The carrier sectors of the phase

The phase can only be handed to a conserved quantity: only a quantity unchanged by the evolution can keep the outcome of an event to the end of the process. In operator language, the basis of the record states must lie in the center of the target of some φ-preserving conditional expectation (the classicalized quantum number must commute with all observables of that level). By the inventory of §A.2, the candidates within the closed system are only:

- the valuation quantum number (the center of the shell algebra, full-ring level, see §A.7 Note 3);
- the spectral-class quantum number (the center of the centralizer);
- the ι-symmetry quantum number (the center of the even-difference level);
- the ensemble (the trivial subalgebra $\mathbb{C}$, conditional expectation $E_{\mathbb{C}}(A) = \varphi(A)\mathbf{1}$).

There is no other location that can carry the phase. The mode-diagonal level does not enter the list: the admissible operation of that level is amplitude-level projection, the post-projection state retains its phase, no phase transfer occurs, and no decoherence event is constituted.

A.4.3 The counting law

Decoherence occurs if and only if there exists a quantum-number operator Z belonging to one of the centers above, commuting with the final-state projection, with the map from intermediate values to Z-eigenvalues injective. Injectivity guarantees that the destinations of the phase are distinguishable (§4.1, case 1); if any condition fails, the situation degenerates to §4.1, case 2. Hence

$$\boxed{\text{number of decoherence events} = \text{number of independent classical quantum numbers of the final sta}}$$

Each decoherence event classicalizes exactly one quantum number. Repeated projections of the same quantum number: conditional expectations are idempotent, $E \circ E = E$; a second projection produces no new event and no second normalization factor. Checks: at the cut of an open path with $j \geq 1$, the classicalized quantum number is not the generation index itself (ρ_j is a weight, an intensity-level object from the start, and carries no phase) but the spectral-class quantum number of the shared endpoint of the decoherent segments (item two of the list in §A.4.2): different segment endpoints belong to different conjugacy classes, injectivity holds under ND1, one cut classicalizes exactly one quantum number, the count is one, in agreement bit for bit with the assignment of §A.3.3; paths displaying no quantum number do not decohere; each ensemble relay (the action of $E_{\mathbb{C}}$) corresponds to one independent phase transfer.

§A.5 Derivation of the eighteen rules

Given the state φ, the two inputs, and the residence character χ_f, the factorization of the evaluation matrix element is uniquely determined by the three answers of §A.2–§A.4. The derivation holds for all odd primes; the table below is stated in the notation of the main text, where the quantities tied to the physical channel (d_1, d_2, N, λ, α_c, α_o, δ, r, etc.) take their $p = 7$ values as identified in the main text, and for general p stand for the corresponding quantities of the respective channels. The eighteen rules of the main text follow item by item:

Rule	Content	Derivation	Basis of uniqueness

R1	closed path: all stations $C=2$	§A.3 case 2	final state has no lattice information; global closure unique
R2	open path: last position $C=1$, others $C=2$	§A.3 cases 3–4	expanded below
R3	decoherent segments close independently; shared endpoint $C=1{+}1$	§A.4.1	orthogonal record states kill inter-segment cross terms
R4	repeated passages add covering numbers	§A.3+§A.4	each passage enters the decomposition independently
R5	one share of background normalization $1/\lvert\zeta(\frac{1}{2})\rvert$ per segment	§A.4.3	expanded below
R6	decoherent-segment vertex exemption $d_1(1/\sqrt{d_1})^2=1$	§A.4.1	inclusive summation exactly restores the conjugate copies
R7	$a=0$ distinguishable: $q^b/\lvert\zeta(\frac{1}{2})\rvert$	§A.2.3 level 1 + Input 1	distinguishable quantum numbers count at intensity level
R8	$a=0$ indistinguishable: $1/\sqrt{q^b}$	§A.2.3 level 1 + Input 1	invariant subspaces project at amplitude level
R9	$a\neq 0$: factor 1	§A.2.3 level 1	selection already made, no residual freedom
R10	conjugate members normalized by $1/\sqrt{G}$	§A.2.3 level 2	normalization of the response vector inside the degenerate block is unique
R11	inclusive counting $\times d_2$ of quantum numbers released by decoherence	§A.4.3	released distinguishable quantum numbers must be summed (color, in the physical channel)
R12	each χ_0 relay $\div N$	§A.4.2	expanded below
R13	$\alpha_c, \alpha_o, \lambda$ at corrected level	Input 2 + §A.4	dwell and emission are real processes with phase already transferred
R14	internal inputs $\delta, r, \alpha^{(0)}$ at Born level	Input 2	virtual intermediate states lie inside the closure
R15	frequency shift $\times(1-\alpha_{\text{eff}})$; exemption clauses	§A.2.3 level 2	expanded below
R16	nodes only at endpoints, intermediate stations identity	§A.2.1	diagonality of K makes intermediate projections identity
R17	three final-state types	§A.3.2	the topological trichotomy is exhaustive
R18	self-loop: closed $\to 1/\alpha_c$, open $\to \alpha_o$	§A.4	expanded below

Five rows of the table involve several upstream links; they are expanded one by one.

R2 (assignment of covering numbers). The final station of an open path is an amplitude-level mode projection, evaluated once there, $C=1$; the remaining stations lie inside the closing squared modulus, each station's response entering the product as a squared modulus, $C = 2$. The alternative assignment (last station also $C = 2$) would require one more intensity-level projection beyond the sharp mode projection, and outside the mode-diagonal level there is no φ-preserving projection location (§A.2.3).

R5 (background normalization). The number of shares is fixed by §A.4.3: each intensity-level closure is an independent decoherence event, and the number of decoherence events equals the number of classicalized quantum numbers. The base is fixed by Input 1: in the definition of the response, $R_k = L(\frac{1}{2}, \chi_k)/\zeta(\frac{1}{2})$, the denominator is the background amplitude at the vertex evaluation point ($\beta = \frac{1}{2}$), and each independent closure carries one share of the denominator of the definition, hence exactly one $1/|\zeta(\frac{1}{2})|$ per segment. An extra closure has no phase to transfer and, by §A.4.1 case 2, equals the identity operation, producing no extra share. The $1/|\zeta(\frac{1}{2})|$ of R7 has the same origin, and q^b is a distinguishable count.

R12 (ensemble relay). A passage through the χ_0 direction is the conditional expectation onto the trivial subalgebra, $E_{\mathbb{C}}(A) = \varphi(A)\mathbf{1}$ (the ensemble item of the §A.4.2 list): re-entering the KMS ensemble means being shared among the N modes, with the phase handed to the ensemble (one independent transfer per passage, §A.4.3), hence exactly one $1/N$ per passage, the starting point not counted.

R15 (frequency shift and its exemptions). Convolution eigenmodes carrying an insertion flow accumulate phase on the degenerate blocks of the centralizer; by the involution structure of the intra-block insertion operator (main text §12), the accumulation grows linearly in units of the dwell period, and the reading is shifted by $(1 - \alpha_{\text{eff}})$. Real-character directions have no degenerate block (§A.2.3, level 2) and no insertion flow: exempt; emergent states are cross-shell superpositions, not single-shell eigenmodes, and lie outside the domain of the evaluation (§A.7 Note 3): exempt. Both exemptions are corollaries.

R18 (self-loop dichotomy). Each dwell period of a closed self-loop ($\chi_s^2 = \bar{\chi}_s$) leaks into the emission channel with probability α_c, and the phase of the period is carried away by the emission channel (the external sector beyond the §A.4.2 list, see §A.7 Note 2), so the period count superposes at intensity level:

$$\sum_{n\geq 0}(1 - \alpha_c)^n = \frac{1}{\alpha_c}.$$

An open self-loop ($\chi_s^2 = \chi_0$) exits into the vacuum and does not return; the transfer is nilpotent in that channel, leaving only the single matrix element α_o. Dwell and emission are real processes with the phase already transferred (R13), the record states are orthogonal by period number (§A.4.1 case 1), the cross terms vanish, and the intensity-level superposition is thereby fixed; superposing at amplitude level instead would require the phase not to have been transferred, contradicting the classification of R13. Numerical cross-check: the amplitude-level dwell weight would be of order $|1/(1-a)|^2 \sim 4/\alpha^2$, several orders of magnitude away from the top-quark mass.

With this, all eighteen rules are derived; what remains is whether the rule set is complete and unique.

§A.6 Completeness and uniqueness

As to completeness, each type of factor in the mass formula belongs to the sector of one conservation law: coherent path evolution (the full algebra), endpoint factors (the mode-diagonal level), correction factors (the zero-frequency level of the modular flow), intergenerational and decoherence factors (the parity-conservation level), shell quantum numbers (grading conservation, at no decoherence cost), and relay factors (the ensemble). These are all the conservation laws there are (the chain is exhausted by §A.2), so the evaluation matrix element has no structural position for any other type of factor. This completeness is at sector level: it asserts that the conservation sectors to which each type of factor belongs are exactly these, not the selection rules within a sector (such as the λ^2 assignment of the four intergenerational cases), which are given in the corresponding sections of the main text.

As to uniqueness, with φ given, the rule set of §A.5 is unique; both inputs are corollaries of the main text (the two modes of superposition come from the modular structure $S = J\Delta^{1/2}$, main text §4; the evaluation level is determined by the temporal structure of the process, main text §6.4) and do not constitute independent conditions. The key alternatives are excluded one by one:

1. Adding decoherence: no phase to transfer, equals the identity operation (§A.4.1);
2. Removing decoherence (open path with $j \geq 1$ kept coherent): ρ_j has nowhere to appear (§A.3.1), contradicting the existence of the intergenerational hierarchy;
3. Relocating decoherence (outside the parity-conservation level): no φ-preserving conditional expectation exists there (§A.2.3–.4);
4. Amplitude-level superposition of dwell: phase transferred $\Rightarrow$ orthogonal records $\Rightarrow$ intensity-level superposition forced (§A.4.1 case 1 + R13); amplitude superposition requires the phase not transferred, contradicting dwell being a real process (R18 expansion; the top-quark mass is a numerical cross-check, not the basis of exclusion);
5. Adding corrections on real-character directions: the centralizer has no degenerate block there (§A.2.3 level 2);
6. Odd-character entry: the matrix elements of the generation projection vanish identically on odd-difference cells (the computation of §A.2.3 level 3).

The above completeness and uniqueness hold within the following boundary conditions.

§A.7 Boundary conditions

1. Nondegeneracy conditions. The conjugate degeneracy $w_k = w_{\bar{k}}$ is a functional-equation identity, valid for all p. For a given p, ND1 and ND2 are finite systems of strict inequalities and are provable by interval arithmetic: each w_k is computed as an interval with a rigorous error bound ($L(\frac{1}{2}, \chi)$ evaluated through the finite Hurwitz-ζ sum, with the error radius controlled throughout), and disjointness of two intervals proves the inequality. Verified at a working

precision of about 120 digits (error radii below 10^{-100}): $p = 7$, ND1 minimal gap ≥ 0.0899, ND2 minimal gap ≥ 0.0150, closest approach to 1 at least 0.109; $p = 11$, ND1 ≥ 0.0573, ND2 inter-class ≥ 0.0087, closest approach to 1 at least 0.055 (ratios internal to a conjugate pair are identically 1, belong to centralizer cells, and do not enter ND2); $p = 13$, ND1 ≥ 0.0101, ND2 ≥ 0.0022, closest approach to 1 at least 0.052. The gaps exceed the error radii by more than ninety orders of magnitude and the intervals are pairwise disjoint, so ND1 and ND2 are proved, in the computer-assisted sense, for $p = 7, 11, 13$; any other given p can be checked case by case in the same way. The chain dimensions, the dimension of the center of the centralizer $(2 + m_c)$, and the commutant dimensions of §A.2.4 (single-point $5, 17, 26$; all-a scalar) are verified for all three examples $p = 7, 11, 13$; the verification uses the commutant criterion, since naive floating-point rank iteration underestimates the generated rank and is not used as evidence. No a priori theorem covers the transcendence of the L-values; but the conditions above are inequalities, not equalities, their validity does not depend on transcendence, and disjointness of the intervals constitutes the proof.

2. External sector. The phase-destination list of §A.4.2 is complete within the closed system; in R18 the phase of the dwell period is carried away by the channel sector, which lies outside the closed system. Its existence is a corollary rather than an independent input: the KMS state of the full system factorizes along the primes—the operator form of $\zeta(\beta) = \prod_q (1 - q^{-\beta})^{-1}$ is $\varphi_\beta = \bigotimes_q \varphi_{\beta,q}$—so fixing p amounts to a partial trace over the remaining prime factors, and the complement of the closed system exists automatically and is infinite-dimensional; records stored in the complement commute with all evaluation operators inside the closed system, are strictly preserved to the end of the process, and infinite dimensionality excludes recurrence. The transfer of the phase is carried by the nonlinear resonance equation of §7.6 of the main text: its structure constants $f_{ijk} = \delta_{i+j,k}$ are the character multiplication table, the vertex strength $\sqrt{\alpha_k}$ is given by §7.4 of the main text, and the leak probability per dwell period is its squared modulus α. The choice of channel, however, is a corollary: the intensity-level series of the closed self-loop requires the record of each period to be kept to the end of the process (§A.4.2), so the emitted quantum must be massless and able to escape; a massive quantum loses the record, degenerating to §A.4.1 case 2, and the series fails; in the boson spectrum (main text §14) the only massless quanta are the photon and the gluons, and the gluons are confined, with no record state that can be kept, so α_c takes the electromagnetic coupling, uniquely. An open self-loop emits once into the vacuum, has no record-keeping requirement, is not subject to the massless selection, and its channel is fixed by conservation of the transition character (the channel character of $\chi_3 \to \chi_0$ is χ_3, the selection rule of §7 of the main text), so α_o takes the weak coupling.

3. Full-ring extension (valid for all K at once). The full ring $L^2(\mathbb{Z}/p^K)$ decomposes into shells by valuation, $|S_v| = \varphi(p^{K-v})$, and $\sum_v \varphi(p^{K-v}) + 1 = p^K$ is an identity ($\sum_{d|N} \varphi(d) = N$): the dimension decomposition needs no induction. Shell preservation holds operator by operator: multiplication by units prime to p, the modular flow, ι, complex conjugation, the insertion operator, and

the conditional expectations targeting the subalgebras they generate are all shell-diagonal; the only cross-shell candidate is the multiplication-by-p operator, which is not in the list of evaluation operations—this is the premise of the present note. Consequently different valuations are separated by superselection: cross-shell phases are not observable, there is no phase to transfer, and the clause "at no decoherence cost" follows; emergent states are cross-shell superpositions, not single-shell eigenmodes, lie outside the domain of the evaluation, and the exemption of R15 follows. As for the ND certificates needed for the chain to recur within each shell: the operator content of the evaluation matrix elements lives entirely in the $v = 0$ shell (the other shells enter the energy-scale formula only through the mode count N), so the certificate needed is the one already verified at the unit-group level. Numerics: $K = 2$ ($42 + 6 + 1 = 49$, orthonormality to 10^{-15}); $K = 3$ ($294 + 42 + 6 + 1 = 343$, orthonormality to 10^{-14}, the cross-shell norms of unit multiplication and ι exactly zero, the multiplication-by-p operator the unique shell shifter).

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